

# Hahn-Valued Measures and Probability

Strong atomicity, coefficientwise extension,  
and hidden negative mass

Merged research report from three manuscripts · 22 September 2026

## Abstract

We study measures with values in  $\mathbb{R}((t^\Gamma))$  and  $\mathbb{C}((t^\Gamma))$ , and through normal forms in the surreal and surcomplex numbers, under two meanings of countable additivity. A *coefficientwise* Hahn measure is a Hahn series of ordinary finite signed or complex measures with one well-ordered exponent support, disjoint unions being added exponent by exponent by ordinary absolutely convergent sums. A *strong* Hahn measure requires the masses of every disjoint sequence of events to be strongly Hahn summable.

*Strong class.* On every countably separated space a strong Hahn measure is a strongly summable family of point masses; neither a common support nor atomicity is assumed, both follow. It extends uniquely to all subsets, a strong probability has finitely supported standard part, and an invariant one lives on finite orbits. Coherent cylinder data on a countable product of finite sets extend strongly exactly when their combined support is well ordered and each exponent is active on boundedly many cylinders per level; independent finite-state coordinates extend exactly when finitely many residue laws are random and the individual rare-outcome probabilities are strongly summable, their aggregates not sufficing. The strong moment problem on an ordinary real sample set has an exact criterion. Positivity is separate: one weight pattern with support of order type  $\omega + 1$  gives a normalized strong signed measure whose cylinder masses on Cantor space, or whose integrals of nonzero nonnegative polynomials on  $[0, 1]$ , are all positive, yet whose mass at one point is  $-t$ ; in the strong class  $\omega + 1$  is the least order type allowing this. The interval example has explicit Hankel determinant valuations and positive Gaussian quadratures at non-Archimedean nodes.

*Coefficientwise class.* A transfinite null-ideal criterion decides positivity, and two exponents already allow positive cylinders with a negative singleton. Independent Bernoulli coordinates perturbed by infinitesimals on one common well-ordered exponent set extend exactly under a weighted  $\ell^2$  condition on every exponent row around baselines uniformly separated from 0 and 1, and under an  $\ell^1$  condition at deterministic baselines, positively on all Borel sets; an i.i.d. Bernoulli law extends if and only if its bias is real. The threshold  $\omega + 1$  and the finite real shadow hold in the strong class only (Section 2.6).

**Status.** A merged AI-assisted research report with written mathematical proofs and three independent exact finite check suites. The strong-class classification, the extension criteria, the threshold and the coefficientwise product thresholds are offered as proposed original contributions, not as certified priority. This is *not* a first non-Archimedean Kolmogorov extension theorem (Ludkovsky and Khrennikov proved one under different axioms) and not a first non-Archimedean probability theory. No named published conjecture is claimed solved; the proofs have not been independently refereed and nothing here is checked in Lean.

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# 1 The question, the two axioms, and the results

## 1.1 Infinite addition is the real issue

Suppose a coin has probability  $\omega^{-1}$  of success. For every ordinary positive integer  $n$  its proposed independent  $n$ -toss law is perfectly meaningful:

$$P_n(x_1, \dots, x_n) = \prod_{j=1}^n p_j^{x_j} (1 - p_j)^{1-x_j}, \quad x_j \in \{0, 1\}, \quad p_j = \omega^{-1}. \quad (1.1)$$

Every weight is nonnegative, the weights sum to one, and deleting the last coordinate recovers the preceding law. Does this consistent family define a probability on infinite sequences?

There is no answer until one says what the probabilities of countably many disjoint events add up to. Replacing real probabilities by nonnegative surreal numbers does not settle this. Surreal normal forms carry one exact addition rule: a family is summable when the union of the supports of its members is well ordered and only finitely many members contribute to each monomial [12]. The rule is exact and useful in formal algebra, but it is far more restrictive than convergence of a real series: the ordinary convergent series  $\sum_{n \geq 1} 2^{-n}$  is not a strong Hahn sum when its terms are read as constant series. A second, more permissive rule adds coefficient by coefficient, by ordinary absolutely convergent sums of real or complex numbers, and asks nothing of the family as a whole.

This report determines what each of the two rules allows. Under the first rule the answer is rigid and complete: every measure is a strong sum of point masses, and the extension, moment and product problems have exact criteria. Under the second rule diffuse laws survive, and the product problem has sharp sequence-space thresholds. In both, positivity of all finite tests does not imply positivity of the measure, but for different reasons and at different support sizes.

## 1.2 The two axioms and the plan of the report

Throughout,  $\Gamma$  is a set-sized ordered abelian group; no divisibility is assumed except in the one place, Gaussian quadrature, where it is stated. We use  $K_{\mathbb{R}} = \mathbb{R}((t^{\Gamma}))$  and  $K_{\mathbb{C}} = \mathbb{C}((t^{\Gamma}))$ , with increasing supports, so that a positive exponent is an infinitesimal scale.

Two definitions are set side by side in Part I (Definitions 2.6 and 2.7):

- a *coefficientwise Hahn measure*  $\mu = \sum_{\gamma \in W} \mu_{\gamma} t^{\gamma}$  has one well-ordered exponent set  $W$  and ordinary finite signed or complex measures  $\mu_{\gamma}$ ; for disjoint events,  $[t^{\gamma}] \mu(\bigcup_n A_n) = \sum_n \mu_{\gamma}(A_n)$  is an ordinary absolutely convergent sum;
- a *strong Hahn measure* satisfies, for every disjoint sequence of events, that the family  $(\mu(A_n))_n$  is strongly Hahn summable with strong sum  $\mu(\bigcup_n A_n)$ .

The first framing, the same Kolmogorov and moment questions asked under the two axioms, is that of source 15. On countably separated spaces the strong Hahn measures are exactly the coefficientwise Hahn measures whose coefficient measures are finitely atomic (Corollary 4.5); so the strong class is a subclass, and a very small one.

The report has two spines. Parts II–IV treat the strong class: classification (sources 10 and 11, independently), extension on product spaces (source 10), and the moment problem with its hidden negative mass (source 11, with the Cantor-space form from source 10). Part V treats the coefficientwise class (source 15). Part VI collects the scope, the non-claims of all three sources, the open questions, the three check suites and the provenance.

### 1.3 Principal results

- A. Strong atomicity with automatic common support** (Theorem 4.2; sources 10 and 11). On a countably separated measurable space, a real or complex strong Hahn measure has the form  $\mu(A) = \sum_{x \in A}^H w_x$  for one strongly summable family  $(w_x)$ ; it extends uniquely to all subsets; it is positive exactly when every  $w_x \geq 0$ . A map is a strong Hahn measure exactly when each of its coefficient functions is a finite combination of point masses (Corollary 4.3); no common support is assumed anywhere. Invariant strong probabilities live on finite orbits (Theorem 6.1, source 15, here on countably separated spaces).
- B. Strong extension on product spaces** (source 10). Coherent cylinder data on  $\prod_n E_n$ ,  $E_n$  finite, have a strong extension exactly when the combined support is well ordered and each exponent is active on a bounded number of cylinders per level (Theorem 8.2). Independent finite-state coordinates have a positive strong extension exactly when finitely many coordinates are random at the residue level and the individual rare-entry probabilities are strongly summable (Theorem 10.1); testing aggregated rare probabilities is not enough (Theorem 11.1).
- C. The strong moment problem and hidden negative mass** (source 11, with source 10). A moment sequence in  $K_{\mathbb{R}}$  comes from a positive strong probability on an ordinary real sample set exactly under three conditions: a common well-ordered support, finite signed atomic coefficient rows, and nonnegative reassembled point weights (Theorem 12.2). One weight pattern (Definition 14.1) gives, on Cantor space, nonnegative cylinder masses with a negative singleton (Proposition 14.2), and on  $[0, 1]$  a functional strictly positive on every nonzero Hahn polynomial nonnegative on  $[0, 1]$  whose unique strong representing measure has mass  $-t$  at 0 (Theorem 14.3). In the strong class the least support order type allowing this is  $\omega + 1$ , for cylinder tests and for polynomial tests (Theorem 15.2). Hankel determinant valuations (Theorem 16.2) and Gaussian quadratures with non-Archimedean nodes (Theorems 17.1 and 17.3) are computed exactly.
- D. The coefficientwise class** (source 15). Signed extension of cylinder data is decided per exponent by total variation (Theorem 19.1); positivity by the transfinite null-ideal criterion (Theorem 19.2; also proved independently by source 18 of the same batch). Independent Bernoulli perturbations extend exactly under a weighted  $\ell^2$  row condition at interior baselines (Theorem 20.1), an  $\ell^1$  row condition at deterministic baselines (Theorem 21.1), and their combination (Theorem 22.1); an i.i.d. Bernoulli law extends if and only if its bias is real (Corollary 23.1).

### 1.4 Sources, and where the merge had to choose

The report is built from three manuscripts, written independently on 22 September 2026 against the same repository snapshot

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of VladimirReshetnikov/Surreal [11]. None of the three read the other two. Their batch numbers are kept as source numbers; the programs and data files in this directory carry them as prefixes.

| No. | Manuscript                                                                                                                                                                  | Contribution to this report                                                                                                                                                                                                                                                                                                                                                                             |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10  | <i>Strong Hahn Measures on Surreal Workspaces: atomicity, exact extension criteria, and hidden positivity obstructions</i> (base)                                           | Setting and scope statements; scalar lemma; atomic classification and its corollaries; countable-separation counterexample; cylinder criterion; Cantor-space hidden negative atom and the cylinder form of the $\omega + 1$ threshold; labelled-word calculus; independent-coordinate criterion; coarse-graining counterexample; surreal transfer and exponent enlargement; integration and its limits. |
| 11  | <i>Positive Moment Functionals with Invisible Negative Mass: Strong Hahn Measures over the Surreals</i>                                                                     | The same classification, with the deterministic common-support lemma used here; moment criterion; finite visibility; valuation barriers; the interval example and its Hausdorff inequalities; the polynomial form of the $\omega + 1$ threshold; Hankel determinants and Jacobi scales; Gaussian quadrature with non-Archimedean nodes; Hermitian moment forms; transitive invariance.                  |
| 15  | <i>Infinite Products of Surreal Probabilities: sharp <math>\ell^1/\ell^2</math> extension thresholds, transfinite positivity, and rigidity of Hahn countable additivity</i> | The two-axiom framing; the coefficientwise extension test; the null-ideal positivity criterion; the positive-cylinder counterexample with two exponents; interior $\ell^2$ , boundary $\ell^1$ and mixed thresholds; the i.i.d. corollary and examples; the near-boundary example; complex versions; conditioning; finite-orbit rigidity and exchangeability.                                           |

Where the sources overlap the merge prints one statement and names every source that proved it. The choices were these.

- (1) *Base.* Source 10 has the weakest hypotheses on the strong spine (any countably separated space; no common support, divisibility, bounded variation or compactness) and the most careful scope statement, and is the base. Its text, labels and notation were rewritten with the prefix `meas`.
- (2) *Framing.* Part I adopts source 15’s two definitions side by side. Source 15 called its class “coefficientwise-regular” and defined it on compact metric spaces; here it is called *coefficientwise* and defined on an arbitrary measurable space, while every theorem keeps the space hypotheses under which its source proved it.
- (3) *Classification.* Source 10’s Theorem “Atomicity with automatic common support” and source 11’s “Atomic classification and unique extension” are the same theorem with the same scalar lemma; it is printed once (Theorem 4.2). The common support is proved by source 11’s deterministic row-selection argument (Lemma 4.1); source 10’s randomized selection lemma is kept as a second route (Remark 4.4). Source 10’s sentence that the randomized argument is needed for the coefficientwise criterion is withdrawn, since the deterministic argument proves that criterion as well. Source 15’s strong classification (for coefficientwise measures on compact metric spaces with a common support *assumed*) is the special case recorded in Corollary 4.5; it is not presented as more general.
- (4) *Duplicates printed once.* The scalar lemma (10, 11); the finitely supported standard part (10, 11); uncountably many infinitesimal atoms (10, 11, 15); integration of scalar functions (10, 11); the Neumann support lemma (10, 15); the weight pattern of the hidden negative atom (10 and 11 use identical weights at different sample points); the  $\omega + 1$  threshold (10 for cylinder tests, 11 for polynomial tests: Theorem 15.2 has one part for each); exponent enlargement (10, 11, 15).
- (5) *Strong results of source 15 re-derived.* Finite-orbit rigidity and exchangeability were proved by source 15 for coefficientwise measures on compact metric spaces with a common support; they are re-derived here from Theorem 4.2 on countably separated spaces, and they subsume

source 11’s transitive-invariance corollary. Source 15’s finite-row criterion at a deterministic baseline is the common-support special case of source 10’s Bernoulli criterion (Remark 21.2).

- (6) *Null ideal.* The positivity criterion is stated canonically as Theorem 19.2, on an arbitrary measurable space, with the negative-part and control-measure forms in which source 18 of the same batch proved it independently for real regular Borel measures on compact Hausdorff spaces. Source 18 is placed as the separate report `surcomplex/hahn-herglotz-z-positivity`; none of its other material is imported.
- (7) *Citations instead of proofs.* Strong summability and the  $n/(n+1)$  example are cited from the foundations report, the Neumann lemma from its positive-support section, and the finite spectral theorem that source 11 reproved from the spectral-theory report (Section 1.5).

## 1.5 Position relative to the collection

**Foundations.** Strong summability is the collection’s definition `found:eq:summability`; the example  $\sum_n t^{n/(n+1)}$ , strongly summable although its partial sums do not converge in the valuation topology, is `found:ex:boundedrankone`; the Hahn–Neumann support lemma is `found:sub:positivesupport`; the sign convention  $t^\gamma = \omega^{-\gamma}$  is `found:eq:workspaces`; and localization of a set of surreal numbers in a set-sized Hahn workspace is `found:thm:workspace`, all in `foundations-and-computation/foundations` [12]. None of these is claimed here.

**Physics.** The report `physics/surreal-scalars-and-spacetime` [13] already observes, in `phys:ex:summable` and in its subsection `phys:sub:probability` (“Probability requires a summation theory, not just positive scalars”), that a countable family assigning the same positive infinitesimal to each outcome is not strongly summable, so that countably additive non-Archimedean probability is not free; its scope table lists a probability theory among the things that “remain to be supplied”. That observation is prior and is not claimed here. Theorem 4.2 turns it into a structure theorem: under strong countable additivity every single exponent is carried by finitely many points, so no exponent can be shared by infinitely many disjoint events. At the pinned snapshot source 15 reported that indexed repository searches for `probability` returned no matches; as a statement about the content of the collection that was not correct, since `phys:sub:probability` was present. Source 15’s other statement, that the catalogue listed no probability report, was correct at the pin. The collection now contains this report and the Herglotz report described below. The physics report poses no open problem that this report answers; it asks that some summation theory be chosen, and this report works out two.

**Spectral theory.** Source 11 reproved the finite spectral theorem over a real closed field for its Gaussian quadrature. The collection already has it: Theorem `thm:spectral` of `surcomplex/spectral-theory` [14] over any set-sized real closed field, and `spec:thm:hermitian` for Hermitian matrices over  $\mathbb{C}((t^\Gamma))$  at *any* set-sized  $\Gamma$ , with a real orthogonal diagonalizer for real symmetric input. We cite them (Section 17). The same report states that “no infinite tensor product, measure-valued probability theory . . . is supplied” and that it has no spectral measures; Section 18 shows, in the strong class, that a natural positive scalar spectral measure need not exist for a positive-definite Hermitian moment form.

**Dynamics.** The report `surcomplex/dynamics-and-normal-forms` [15] constructs a canonical invariant density on the torus (`dyn:thm:qpdensity`) and records, as its non-claim N75, that this is not a countably additive **No**-valued probability measure on arbitrary subsets. Source 15’s finite-orbit theorem (Theorem 6.1) proves that refusal is forced in the strong class: when every orbit of a group acting by Borel automorphisms on a countably separated space, such as the

ordinary torus, is infinite, there is no strongly countably additive invariant Hahn probability, neither on Borel sets nor on all subsets (Remark 6.5).

**Herglotz positivity.** Source 18 of the same batch, placed as `surcomplex/hahn-herglotz-positivity` [16], works in the coefficientwise class on the circle with Fourier moments and Toeplitz tests. It proved the null-ideal criterion independently (Theorem 19.2 is the canonical statement) and gives the Toeplitz counterpart of the two-exponent counterexample of Example 19.5.

**Prony reconstruction and Markov generators.** The word “moment” has other senses in the collection (Convention 2.11). The report `surcomplex/prony-reconstruction-at-sur-real-scales` [17] reconstructs a finite weighted configuration with Hahn-valued nodes from its first power moments; it involves no measure. The report `surreal/markov-generators-at-every-scale` [18] studies finite-state Hahn rate matrices and excludes path measures; the extension of Markov path laws to infinite sequences is not treated there or here (Question 26.9).

## 1.6 Position relative to the literature

Non-Archimedean probability is not one axiomatic theory, and nothing here refutes an existing one. Benci, Horsten and Wenmackers use, among other axioms, a homomorphism from functions on finite subsets to a non-Archimedean field together with a generalized limit [1]; their continuity axiom is different from finite contribution at every Hahn exponent in every disjoint countable family, so the atomicity theorem does not apply to their theory merely because its values are non-Archimedean. Brickhill and Horsten relate non-Archimedean and lexicographic probabilities [2]; lexicographic probability is not claimed here. Bottazzi constructs a real-valued measure on non-Archimedean extensions of  $\mathbb{R}$  [4], and Kaiser develops semialgebraic measure and integration over non-Archimedean real closed fields [5]; both change the sample field or the target and are different from a Hahn-valued measure on an ordinary Borel space.

Ludkovsky and Khrennikov already prove a non-Archimedean analogue of the Kolmogorov extension theorem. Their measures live on a covering ring, with ultrametric boundedness and continuity along shrinking families, and their extension theorem concerns a measure-dependent completion of a cylinder ring [3, Definitions 2.1–2.4 and Theorem 2.15.2]. Neither the coefficient-local finiteness on the full Borel  $\sigma$ -algebra used in the strong class nor the per-exponent finite-variation rule of the coefficientwise class is their axiom. Accordingly the phrase “a first non-Archimedean Kolmogorov theorem” would be incorrect for this report, for the strong extension theorems of Part III and equally for the coefficientwise extension theorems of Part V. (Source 10 made this attribution; source 15 did not cite Ludkovsky and Khrennikov, and the attribution is extended to its theorems here.)

Classical inputs are used and credited: the Hahn–Neumann support lemma [8], Gonshor’s normal forms [9] and the Hahn-field facts reviewed by Bournez and Guilmant [10], the Riesz–Markov representation theorem, Kakutani’s square-summability criterion for real infinite products [6], and the classical moment theory on an interval [7].

## Part I

# The two additivity axioms

## 2 Hahn fields, strong sums, and two kinds of countable addition

### 2.1 Hahn fields and the surreal interpretation

Throughout,  $\mathbb{N} = \{1, 2, 3, \dots\}$ ; nonnegative indices are written explicitly. Let  $\Gamma$  be a set-sized linearly ordered abelian group, written additively, and let  $k$  be  $\mathbb{R}$  or  $\mathbb{C}$ . Put

$$K = k((t^\Gamma)) = \left\{ \sum_{\gamma \in S} a_\gamma t^\gamma : S \subseteq \Gamma \text{ well ordered, } a_\gamma \in k \right\}, \quad K_{\mathbb{R}} = \mathbb{R}((t^\Gamma)), \quad K_{\mathbb{C}} = \mathbb{C}((t^\Gamma)).$$

Supports are well ordered in the *increasing* order. We write  $[t^\gamma]f$  for the coefficient of  $t^\gamma$  in  $f$  and  $\text{supp } f = \{\gamma : [t^\gamma]f \neq 0\}$ . For  $f \neq 0$  put  $v(f) = \min \text{supp } f$  and  $\text{lc}(f) = [t^{v(f)}]f$ ;  $v(0) = +\infty$ . A nonzero real Hahn series is positive exactly when its leading coefficient is positive. Thus  $t^\gamma > 0$  always, and  $t^\gamma$  is smaller than every positive real number when  $\gamma > 0$ . The coefficient field is embedded as the constant series. When  $k = \mathbb{R}$ ,

$$\mathcal{O} = \{f : v(f) \geq 0\}, \quad \mathfrak{m} = \{f : v(f) > 0\}, \quad \text{st}(f) = [t^0]f \quad (f \in \mathcal{O});$$

“finite” means bounded in absolute value by an ordinary real number, and  $\mathfrak{m}$  is the set of infinitesimals, 0 included. Conjugation on  $K_{\mathbb{C}}$  acts on coefficients. *No divisibility assumption on  $\Gamma$  is made anywhere, except in the finite Gaussian quadrature theorem of Section 17, which needs a real closed field and says so.*

**Convention 2.1** (Surreal normal forms). Fix an order-preserving additive embedding  $\iota : \Gamma \rightarrow \mathbf{No}$ ; when  $\Gamma$  is a subgroup of  $\mathbf{No}$ ,  $\iota$  is the inclusion. The normal-form map is

$$\sum_{\gamma \in S} a_\gamma t^\gamma \mapsto \sum_{\gamma \in S} a_\gamma \omega^{-\iota(\gamma)}. \quad (2.1)$$

The minus sign is essential: increasing well-ordered Hahn exponents become decreasing well-based surreal exponents. This is the collection’s convention  $t^\gamma = \omega^{-\gamma}$  (`found:eq:workspaces` [12]). For real coefficients (2.1) is an ordered field embedding [9, 10]; applying it to real and imaginary parts embeds  $K_{\mathbb{C}}$  in  $\mathbf{No}[i]$ . It preserves the designated strong sums, not an arbitrary notion of topological convergence. Every theorem below is proved in a set-sized Hahn workspace first and transported afterwards; no sample space of all surreal numbers and no proper-class sum is used.

### 2.2 Strong summability

**Definition 2.2** (Strong Hahn summability). A set-indexed family  $(f_i)_{i \in I}$  in  $K$  is *strongly summable* if

$$\bigcup_{i \in I} \text{supp}(f_i) \text{ is well ordered} \quad \text{and} \quad \#\{i : [t^\gamma]f_i \neq 0\} < \infty \quad (\gamma \in \Gamma).$$

Its strong sum  $\sum_{i \in I}^H f_i$  has coefficient  $\sum_i [t^\gamma]f_i$  at  $\gamma$ , a finite sum. Zero members are permitted, in any number.

This is `found:eq:summability` [12]. The *union of the supports* of the members appears in the definition, not merely the support of the sum after cancellation.

**Lemma 2.3** (Regrouping, products, and positivity). *Subfamilies and regroupings of strongly summable families are strongly summable, with the same sum. If  $(f_i)$  and  $(g_j)$  are strongly summable, then  $(f_i g_j)_{(i,j)}$  is strongly summable and*

$$\left(\sum_i^H f_i\right)\left(\sum_j^H g_j\right) = \sum_{i,j}^H f_i g_j;$$

*the same holds for any fixed finite number of factors. A strong sum of nonnegative real Hahn series is nonnegative; it is positive when at least one summand is positive, and then*

$$v\left(\sum_i^H f_i\right) = \min_{f_i \neq 0} v(f_i).$$

*Proof.* Subfamilies are immediate. For a regrouping, a fixed coefficient occurs in only finitely many original summands, hence in finitely many groups, and the support stays inside the original well-ordered union.

For products: if  $S, T$  are well-ordered subsets of an ordered abelian group,  $S + T$  is well ordered and each element has only finitely many representations  $s + t$  with  $(s, t) \in S \times T$ . Indeed, infinitely many representations would give an increasing sequence of distinct  $s$  and hence a decreasing sequence of  $t$ ; and a decreasing sequence in  $S + T$  would, after passing to a subsequence with nondecreasing  $S$ -coordinates (a sequence in a well order has an infinite nondecreasing subsequence), give a decreasing sequence of  $T$ -coordinates. At each pair of exponents the two families supply only finitely many indices. This proves strong summability of the product family and coefficientwise distributivity; induct on the number of factors.

For positivity, let  $\alpha$  be the least exponent in the union of the supports of the nonzero summands; it exists because that union is well ordered. Every summand contributing at  $\alpha$  has its leading term there, with positive coefficient. There are finitely many of them and their sum is positive, and no smaller exponent occurs. This also gives the valuation formula.  $\square$

**Example 2.4** (Three distinct summation phenomena). For  $\Gamma = \mathbb{Q}$ , the family  $(t^n)_{n \geq 1}$  is strongly summable. So is  $(t^{n/(n+1)})_{n \geq 1}$ , although its finite partial sums do not approach its strong sum in the valuation topology: every remainder has valuation less than 1 (this is `found:ex:boundedrankone` [12]). The family  $(2^{-n}t)_{n \geq 1}$  is not strongly summable, because infinitely many members contribute at exponent 1, although the real series of its coefficients converges; for the same reason the ordinary convergent series  $\sum_{n \geq 1} 2^{-n} = 1$  is not a strong sum of constant series. Finally  $(t^{1/n})_{n \geq 1}$  fails the support condition.

No argument in this report replaces a strong sum by a topological limit of partial sums, in the valuation topology or in the surreal fine topology. All exchanges of summation in the strong class are licensed by Definition 2.2 and Lemma 2.3.

### 2.3 Measurable spaces

**Definition 2.5** (Countable separation). A measurable space  $(X, \Sigma)$  is *countably separated* if some sequence  $D_1, D_2, \dots \in \Sigma$  separates points: for distinct  $x, y$  there is an  $n$  with exactly one of  $x, y$  in  $D_n$ .

Each singleton of such a space is measurable (intersect, over  $n$ , the side of  $D_n$  containing the point), so every countable subset is measurable. Standard Borel spaces, countable products of finite discrete spaces, and every subset of  $\mathbb{R}$  with its relative Borel  $\sigma$ -algebra are countably separated. Countable separation does not mean that  $X$  is countable. We use ordinary set theory with choice; every measurable space and every exponent group in this report is a set.

## 2.4 Coefficientwise and strong Hahn measures

For any map  $\mu : \Sigma \rightarrow K$  and  $\gamma \in \Gamma$  we call

$$\mu_\gamma(A) = [t^\gamma]\mu(A) \quad (A \in \Sigma)$$

the *coefficient measure* (or coefficient function, before any additivity is known) of  $\mu$  at  $\gamma$ . The two axioms are the following.

**Definition 2.6** (Coefficientwise Hahn measure). A *coefficientwise Hahn measure* on  $(X, \Sigma)$  is a map  $\mu : \Sigma \rightarrow K$  for which there is one set-sized well-ordered set  $W \subseteq \Gamma$  such that every coefficient measure  $\mu_\gamma$  is a countably additive signed measure of finite total variation (if  $k = \mathbb{R}$ ) or a complex measure (if  $k = \mathbb{C}$ ), and  $\mu_\gamma = 0$  for  $\gamma \notin W$ :

$$\mu(A) = \sum_{\gamma \in W} \mu_\gamma(A) t^\gamma \quad (A \in \Sigma). \quad (2.2)$$

Zero coefficient measures are allowed.

For pairwise disjoint events  $A_n$ , Definition 2.6 gives

$$[t^\gamma]\mu\left(\bigcup_n A_n\right) = \sum_n \mu_\gamma(A_n), \quad \sum_n |\mu_\gamma(A_n)| \leq \|\mu_\gamma\|_{\text{TV}}, \quad (2.3)$$

ordinary absolutely convergent real or complex sums, where  $|\nu|$  denotes total variation and  $\|\nu\|_{\text{TV}} = |\nu|(X)$ . Equation (2.3) does *not* say that the family  $(\mu(A_n))_n$  is strongly summable. When such a sum over a countable family is formed exponent by exponent we write  $\sum^{\text{coef}}$ , and  $\prod^{\text{coef}}$  for the corresponding infinite products; these are never strong sums unless that is proved.

**Definition 2.7** (Strong Hahn measure). A *strong Hahn measure* on  $(X, \Sigma)$  is a map  $\mu : \Sigma \rightarrow K$  with  $\mu(\emptyset) = 0$  such that every pairwise disjoint sequence  $(A_n)_{n \geq 1}$  in  $\Sigma$  satisfies

$$(\mu(A_n))_{n \geq 1} \text{ is strongly summable, and } \mu\left(\bigcup_{n \geq 1} A_n\right) = \sum_{n \geq 1}^H \mu(A_n). \quad (\text{SH})$$

Finite additivity follows by adjoining empty sets. Neither a common support for the values  $\mu(\Sigma)$  nor any continuity for the valuation topology is imposed, and no ordinary countable sum at a coefficient is hidden in (SH): at each exponent only finitely many terms are nonzero. Source 10 and source 11 use Definition 2.7 as stated. Source 15 called a measure *strongly countably additive* when it is coefficientwise *and* satisfies (SH). On countably separated spaces the two definitions agree: a strong Hahn measure is automatically coefficientwise (Corollary 4.5). For instance, the ordinary probability  $\sum_n 2^{-n} \delta_n$  on  $\mathbb{N}$ , read with constant values, is a coefficientwise Hahn measure and not a strong one.

**Definition 2.8** (Positivity and probability). A real-valued ( $k = \mathbb{R}$ ) coefficientwise or strong Hahn measure is *positive* if  $\mu(A) \geq 0$  for every  $A \in \Sigma$ . A positive measure of either class with  $\mu(X) = 1$  is a *coefficientwise Hahn probability*, respectively a *strong Hahn probability*. Complex-valued measures carry no order and no probability interpretation.

*Remark 2.9* (Coefficientwise probabilities have nonnegative exponents). A coefficientwise Hahn probability takes values in  $[0, 1]_{K_{\mathbb{R}}}$ . A nonnegative Hahn series bounded by 1 has no negative leading exponent, hence no negative exponent at all; so every coefficient measure at a negative exponent vanishes,  $\mu_0$  is an ordinary probability, and  $\mu_\gamma(X) = 0$  for  $\gamma \neq 0$ . One may therefore take  $W \subseteq \Gamma_{\geq 0}$  with  $0 \in W$ . The same holds for strong Hahn probabilities.

*Remark 2.10* (One meaning of “strong”). “Strong” refers only to Definition 2.2 and to (SH). It is not any of the several unrelated notions called strong additivity in vector-measure theory, not the formal-power-series maps preserving strong sums that the collection’s Lean ledger calls strongly additive, and not a strong operator topology.

## 2.5 Terminology

**Convention 2.11** (One meaning per word). The three sources and the rest of the collection use several words in more than one sense. In this report:

- (a) *Moment* means a power moment  $m_n = \int x^n d\mu$  of a measure, or  $L(X^n)$  of a linear functional, on an *ordinary real* sample set (Part IV). Where source 15 spoke of fourth moments of a random variable we write fourth-power expectations  $\mathbb{E}_0 Z^4$ . Other senses in the collection are different objects: Fourier moments on the circle in the Herglotz report [16], power moments of a finite configuration with Hahn-valued nodes in the Prony report [17], and contour, residue and inverse moments in the analysis, finite-deformation and infinite-dimensional spectral reports.
- (b) *Positive*. A real Hahn series  $f$  is positive,  $f > 0$ , when  $\text{lc}(f) > 0$ , and nonnegative,  $f \geq 0$ , when  $f > 0$  or  $f = 0$ . A *positive measure* is one with  $\mu(A) \geq 0$  for all events (Definition 2.8); the word never means that nonempty events have positive mass. A linear functional  $L$  on a class of real test functions is *positive* if  $L(f) \geq 0$  for every  $f \geq 0$  in the class, and *strictly positive* if  $L(f) > 0$  for every nonzero  $f \geq 0$  in the class; “ $f \geq 0$ ” is always tested at the ordinary sample points. A Hermitian form is *positive definite* if  $\langle P, P \rangle > 0$  for  $P \neq 0$ . Exponents in  $\Gamma_{>0}$  are called positive exponents.
- (c) *Infinitesimal family*. A family of Hahn series each of whose nonzero members has positive valuation. (Source 10 called this a positive-order family.)
- (d) *Strong* has the single meaning of Remark 2.10. *Coefficientwise* names the class of Definition 2.6 and the  $\sum^{\text{coef}}$  operations; the coefficient measure  $\mu_\gamma$  exists in both classes. We never write a bare “Hahn measure” or “Hahn probability” for a statement that holds in only one class.
- (e) *Regular* has no meaning of its own here. Source 15’s “coefficientwise-regular” is our “coefficientwise”; its adjective was about the coefficient measures and never meant Brickhill and Horsten’s regularity (every nonempty event has positive probability [2]). Finite Borel measures on compact metric spaces are automatically regular in the Radon sense; that sense is used only inside the Riesz–Markov step of Theorem 19.1 and when source 18’s setting is described.
- (f) *Atom*. A point atom of a measure is a point  $x$  with  $\mu(\{x\}) \neq 0$ ; a coefficient atom at  $\gamma$  is a point atom of  $\mu_\gamma$ . An ordinary measure is *atomless* if it has no point atoms.

## 2.6 Two traps: what holds only in the strong class

Two pairs of results look contradictory until the class of the measure is named. Both apparent contradictions disappear with the qualifier, and each becomes a false sentence without it.

**Trap 1: the threshold  $\omega + 1$  versus two exponents.** In the strong class, if the global support of a measure has order type at most  $\omega$ , then nonnegative cylinder masses (on Cantor space) or nonnegative integrals of real polynomial squares (on a real sample set) force every event to be nonnegative, and  $\omega + 1$  is the least order type at which a negative event can hide behind nonnegative finite tests (Theorem 15.2, sources 10 and 11). In the coefficientwise class, on the *same* binary Cantor space and with the *same* cylinder tests, the measure  $P_0 + t^\alpha(\delta_y - \delta_x)$  of Example 19.5 (source 15) has support  $\{0, \alpha\}$  of order type 2, strictly positive cylinder masses and  $\mu(\{x\}) = -t^\alpha$ . There is no contradiction. The leading coefficient  $P_0$  of that example is atomless, and in the strong class every coefficient measure is finitely atomic (Theorem 4.2); so the example is not strong. The mechanisms differ: a coefficientwise measure can hide negative

mass on a set that is null for all earlier coefficients (Theorem 19.2), while a strong measure has no such null sets beyond finitely many points per exponent and can hide negative mass only behind infinitely many earlier atoms. The correct statement is: *the least support order type at which nonnegative finite tests coexist with a negative event is  $\omega + 1$  in the strong class and 2 in the coefficientwise class* (Remark 19.6).

**Trap 2: the finite real shadow.** The standard part of a positive strong Hahn probability is an ordinary probability with finite support (Corollary 4.6, sources 10 and 11). The standard part of a coefficientwise Hahn probability can be an atomless Bernoulli product (Theorem 20.1) or have infinitely many atoms (Example 24.3). The sentence “a Hahn probability has a finitely supported standard part” is false without the word “strong”. The same trap concerns product laws: the fair-coin product is a coefficientwise Hahn probability (with constant values), while its cylinder data have no strong extension at all, signed or positive (Theorem 8.2: the width at exponent 0 is  $2^n$  at depth  $n$ ); an i.i.d. Bernoulli law with real bias  $p$  extends coefficientwise for every  $p \in [0, 1]$  (Corollary 23.1) and strongly only for  $p \in \{0, 1\}$  (Corollary 10.3).

A third, milder confusion concerns symmetry. Ordinary invariant probability on the circle, embedded with constant values, is a coefficientwise Hahn probability; for an irrational rotation there is no strong one (Remark 6.4).

## Part II

# Strong Hahn measures: classification

### 3 The coefficient-atomicity mechanism

We first isolate an elementary fact about scalar measures. A detailed proof is needed because no positivity of coefficient measures can be assumed: a positive Hahn series may have many negative later coefficients, so the coefficient measures of a positive Hahn measure are in general signed.

**Lemma 3.1** (Infinite Boolean algebras contain disjoint sequences). *An infinite Boolean algebra contains a countably infinite family of pairwise disjoint nonzero elements.*

*Proof.* If it has infinitely many atoms, choose countably many distinct atoms. Otherwise it has finitely many atoms. If their join is the unit, every element is a join of a subfamily of these atoms and the algebra is finite. Thus their complement is nonzero and has no atom below it. Split this complement into two nonzero parts, keep one and set the other aside; the kept part still has no atom below it. Repetition yields the desired pairwise disjoint nonzero parts. The case with no atoms is the same argument started from the unit.  $\square$

**Lemma 3.2** (Disjoint-finite scalar measures are finitely point-atomic; sources 10 and 11). *Let  $(X, \Sigma)$  be countably separated and let  $\lambda : \Sigma \rightarrow k$  be countably additive. Suppose that for every disjoint sequence  $(A_n)$  only finitely many  $\lambda(A_n)$  are nonzero. Then*

$$\lambda(A) = \sum_{x \in F \cap A} c_x \quad (A \in \Sigma) \quad (3.1)$$

for a finite set  $F \subseteq X$  and nonzero  $c_x \in k$ ; the empty sum is allowed. After zero coefficients are omitted,  $F$  and the  $c_x$  are unique.

*Proof.* Call  $A \in \Sigma$  *hereditarily null* (source 11 says *restriction-null*) if  $\lambda(B) = 0$  for every measurable  $B \subseteq A$ ; this is stronger than  $\lambda(A) = 0$ , which cancellation can produce on a set carrying nonzero signed mass. The family  $\mathcal{N}$  of hereditarily null sets is a  $\sigma$ -ideal: for  $B \subseteq \bigcup_n A_n$  with each  $A_n \in \mathcal{N}$ , the disjoint pieces  $B \cap (A_n \setminus \bigcup_{j < n} A_j)$  have value zero, and countable additivity gives  $\lambda(B) = 0$ .

Consider the quotient Boolean algebra  $\Sigma/\mathcal{N}$ . If it were infinite, Lemma 3.1 would supply disjoint nonzero classes. Choose representatives  $C_n$  and replace each by  $C_n \setminus \bigcup_{j < n} C_j$ ; the classes stay nonzero because the removed intersections lie in  $\mathcal{N}$ , and the representatives are now disjoint. Each contains a measurable  $B_n$  with  $\lambda(B_n) \neq 0$ , contradicting the hypothesis. So the quotient is finite.

If the quotient is zero,  $\lambda = 0$ . Otherwise choose disjoint representatives  $A_1, \dots, A_r$  of its atoms whose complement is hereditarily null. For measurable  $B \subseteq A_j$ , either  $B$  or  $A_j \setminus B$  is hereditarily null, so  $\lambda(B)$  is 0 or  $c_j := \lambda(A_j)$ ; and  $c_j \neq 0$ , for otherwise  $A_j$  would be hereditarily null.

Let  $(E_n)$  be a countable separating family. For each  $j, n$  choose the side  $D_{j,n}$  of  $E_n$  for which  $A_j \setminus D_{j,n}$  is hereditarily null. The set  $A_j \cap \bigcap_n D_{j,n}$  has hereditarily null complement in  $A_j$ ; it is nonempty because  $c_j \neq 0$ , and it has at most one point because the  $E_n$  separate points. Write it  $\{x_j\}$ . Then  $\lambda(\{x_j\}) = c_j$  and  $A_j \setminus \{x_j\}$  is hereditarily null, which proves (3.1). Evaluating on singletons proves uniqueness.  $\square$

The proof uses only a Boolean-algebra observation, countable additivity and countable separation. It does not hide a bounded-variation assumption for a signed or complex coefficient measure, and it deliberately treats signed and complex measures: a proof using positivity of each coefficient would not be valid.

*Remark 3.3* (Second route in the compact metric case; source 15). For a finite signed or complex Borel measure  $\nu$  on a compact metric space, source 15 proves directly that  $\nu$  is supported on a finite set if and only if there is no sequence of pairwise disjoint Borel sets  $A_n$  with  $\nu(A_n) \neq 0$  for every  $n$ . If  $\nu$  is not finitely supported, the support of  $|\nu|$  is an infinite compact set; choose distinct support points  $x_n \rightarrow x$  with  $x_n \neq x$  and  $d(x_{n+1}, x) < d(x_n, x)/4$ ; the balls of radius  $d(x_n, x)/8$  about  $x_n$  are pairwise disjoint and have positive total variation, and each contains a Borel set of nonzero  $\nu$ -measure. This is the special case of Lemma 3.2 with finite total variation and a metric; the lemma needs neither.

**Proposition 3.4** (Finite coefficient measures). *If  $\mu$  is a strong Hahn measure on a countably separated space, then for every  $\gamma \in \Gamma$  the coefficient measure  $\mu_\gamma$  is a finite linear combination of Dirac measures: there are a finite set  $F_\gamma \subseteq X$  and nonzero  $c_{\gamma,x} \in k$  with*

$$\mu_\gamma(A) = \sum_{x \in A \cap F_\gamma} c_{\gamma,x} \quad (A \in \Sigma). \quad (3.2)$$

*With  $w_x = \mu(\{x\})$  one has  $c_{\gamma,x} = [t^\gamma]w_x$  and  $F_\gamma = \{x : [t^\gamma]w_x \neq 0\}$ .*

*Proof.* For a disjoint sequence, (SH) says that the coefficient values at  $\gamma$  are nonzero at only finitely many indices and that the coefficient of the union is their finite sum. Thus  $\mu_\gamma$  is countably additive as an ordinary scalar measure and satisfies Lemma 3.2. Evaluating (3.2) on singletons identifies  $c_{\gamma,x}$  and  $F_\gamma$ .  $\square$

## 4 Automatic common support and the atomic classification

At this point each  $F_\gamma$  is finite and each  $w_x$  is a Hahn series. A common well-ordered support does *not* follow by taking a union of well-ordered sets. The following deterministic argument,

from source 11, is the additional step. It uses only finite coefficient atomicity and the fact that  $\mu(A)$  is a Hahn series for every countable  $A$ ; it never uses (SH).

**Lemma 4.1** (Automatic common well-order; source 11). *Let  $(X, \Sigma)$  be countably separated and let  $\mu : \Sigma \rightarrow K$  be any map whose coefficient functions all have the finite atomic form (3.2). Put  $w_x = \mu(\{x\})$ . Then*

$$S = \bigcup_{x \in X} \text{supp } w_x$$

*is well ordered, and  $S = \bigcup_{A \in \Sigma} \text{supp } \mu(A)$ .*

*Proof.* By (3.2),  $F_\gamma = \{x : [t^\gamma]w_x \neq 0\}$  and every exponent of every  $\mu(A)$  is an exponent of some  $w_x$  with  $x \in A$ ; this proves the last equality. Suppose  $S$  is not well ordered and choose (in ZFC a linear order that is not well founded contains one) a strictly decreasing sequence  $\gamma_1 > \gamma_2 > \dots$  in  $S$ . Each finite row  $F_{\gamma_n}$  is nonempty, and each point  $x$  belongs to only finitely many rows, since otherwise  $\text{supp } w_x$  would contain a decreasing sequence.

Pass inductively to a subsequence whose rows are pairwise disjoint: after finitely many rows have been chosen their union is finite, each point in it lies in only finitely many of the remaining rows, and so some later row avoids the union. Choose one point  $x_k$  in the  $k$ th selected row, of exponent  $\gamma_{n_k}$ , and put  $A = \{x_k : k \geq 1\}$ , a measurable set. By (3.2) the coefficient of  $\mu(A)$  at  $\gamma_{n_k}$  is exactly  $c_{\gamma_{n_k}, x_k} \neq 0$ , because no other chosen point lies in that row. So  $\text{supp } \mu(A)$  contains a strictly decreasing sequence, contradicting  $\mu(A) \in K$ .  $\square$

**Theorem 4.2** (Atomic classification with automatic common support; sources 10 and 11). *Let  $(X, \Sigma)$  be countably separated and let  $\mu : \Sigma \rightarrow K$  be a strong Hahn measure. Put  $w_x = \mu(\{x\})$ . Then:*

- (i) *the family  $(w_x)_{x \in X}$  is strongly summable;*
- (ii) *for every  $A \in \Sigma$ ,*

$$\mu(A) = \sum_{x \in A}^H w_x; \tag{4.1}$$

- (iii) *the global support*

$$S(\mu) := \bigcup_{A \in \Sigma} \text{supp } \mu(A) = \bigcup_{x \in X} \text{supp } w_x \tag{4.2}$$

*is well ordered, and at every exponent only finitely many  $w_x$  have a nonzero coefficient;*

- (iv)  *$\mu$  extends uniquely to a strong Hahn measure on  $\mathcal{P}(X)$ , namely by (4.1); the extension is additive for every set-indexed disjoint family, with the sum taken strongly.*

*Conversely, every strongly summable family  $(w_x)_{x \in X}$  defines by (4.1) a strong Hahn measure on all subsets of  $X$ . For real coefficients such a measure is positive if and only if every  $w_x \geq 0$ . Neither a common support nor atomicity is assumed; both are conclusions.*

*Proof.* Proposition 3.4 gives (3.2) for every  $\gamma$ , and Lemma 4.1 makes  $S(\mu)$  well ordered and proves (4.2). Together, finiteness of each  $F_\gamma$  and the common well order say precisely that  $(w_x)$  is strongly summable, and (3.2) proves (4.1) coefficient by coefficient. This gives (i)–(iii).

Conversely, given a strongly summable family, every subfamily is summable, so (4.1) is defined on every subset. For a disjoint family of subsets (countable or set-indexed), summing the masses is a regrouping of the weight family according to the subset containing each index; by Lemma 2.3 the regrouped family is strongly summable with the same sum. This proves (SH) and the arbitrary additivity in (iv). Any other strong extension to  $\mathcal{P}(X)$  has the same singleton masses, and  $(X, \mathcal{P}(X))$  is still countably separated, so the classification just proved, applied on  $\mathcal{P}(X)$ , forces the same formula; this is uniqueness. Positivity of  $\mu$  gives  $w_x = \mu(\{x\}) \geq 0$ ; conversely nonnegative weights have nonnegative strong sums by Lemma 2.3.  $\square$

The extension in (iv) is stronger than countable additivity, but it depends on the very restrictive summability axiom. It is not an extension theorem for ordinary Lebesgue measure (Example 5.3).

**Corollary 4.3** (Coefficientwise criterion without a support hypothesis; source 10). *For an arbitrary map  $\mu : \Sigma \rightarrow K$  on a countably separated space the following are equivalent:  $\mu$  is a strong Hahn measure; each coefficient function  $A \mapsto [t^\gamma]\mu(A)$  is a finite linear combination of point masses. In the second condition no common support and no strong additivity are assumed in advance.*

*Proof.* Necessity is Proposition 3.4. For sufficiency, Lemma 4.1 uses only finite coefficient atomicity and the existence of the Hahn value  $\mu(A)$ , so it supplies the common well order without (SH); the family  $(w_x)$  is then strongly summable, (3.2) gives (4.1), and the converse clause of Theorem 4.2 proves (SH).  $\square$

*Remark 4.4* (Second routes to the common support; source 10). Source 10 obtained the common support from a randomized selection lemma: if  $D$  is countable and  $L_n(A) = \sum_{x \in F_n \cap A} c_{n,x}$ ,  $A \subseteq D$ , with  $F_n$  finite and some  $c_{n,x} \neq 0$  for each  $n$ , then some  $A \subseteq D$  has  $L_n(A) \neq 0$  for infinitely many  $n$ . Choose the membership bits of  $A$  independently with ordinary probability  $\frac{1}{2}$  (realized, after enumerating  $D$ , by the binary digits of a uniform number in  $[0, 1)$ ). Conditioning on all bits in  $F_n$  except one with nonzero coefficient shows  $\mathbb{P}(L_n(A) \neq 0) \geq \frac{1}{2}$ ; hence every tail event  $\bigcup_{n \geq m} \{L_n(A) \neq 0\}$  has probability at least  $\frac{1}{2}$ , and by continuity from above  $L_n(A) \neq 0$  infinitely often with probability at least  $\frac{1}{2}$ . Applied with  $F_n = F_{\gamma_n}$  along a decreasing sequence of exponents, this contradicts  $\mu(A) \in K$  exactly as in Lemma 4.1. Only an ordinary real product probability is used; it must not be instantiated with the Hahn measure under construction. For a measure already known to be strong there is a third route: along a decreasing sequence  $\gamma_n$  pick  $x_n \in F_{\gamma_n}$ ; infinitely many repetitions of one point would put a decreasing sequence into one  $\text{supp } w_x$ , and otherwise countably many distinct singletons form a disjoint family whose masses are strongly summable by (SH), which is impossible. Source 10 stated that the randomized argument is needed for the coefficientwise criterion, where this third route is unavailable; the deterministic Lemma 4.1 shows that it is not needed.

**Corollary 4.5** (The two axioms on countably separated spaces). *For a map  $\mu : \Sigma \rightarrow K$  on a countably separated space the following are equivalent:*

- (a)  $\mu$  is a strong Hahn measure (Definition 2.7);
- (b)  $\mu$  is a coefficientwise Hahn measure and every coefficient measure  $\mu_\gamma$  is supported on finitely many points;
- (c)  $\mu$  is a coefficientwise Hahn measure satisfying (SH) (source 15's “strongly countably additive”).

In that case one may take  $W = S(\mu)$  in (2.2).

*Proof.* (a) $\Rightarrow$ (b): by Theorem 4.2 each  $\mu_\gamma$  is a finite combination of Dirac masses, hence countably additive of finite total variation, and  $S(\mu)$  is a common well-ordered support. (b) $\Rightarrow$ (a) is Corollary 4.3, since a finite signed or complex measure supported on a finite set  $F$  is  $\sum_{x \in F} \mu_\gamma(\{x\})\delta_x$ . (c) $\Rightarrow$ (a) is trivial and (a) $\Rightarrow$ (c) follows from (a) $\Rightarrow$ (b).  $\square$

Source 15 proved (b) $\Leftrightarrow$ (c) as its Theorem “Finite-atomic coefficient classification”, for coefficientwise measures on a compact metric space, with the common support  $W$  assumed and the scalar step done by Remark 3.3. That theorem is the special case of Corollary 4.5 for such spaces; it is not more general, and its proof, which needs the common  $W$  and finite total variation, is not used for the general statement.

**Corollary 4.6** (Finite real shadow of a strong probability; sources 10 and 11). *Let  $\mu$  be a strong Hahn probability on a countably separated space. Every event mass lies in  $[0, 1]$ , every point weight has valuation at least 0, and the standard part is an ordinary probability with finite point support:*

$$\text{st } \mu(A) = [t^0]\mu(A) = \sum_{x \in A \cap F_0} c_{0,x}, \quad c_{0,x} > 0, \quad \sum_{x \in F_0} c_{0,x} = 1.$$

*Proof.* Finite additivity and positivity give  $0 \leq \mu(A) \leq 1$ , so no event mass has a negative exponent. At exponent 0 only the finitely many points of  $F_0$  contribute; each  $c_{0,x}$  is the leading coefficient of the nonnegative weight  $w_x$ , hence positive; and their sum is  $[t^0]\mu(X) = 1$ .  $\square$

Thus an ordinary diffuse real probability, read with constant Hahn values, is not a strong Hahn probability, and neither is an ordinary probability with infinitely many atoms: both fail finite activity at exponent 0. This is a theorem about the stated axiom, not a defect of ordinary measures, and it is false for coefficientwise probabilities (Section 2.6, Trap 2).

**Corollary 4.7** (Cardinality of the atomic set; source 10). *If  $W_\mu = \{x : w_x \neq 0\}$ , then  $W_\mu = \bigcup_{\gamma \in S(\mu)} \{x : [t^\gamma]w_x \neq 0\}$ , a union of finite sets. Consequently  $W_\mu$  is at most countable when  $S(\mu)$  is, and has cardinality at most  $\max(\aleph_0, |S(\mu)|)$  in general.*

*Remark 4.8* (Uncountably many infinitesimal atoms; sources 10, 11 and 15). Atomicity is finite support *per coefficient*, not finiteness or countability of the whole law. Let  $\kappa$  be any set ordinal and  $X = \{x_*\} \cup \{x_\xi : \xi < \kappa\}$  with distinct points. The weights

$$w_{x_\xi} = \omega^{-(\xi+1)}, \quad w_{x_*} = 1 - \sum_{\xi < \kappa}^H \omega^{-(\xi+1)} \quad (4.3)$$

form a strongly summable family of positive elements with sum 1; the latter sum is a legitimate infinitesimal normal form. They define a positive strong probability on all subsets of  $X$  whose real shadow is the single atom at  $x_*$  and which has  $\kappa$  further positive infinitesimal atoms. For a countably separated instance with uncountably many atoms, apply this to a well ordering of  $\mathbb{R}$  and restrict to the Borel sets. More generally, for any well-ordered family of distinct positive exponents  $(\gamma_\alpha)$ , distinct points  $x_\alpha$  and an anchor  $a$ , the weights  $w_{x_\alpha} = t^{\gamma_\alpha}$ ,  $w_a = 1 - \sum_\alpha^H t^{\gamma_\alpha}$  work, even for an uncountable index set. The classification says “strongly atomic”, not “countably atomic”, and it is not a classification of all non-Archimedean measures under every possible definition of countable addition.

**Example 4.9** (Countable separation cannot be dropped; source 10). Let  $X$  be uncountable with its countable-cocountable  $\sigma$ -algebra, and put  $\lambda(A) = 0$  for countable  $A$ ,  $\lambda(A) = 1$  for cocountable  $A$ . A disjoint countable family has at most one cocountable member, and if none occurs the union is countable. Thus  $\lambda$  is a strong Hahn measure, even with values in the constant series; yet  $\lambda(\{x\}) = 0$  for every  $x$  while  $\lambda(X) = 1$ . The space is not countably separated: the countable exceptional sides of a countable family of measurable sets have countable union, leaving uncountably many points with one membership pattern.

## 5 Integration and pushforward

**Proposition 5.1** (Pushforward and scalar-valued integration; sources 10 and 11). *Let  $\mu$  have the representation (4.1).*

(i) *Any measurable map  $f : X \rightarrow Y$  has a strong pushforward measure  $f_*\mu(B) = \mu(f^{-1}(B))$ .*

- (ii) For every function  $g : X \rightarrow k$ , with no boundedness or measurability assumption needed after the extension to all subsets,

$$\int g d\mu := \sum_{x \in X}^H g(x) w_x \quad (5.1)$$

is a well-defined Hahn series. It is  $k$ -linear, agrees with  $\mu$  on indicators, and  $[t^\gamma] \int g d\mu = \sum_{x \in F_\gamma} g(x) c_{\gamma, x}$ .

- (iii) If  $k = \mathbb{R}$ ,  $\mu$  is positive and  $g \geq 0$ , then  $\int g d\mu \geq 0$ .
- (iv) If  $X \subseteq \mathbb{R}$ , every polynomial  $P \in K[X]$  is integrable: the family  $(w_x P(x))_x$  is strongly summable, and  $\int P d\mu$  is obtained by expanding  $P$  into finitely many coefficients times ordinary powers of  $x$ .

*Proof.* Inverse images preserve disjoint unions, which proves (i) from (SH); alternatively group the weights over the fibres of  $f$ . Multiplication by  $g(x) \in k$  introduces no new exponent, and at each exponent only the finitely many coefficient atoms contribute, so the family in (5.1) is strong; its coefficient formula gives linearity and the indicator identity, and (iii) follows from Lemma 2.3. For (iv),  $P = \sum_{j \leq d} b_j X^j$  with  $b_j \in K$ ; each family  $(b_j x^j w_x)_x$  is the product of the one-member family  $(b_j)$  with the strong family  $(x^j w_x)_x$ , so Lemma 2.3 applies, and finitely many strong families add.  $\square$

This integral is useful, but it is not Lebesgue integration with coefficients promoted to surreal scalars. Its apparent integrability of every ordinary scalar function comes from finite evaluation at each exponent. It is not a claim that every Hahn-valued function is integrable: arbitrary coefficient supports can destroy strong summability.

**Example 5.2** (Failure for Hahn-valued integrands; source 10). On  $X = \{0, 1, 2, \dots\}$  put  $w_n = t^n$  for  $n \geq 1$  and  $w_0 = 1 - \sum_{n \geq 1}^H t^n$ , a positive strong probability. The Hahn-valued function  $g(n) = t^{-n}$  for  $n \geq 1$ ,  $g(0) = 0$ , has  $g(n)w_n = 1$  for every  $n \geq 1$ , so the family in (5.1) is not strongly summable. Formula (5.1) deliberately assumes that  $g$  takes values in the coefficient field.

**Example 5.3** (An ordinary measure that is not strong; source 10). Lebesgue probability on  $[0, 1]$ , embedded as constant Hahn series, fails (SH): choose disjoint intervals of lengths  $2^{-n}$  (with harmless endpoint conventions); every mass contributes at exponent 0. The ordinary series of lengths converges, but it is not a strong family. So the finite-shadow corollary does not exclude an ordinary real probability theory embedded by a different addition rule; it excludes treating its usual infinite sums as strong sums. That different rule is the coefficientwise class of Part V.

A more permissive theory could allow absolutely convergent real coefficient sums at each exponent, or use topological or generalized-limit additivity. Such theories have different extension and positivity problems, and the theorems of Parts II–IV should not be transferred to them without checking the changed hypotheses. Conversely, replacing (SH) by a weaker rule to recover diffuse probability abandons the exact finite-contribution property that causes the rigidity. The first of these permissive theories, with a common well-ordered support assumed, is Part V.

## 6 Invariant strong probabilities live on finite orbits

Source 15 proved the results of this section for coefficientwise measures on compact metric spaces satisfying (SH), with a common support assumed. Through Theorem 4.2 the same proofs work on every countably separated space with no support hypothesis.

**Theorem 6.1** (Finite-orbit rigidity; source 15). *Let a group  $G$  act on a countably separated space  $(X, \Sigma)$  by bijections  $g$  with  $g(A) \in \Sigma$  and  $g^{-1}(A) \in \Sigma$  for  $A \in \Sigma$ , and let  $\mu$  be a  $G$ -invariant strong Hahn measure:  $\mu(gA) = \mu(A)$ . Then every point with  $w_x \neq 0$  has a finite orbit, and  $w$  is constant on orbits. If  $\mu$  is a strong Hahn probability, then*

$$\mu(A) = \sum_{O \in \mathcal{O}_f}^H w_O \frac{|A \cap O|}{|O|}, \quad w_O \geq 0, \quad \sum_{O \in \mathcal{O}_f}^H w_O = 1, \quad (6.1)$$

where  $\mathcal{O}_f$  is a set of finite orbits and  $(w_O)$  is strongly summable. Conversely every such formula defines a  $G$ -invariant strong Hahn probability, on  $\Sigma$  and on all subsets of  $X$ .

*Proof.* Invariance applied to singletons gives  $w_{gx} = w_x$ . Fix  $\gamma$  and  $x$  with  $[t^\gamma]w_x \neq 0$ . Every point  $gx$  has the same nonzero coefficient at  $\gamma$ , and by Theorem 4.2 only finitely many points do; hence the orbit of  $x$  is finite. Put  $w_O = \sum_{x \in O} w_x = |O| w_x$  for a finite orbit  $O$  on which  $w \neq 0$ . Regrouping the strongly summable point family by orbits (Lemma 2.3) gives (6.1); positivity and normalization pass to  $(w_O)$ . Conversely, split each  $w_O$  equally among the finitely many points of  $O$ ; at any exponent only finitely many orbit weights contribute, each at finitely many points, so the point family is strongly summable, and Theorem 4.2 gives a strong Hahn probability on all subsets, invariant because its weights are constant on orbits.  $\square$

The complex and signed versions hold verbatim: an invariant strong Hahn measure has its point atoms on finite orbits. The words probability and nonnegative are reserved for real coefficients.

**Corollary 6.2** (No transitive invariant strong probability; source 11). *An infinite countably separated space admits no strong Hahn probability invariant under a transitive group of measurable permutations.*

*Proof.* The only orbit is infinite, so by Theorem 6.1 every  $w_x$  vanishes and  $\mu(X) = 0$ . Source 11's direct argument: transitivity makes all singleton weights equal; if the common weight is nonzero, a countably infinite family of distinct singletons contributes its leading coefficient infinitely often, violating strong summability, and if it is zero, Theorem 4.2 makes the total mass zero.  $\square$

**Corollary 6.3** (Exchangeability collapses; source 15). *For a finite alphabet  $E$ , every exchangeable strong Hahn probability on  $E^\mathbb{N}$  is a mixture of constant sequences:*

$$\mu = \sum_{a \in E} q_a \delta_{(a,a,\dots)}, \quad q_a \in K_\mathbb{R}, \quad q_a \geq 0, \quad \sum_{a \in E} q_a = 1. \quad (6.2)$$

*In particular no nondegenerate i.i.d. Bernoulli law, even one with ordinary real bias, is a strong Hahn probability.*

*Proof.* Exchangeability is invariance under the group of finite coordinate permutations. Its only finite orbits are the constant sequences: in a nonconstant sequence some symbol occurs infinitely often and a different symbol occurs somewhere, and swapping that occurrence with infinitely many positions of the first symbol gives infinitely many distinct images. Theorem 6.1 gives (6.2). For a binary mixture of the two constant paths, the event  $X_1 \neq X_2$  has probability zero, whereas its independent Bernoulli probability is  $2p(1-p) > 0$  for  $0 < p < 1$  in an ordered field.  $\square$

This does not contradict classical exchangeability theory: ordinary diffuse probabilities fail the finite-incidence requirement, so they do not satisfy the strong hypothesis.

*Remark 6.4* (Shifts and rotations; source 15). A strong Hahn probability on  $E^{\mathbb{Z}}$  invariant under the two-sided shift is a strong mixture of uniform measures on finite periodic orbits. For an irrational rotation of the ordinary circle every orbit is infinite, so no invariant strong Hahn probability exists. Ordinary invariant circle measure still exists as a coefficientwise Hahn probability concentrated at exponent 0; it simply is not strong.

*Remark 6.5* (The refusal N75 of the dynamics report). The report `surcomplex/dynamics-and-normal-forms` [15] states of its invariant density `dyn:thm:qpdensity` that it is “not a countably additive **No**-valued probability measure on arbitrary subsets” (`dyn:subsec:qpdensity`; non-claim N75 in its ledger). Theorem 6.1 shows that in the strong sense the refused object cannot exist for any symmetry with only infinite orbits: if a group acts on the ordinary torus by Borel automorphisms all of whose orbits are infinite (for instance the powers of a translation by a vector with an irrational coordinate, or the group  $\mathbb{R}$  acting by the flow of a nonzero constant vector field), there is no strongly countably additive invariant Hahn or surreal probability on the Borel sets, and hence none on all subsets. This concerns invariance under an ordinary group action; the invariance statement of `dyn:thm:qpdensity` is the identity  $I_{\mathcal{X}}(\mathcal{X} \cdot \nabla f) = 0$  for a Hahn vector field, and nothing here strengthens or restates it. In the coefficientwise class the density itself gives a positive object. For real data it is 1 plus a series of positive support whose coefficients are real-analytic, hence bounded, on the real torus, and whose mean is 1; integrating its coefficients against Haar measure  $m$  gives a coefficientwise Hahn probability with  $\mu_0 = m$  and  $\mu_{\gamma} \ll m$  for  $\gamma > 0$ , positive on every Borel set by Corollary 19.4.

## 7 Surreal and surcomplex values

Let  $X$  and  $\Sigma$  be sets and let  $\mu : \Sigma \rightarrow \mathbf{No}$  or  $\mathbf{No}[i]$  be a function on that set domain. Work in a class theory with replacement, or in a sufficiently large universe for the set-sized data. The range is a set, each of its normal forms has set-sized support, and so the union of the supports of all values is a set, whether or not it is well based. The additive subgroup of **No** generated by the negatives of all these exponents is a set-sized ordered abelian group  $\Gamma$ , and every value of  $\mu$  is represented in  $K_{\mathbb{R}}$ , or in  $K_{\mathbb{C}}$  after combining the two coordinate supports. This is the localization of `found:thm:workspace` [12], here without passing to a divisible hull. The strong-additivity axiom is unchanged by the identification, so Theorem 4.2 applies and proves a common well-ordered support as a theorem, rather than assuming it during localization.

**Corollary 7.1** (Actual surreal atomicity; source 10). *On a countably separated set-sized measurable space, every **No**-valued or **No**[ $i$ ]-valued measure that is countably additive by canonical strong normal-form sums is a strong sum of its singleton masses. Every normal-form coefficient is carried by finitely many points, and all event masses have a common admissible support. A positive normalized surreal measure of this kind has a finitely supported ordinary real standard part.*

No topology on the proper class **No** is required. The sample space is a set, and nothing is asserted about a measure on the full power class of **No**. Exponent enlargement cannot repair the extension obstructions of Part III; see Proposition 11.3.

### Part III

## Strong Hahn measures: extension on product spaces

This part is source 10's. Every measure in it is strong; the coefficientwise counterparts are in Part V.

### 8 An exact extension theorem for cylinder data

Let  $E_n$  be nonempty finite sets and

$$\Omega = \prod_{n \geq 1} E_n, \quad T_n = \prod_{j=1}^n E_j, \quad T_0 = \{\emptyset\}.$$

For a word  $s \in T_n$ ,  $[s]$  denotes its cylinder. The product Borel  $\sigma$ -algebra  $\mathcal{B}(\Omega)$  is countably separated and generated by the cylinders. Suppose  $m_s \in K$  are *coherent*:

$$m_s = \sum_{a \in E_{n+1}} m_{sa} \quad (s \in T_n). \quad (8.1)$$

This gives a finitely additive law on the cylinder algebra; it does not yet define masses of arbitrary Borel events. For each exponent define the *active width* at depth  $n$  and the combined support by

$$r_\gamma(n) = \#\{s \in T_n : [t^\gamma]m_s \neq 0\}, \quad S_T = \bigcup_{n \geq 0} \bigcup_{s \in T_n} \text{supp}(m_s). \quad (8.2)$$

**Lemma 8.1** (Bounded-width coherent scalar trees). *Let  $c_s \in k$  satisfy (8.1). If the number of nonzero  $c_s$  at each level is bounded independently of the level, then there are finitely many distinct paths  $x_1, \dots, x_r \in \Omega$  and nonzero constants  $d_1, \dots, d_r$  with*

$$c_s = \sum_{j: x_j \in [s]} d_j \quad \text{for every word } s. \quad (8.3)$$

*This representation is unique.*

*Proof.* Each nonzero parent has at least one nonzero child, and children of different parents are distinct, so the widths form a nondecreasing sequence of integers. A bounded such sequence is eventually constant, say from level  $N$  on. At every later level each nonzero parent has exactly one nonzero child and every zero parent has only zero children; an extra nonzero child, or nonzero children under a zero parent, would increase the width. The nonzero nodes at level  $N$  therefore continue along finitely many unique paths, and coherence makes the coefficient constant along each path. These paths and coefficients represent all levels  $n \geq N$ , and summing descendants gives the earlier levels. For uniqueness, a sufficiently deep level separates any finite collection of distinct paths, and its cylinder values recover the coefficients.  $\square$

**Theorem 8.2** (Exact strong cylinder-extension criterion; source 10). *The coherent family  $(m_s)$  has a strong Hahn measure extension to  $\mathcal{B}(\Omega)$  if and only if*

$$S_T \text{ is well ordered}, \quad (C1)$$

$$\sup_{n \geq 0} r_\gamma(n) < \infty \quad \text{for every } \gamma \in \Gamma. \quad (C2)$$

*When it exists the extension is unique, its global support is  $S_T$ , and it extends uniquely to all subsets of  $\Omega$ .*

*Proof.* Suppose  $\mu$  is an extension. By Theorem 4.2 its global support is well ordered and each coefficient measure has finitely many point atoms. At a fixed level each nonzero coefficient cylinder contains at least one of these atoms, and the cylinders of one level are disjoint, so their number is bounded by the number of coefficient atoms. This proves (C1) and (C2). A nonzero finite atomic coefficient measure is nonzero on some cylinder (separate its finitely many atoms), so the global support is exactly  $S_T$ .

Conversely, for each  $\gamma \in S_T$  apply Lemma 8.1 to  $c_s = [t^\gamma]m_s$  and obtain a finite atomic scalar measure  $\mu_\gamma$  on  $\Omega$ . For any  $A \subseteq \Omega$  put

$$\mu(A) = \sum_{\gamma \in S_T} \mu_\gamma(A) t^\gamma. \quad (8.4)$$

Its support lies in the well-ordered  $S_T$ , so it is a Hahn series. For a disjoint sequence of subsets, at a fixed  $\gamma$  only finitely many members meet the finite atomic support of  $\mu_\gamma$ ; the family of masses is strong, and finite coefficient additivity gives (SH). Formula (8.3) gives the prescribed cylinder values.

Any strong extension has finite atomic coefficient measures by Theorem 4.2, and each is determined by its cylinder values by the last argument of Lemma 8.1; this also excludes exponents outside  $S_T$ . The all-subsets assertion is Theorem 4.2(iv).  $\square$

*Remark 8.3* (What the criterion does and does not bound). The width bound in (C2) may depend on  $\gamma$ ; no uniform bound on the number of all atoms is required, and the stabilization level in Lemma 8.1 may also depend on  $\gamma$ . These two nonuniformities are essential: they permit infinitely many infinitesimal scales, and they are responsible for the positivity obstruction of Section 14.

**Corollary 8.4** (Exact positivity test; source 10). *Assume (C1)–(C2),  $k = \mathbb{R}$  and  $m_\emptyset = 1$ . Reconstruct the coefficient measures as above and put*

$$w_x = \sum_{\gamma \in S_T} \mu_\gamma(\{x\}) t^\gamma. \quad (8.5)$$

*There is a positive strong probability extension if and only if  $w_x \geq 0$  for every  $x \in \Omega$ .*

*Proof.* This is the positivity clause of Theorem 4.2 together with the uniqueness in Theorem 8.2.  $\square$

Nonnegativity of all cylinder masses is *not* a substitute for the test: Proposition 14.2 gives nonnegative cylinders whose unique strong extension has a negative point weight.

## 9 Labelled support calculus for infinite products

The independent-product theorem must control both the support and the number of labelled contributions at a fixed exponent. We state the classical support result in the form needed.

**Lemma 9.1** (Neumann support lemma, classical; stated by sources 10 and 15). *Let  $S \subseteq \Gamma_{>0}$  be well ordered. The additive monoid*

$$S^* = \{s_1 + \cdots + s_m : m \geq 0, s_j \in S\}$$

*is well ordered, and for each  $\rho \in \Gamma$  only finitely many ordered finite words  $(s_1, \dots, s_m)$  in  $S$  have sum  $\rho$ . In particular only finitely many pairs  $(\alpha, \beta) \in S^* \times S^*$  satisfy  $\alpha + \beta = \rho$ .*

This is the positive-support form of Neumann's lemma [8], and it is the collection's **found: sub:positivesupport** [12]; we import it. It holds at arbitrary rank: word lengths are not bounded by an Archimedean comparison with a least positive scale. Source 15 recalls one standard proof: finite words over a well-ordered alphabet are well-quasi-ordered by increasing subsequence embedding, and a proper embedding strictly increases the sum since all letters are positive; so there is neither an infinite strictly decreasing sequence of sums nor infinitely many distinct words of one sum. The pair assertion follows by concatenating representations of the two summands.

*Remark 9.2* (A fixed exponent, not a cofinal decay; source 15). The lemma concerns contributions to a *fixed* exponent, not the number of exponents below a bound. If  $0 < \alpha$  is infinitesimal relative to  $\beta$  in  $\Gamma$ , all  $n\alpha$  lie below  $\beta$ . No argument here infers that the powers of  $t^\alpha$  are cofinal in the valuation topology.

**Lemma 9.3** (Labelled words and subproducts). *Let  $(a_i)_{i \in I}$  be a strongly summable infinitesimal family. Then the family of labelled products*

$$(a_{i_1} \cdots a_{i_m})_{m \geq 0, (i_1, \dots, i_m) \in I^m} \quad (9.1)$$

*is strongly summable, where zero products may be discarded and the empty word occurs once. Any subfamily of these labelled products is therefore strongly summable, including the products indexed by finite subsets of  $I$ .*

*Proof.* Let  $S$  be the union of the (positive) supports of the  $a_i$ . The support of every product lies in  $S^*$ . At a fixed exponent  $\rho$ , Neumann's lemma gives finitely many exponent words  $(s_1, \dots, s_m)$  summing to  $\rho$ ; for each letter  $s_j$  only finitely many labels  $i_j$  contribute a nonzero coefficient, by strong summability of  $(a_i)$ . So only finitely many labelled coefficient words occur at  $\rho$ , and in particular only finitely many nonzero products contribute there.  $\square$

Infinitely many zero labels cause no problem in (9.1): every nonempty product containing one is the zero series, which contributes at no exponent, and the single empty product is 1.

**Proposition 9.4** (Infinite products of infinitesimal perturbations). *For a strongly summable infinitesimal family  $(q_n)_{n \geq 1}$ , define*

$$U = \prod_{n \geq 1}^H (1 - q_n) := \sum_{F \in \text{Fin}(\mathbb{N})}^H (-1)^{|F|} \prod_{n \in F} q_n. \quad (9.2)$$

*Then  $U \in 1 + \mathfrak{m}$  is a unit. The families*

$$\left( \frac{q_n}{1 - q_n} \right)_n \quad \text{and more generally} \quad \left( \frac{a_{n,j}}{1 - q_n} \right)_{n,j} \quad (9.3)$$

*are strongly summable whenever the infinitesimal family  $(a_{n,j})_{n,j}$  is strongly summable and  $q_n = \sum_j a_{n,j}$  with finitely many  $j$  at each  $n$ . All finite-factor product identities used with (9.2) remain valid coefficientwise.*

*Proof.* Equation (9.2) is a signed subfamily of the labelled words of Lemma 9.3; its nonconstant support is positive, so  $U \in 1 + \mathfrak{m}$ . The geometric series  $(1 - q_n)^{-1} = \sum_{m \geq 0}^H q_n^m$  is valid by the same lemma. Expanding  $a_{n,j} \sum_m q_n^m$  gives labelled infinitesimal words whose first letter carries the distinguished index  $(n, j)$ ; at a fixed exponent there are finitely many such words and labels, which proves (9.3). Every product identity can be checked at a fixed exponent, where the relevant words and labels form a finite set, and it reduces to the identity for finitely many factors.  $\square$

**Lemma 9.5** (Recovering rare entries from first-error masses). *Let  $p_{n,j} \in \mathfrak{m}$  be real Hahn series (zero allowed), with finitely many indices  $j$  at each  $n$ , and put*

$$q_n = \sum_j p_{n,j}, \quad u_n = 1 - q_n, \quad h_{n,j} = p_{n,j} \prod_{k < n} u_k. \quad (9.4)$$

*Then  $(p_{n,j})_{n,j}$  is strongly summable if and only if  $(h_{n,j})_{n,j}$  is.*

*Proof.* If  $(p_{n,j})$  is strong, expanding the finite products in (9.4) gives labelled words whose first letter has the distinguished label  $(n, j)$ , and Lemma 9.3 makes  $(h_{n,j})$  strong. Conversely, finite telescoping gives

$$\prod_{k < n} u_k = 1 - H_{<n}, \quad H_{<n} = \sum_{k < n} \sum_j h_{k,j}. \quad (9.5)$$

Each nonzero  $h_{n,j}$  has positive valuation, so

$$p_{n,j} = h_{n,j}(1 - H_{<n})^{-1} = \sum_{m \geq 0}^H h_{n,j} H_{<n}^m. \quad (9.6)$$

Expand  $H_{<n}^m$  into labelled words: the first label  $(n, j)$  is again distinguished and all later coordinate labels are smaller than  $n$ . At each exponent Lemma 9.3 leaves finitely many full words, hence finitely many distinguished indices, so (9.6) is strongly summable simultaneously for all  $(n, j)$ . No uniform bound on  $n$  or  $m$  is imposed.  $\square$

## 10 Independent coordinates: a sharp strong extension criterion

Let  $p_{n,a} \in K_{\mathbb{R}}$  satisfy

$$p_{n,a} \geq 0, \quad \sum_{a \in E_n} p_{n,a} = 1, \quad (10.1)$$

so every entry is finite. The independent cylinder law is

$$m_s = \prod_{n=1}^{|s|} p_{n,s_n}. \quad (10.2)$$

Write  $\bar{p}_{n,a} = \text{st}(p_{n,a})$ , an ordinary finite-state probability vector. Call coordinate  $n$  *residually nondeterministic* if at least two entries of this vector are positive, and let  $J$  be the set of such coordinates. For  $n \notin J$  there is a unique  $b_n \in E_n$  with  $\bar{p}_{n,b_n} = 1$ ; every other entry is zero or a positive infinitesimal, and these are the *rare entries*.

**Theorem 10.1** (Exact independent finite-state criterion; source 10). *The independent cylinder law (10.2) has a positive strong Hahn probability extension if and only if*

- (i)  *$J$  is finite, and*
- (ii) *the family of individual rare entries*

$$(p_{n,a})_{n \notin J, a \neq b_n} \quad (10.3)$$

*is strongly summable.*

*The extension is then unique, and it is supported on the paths with only finitely many deviations from the  $b_n$  outside  $J$ .*

## 10.1 Necessity: finite residue branching

*Proof of necessity in Theorem 10.1.* Let  $\mu$  be a positive strong extension. By Corollary 4.6,  $\text{st} \circ \mu$  is a real probability with finitely many point atoms, say at most  $M$ . If there were  $r$  residually nondeterministic coordinates, choose two positive-residue states at each. The corresponding  $2^r$  disjoint finite-coordinate cylinder events have positive constant coefficients by (10.2), so each contains a distinct residue atom and  $2^r \leq M$ . Hence  $J$  is finite.

Project away the coordinates in  $J$  and relabel the others in increasing order; pushforward preserves (SH), so we may assume  $J$  empty. Put  $q_n = \sum_{a \neq b_n} p_{n,a}$  and  $u_n = p_{n,b_n} = 1 - q_n$ ; each  $q_n$  is zero or a positive infinitesimal. For  $a \neq b_n$  let  $B_{n,a}$  be the event that coordinates  $1, \dots, n-1$  take their base states and coordinate  $n$  equals  $a$ . These events are pairwise disjoint as  $(n, a)$  varies, with prescribed masses  $\mu(B_{n,a}) = p_{n,a} \prod_{k < n} u_k = h_{n,a}$ . The family is countable because the alphabets are finite, so (SH) makes  $(h_{n,a})$  strongly summable, and Lemma 9.5 makes  $(p_{n,a})_{a \neq b_n}$  strongly summable.  $\square$

## 10.2 Sufficiency: the finite-deviation formula

For the construction, reorder the complement of the finite set  $J$  as  $1, 2, \dots$  temporarily; the notation keeps the original indices. Define

$$q_n = \sum_{a \neq b_n} p_{n,a}, \quad u_n = 1 - q_n, \quad r_{n,a} = \frac{p_{n,a}}{u_n} \quad (n \notin J, a \neq b_n), \quad U = \prod_{n \notin J}^H u_n.$$

The  $q_n$  form a strong family by grouping (10.3), and Proposition 9.4 applies. Let  $\Omega_{\text{fin}}$  be the set of paths  $x$  for which  $F(x) = \{n \notin J : x_n \neq b_n\}$  is finite, and put

$$W_x = \left( \prod_{j \in J} p_{j,x_j} \right) U \prod_{n \in F(x)} r_{n,x_n} \quad (x \in \Omega_{\text{fin}}), \quad W_x = 0 \quad (x \notin \Omega_{\text{fin}}). \quad (10.4)$$

The product over  $J$  is ordinary and finite: residue randomness is kept exactly at the finitely many exceptional coordinates, not replaced by a chosen state.

*Proof of sufficiency and uniqueness in Theorem 10.1.* The family  $(r_{n,a})$  is strong and infinitesimal by Proposition 9.4. Its products indexed by finite coordinate subsets, with one rare-state label per chosen coordinate, are a subfamily of the labelled words, hence strongly summable. Multiplication by the fixed series  $U$  preserves strong summability (the sum of its support with the common support is well ordered with finitely many contributions per exponent), and so does the finite union over the choices at  $J$  with their fixed finite products. Thus  $(W_x)$  is strong. Each  $u_n$  has leading coefficient 1, so  $U$  has leading coefficient 1 and is positive; every other factor in (10.4) is nonnegative, so every  $W_x \geq 0$ .

For  $n \notin J$ ,

$$1 + \sum_{a \neq b_n} r_{n,a} = 1 + \frac{q_n}{u_n} = u_n^{-1}. \quad (10.5)$$

Summing the finite-deviation products and applying the justified coefficientwise finite-factor identities gives

$$\sum_{\substack{F \subseteq \mathbb{N} \setminus J \text{ finite} \\ a_n \neq b_n \ (n \in F)}}^H \prod_{n \in F} r_{n,a_n} = \prod_{n \notin J}^H \left( 1 + \sum_{a \neq b_n} r_{n,a} \right) = U^{-1}. \quad (10.6)$$

The finite sum over the coordinates in  $J$  of  $\prod_{j \in J} p_{j,x_j}$  is 1, so  $\sum_x^H W_x = 1$  by (10.4)–(10.6). By Theorem 4.2 the weights define a positive strong probability on all subsets. To check a finite-coordinate cylinder, separate the specified coordinates from the unspecified ones in (10.6): at a specified base coordinate the factor is  $u_n$ , at a specified rare coordinate it is  $u_n r_{n,a} = p_{n,a}$ ,

and at an unspecified coordinate the summed contribution is  $u_n(1 + \sum_a r_{n,a}) = 1$ ; the exceptional coordinates satisfy the same marginal identity by (10.1). The cylinder mass is exactly (10.2). Uniqueness follows from Theorem 8.2, and (10.4) gives the concentration on  $\Omega_{\text{fin}}$ .  $\square$

*Remark 10.2* (A precise meaning of concentration). The set  $\Omega_{\text{fin}}$  is countable, because the alphabets and  $J$  are finite and only finitely many other coordinates may vary; it is Borel. Formula (10.4) gives  $\mu(\Omega \setminus \Omega_{\text{fin}}) = 0$  and  $\mu(\Omega_{\text{fin}}) = 1$ . A path with infinitely many deviations can still lie in each of its positive-mass finite cylinders; it is simply not a point atom.

### 10.3 Bernoulli variables

**Corollary 10.3** (One-line Bernoulli criterion; source 10). *Let  $0 \leq p_n \leq 1$  in  $K_{\mathbb{R}}$  prescribe independent Bernoulli coordinates with success probabilities  $p_n$ , and put  $q_n = \min(p_n, 1 - p_n)$ . There is a positive strong Borel product law if and only if  $(q_n)_{n \geq 1}$  is strongly summable.*

*Proof.* A residue Bernoulli law is nondeterministic exactly when  $\text{st}(q_n) > 0$ . Since every  $q_n \geq 0$ , coefficient-local finiteness at exponent 0 makes this happen at only finitely many  $n$  when  $(q_n)$  is strong. Outside those indices  $q_n$  is exactly the rare-entry probability. Conversely, a strong rare-entry tail together with finitely many exceptional  $q_n$  is a strong family. Apply Theorem 10.1.  $\square$

Choose at each coordinate a state of probability  $1 - q_n$  as base state, resolving ties arbitrarily. With the finitely many noninfinitesimal factors separated before the infinite product is formed, the probability of the path whose deviations occur exactly at a finite set  $F$  is

$$\left( \prod_{n \geq 1}^H (1 - q_n) \right) \prod_{n \in F} \frac{q_n}{1 - q_n}; \quad (10.7)$$

the denominators never vanish, since  $q_n \leq \frac{1}{2}$ . In particular an i.i.d. Bernoulli law with real bias  $p$  has a strong extension exactly when  $p \in \{0, 1\}$  (compare Corollary 6.3).

**Example 10.4** (Successes and failures of strong extension; source 10). The probabilities  $p_n = t^n$  admit a strong product law; its all-zero atom is  $\prod_{n \geq 1}^H (1 - t^n)$ . So do  $p_n = t^{n/(n+1)}$ , although the associated partial sums need not converge in the valuation topology. Neither  $p_n = 2^{-n}t$  nor  $p_n = t$  admits one: infinitely many rare entries contribute at exponent 1, and in the first case this persists despite ordinary summability of the real coefficients. The probabilities  $p_n = t^{1/n}$  fail because their supports descend, and  $p_n = \frac{1}{2}$  at infinitely many coordinates fails already at the residue level. Finitely many ordinary nontrivial probabilities together with a strong infinitesimal rare tail do admit an extension.

At a deterministic baseline with perturbations on a common well-ordered exponent set, Corollary 10.3 says that every exponent row must be finitely supported; this is source 15's finite-row criterion, compared with the  $\ell^1$  threshold of the coefficientwise class in Remark 21.2.

## 11 Coarse-graining can hide a genuine obstruction

Theorem 10.1 tests the individual rare entries, not merely their sum at each coordinate. This is necessary because positive Hahn series can have negative higher coefficients. Take

$$\Gamma = \mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}, \quad \alpha = (0, 1), \quad \beta = (1, 0), \quad \gamma_n = \beta + \frac{1}{n}\alpha,$$

so that  $\gamma_1 > \gamma_2 > \dots$  while  $n\alpha < \gamma_n$  for every  $n$ . For  $E_n = \{0, 1, 2\}$  put

$$p_{n,0} = 1 - t^{n\alpha}, \quad p_{n,1} = \frac{1}{2}(t^{n\alpha} + t^{\gamma_n}), \quad p_{n,2} = \frac{1}{2}(t^{n\alpha} - t^{\gamma_n}). \quad (11.1)$$

All three entries are positive and sum to 1; the residue state is 0 with probability 1.

**Theorem 11.1** (Extension may fail after refining a valid binary law; source 10). *The independent three-state cylinder laws (11.1) have no positive strong Borel extension. Their binary coarse-graining, which merges states 1 and 2, has a positive strong Borel extension.*

*Proof.* The coarse-grained rare probability is  $p_{n,1} + p_{n,2} = t^{n\alpha}$ ; this family is strongly summable, so Corollary 10.3 gives a binary strong product law. The individual rare-entry supports contain the strictly decreasing sequence  $(\gamma_n)$ , so (10.3) is not strongly summable and Theorem 10.1 rules out a three-state extension.

There is also a direct witness. Let  $B_n$  be the event that the first  $n - 1$  coordinates are 0 and the  $n$ th is 1. These events are pairwise disjoint, with prescribed masses

$$h_n = \frac{1}{2}(t^{n\alpha} + t^{\gamma_n}) \prod_{k < n} (1 - t^{k\alpha}). \quad (11.2)$$

The coefficient of  $t^{\gamma_n}$  in  $h_n$  is  $\frac{1}{2}$ : contributions from  $t^{n\alpha}$  have first coordinate 0, and  $t^{\gamma_n}$  times a nonempty subproduct has exponent strictly larger than  $\gamma_n$ . So the combined support of the  $h_n$  contains the descending sequence  $(\gamma_n)$ , and this disjoint family violates (SH).  $\square$

*Remark 11.2* (Why positivity does not protect support). The negative coefficient of  $t^{\gamma_n}$  in  $p_{n,2}$  does not make that probability negative, since the smaller exponent  $n\alpha$  has positive leading coefficient. Upon aggregation the two higher coefficients cancel; this is exactly the information lost by testing only total rare probabilities. The example does not contradict preservation of strong measures under pushforward; it shows that a valid coarse law need not admit a prescribed strong refinement.

With  $\iota(a, b) = a\omega + b$ , which is additive and strictly order preserving on  $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ , Convention 2.1 sends

$$t^{n\alpha} \mapsto \omega^{-n}, \quad t^{\gamma_n} \mapsto \omega^{-\omega^{-1/n}}, \quad (11.3)$$

so (11.1) is an explicit system of positive surreal probabilities, not merely a formal example in an unrelated field.

**Proposition 11.3** (Absoluteness under exponent enlargement; source 10). *Let  $j : \Gamma \rightarrow \Delta$  be an order-preserving additive embedding, and transport coherent cylinder values from  $k((t^\Gamma))$  to  $k((u^\Delta))$  by  $t^\gamma \mapsto u^{j(\gamma)}$ . A strong extension exists in the larger field if and only if it exists in the original one. For real coefficients the same holds for positive strong extensions.*

*Proof.* The combined support is carried order-isomorphically onto its image, so (C1) is unchanged; the widths at image exponents are unchanged and those at all other exponents are zero, so (C2) is unchanged. Theorem 8.2 gives the equivalence for signed or complex extensions, and its uniqueness proof shows that an extension in the larger field acquires no extra exponents, since a finite atomic coefficient measure vanishing on every cylinder is zero. The reconstructed singleton weights are exactly the transported original weights, and the ordered embedding preserves their signs.  $\square$

In particular, allowing more surreal exponents does not rescue any failed strong cylinder extension: a proposed surreal extension has set-sized range and can be localized together with the original data in one larger exponent group (Section 7), where the proposition applies. Source 11 makes the same observation for its moment example (Corollary 18.1), and source 15 for the coefficientwise i.i.d. obstruction (after Corollary 23.1).

## Part IV

# Strong Hahn measures: moments, hidden negative mass, and the $\omega + 1$ threshold

This part is source 11's, with source 10's Cantor-space form of the hidden negative atom and of the threshold. Sample points are *ordinary real points*; the values of measures, moments, polynomial coefficients and inner products lie in a Hahn field or in the surreals. Only in the finite Gaussian quadrature theorem are nodes allowed to lie in a real closed Hahn field. Keeping these two sample domains apart is essential to the results.

## 12 An exact strong moment representation criterion

The classical moment problem asks when a sequence  $(m_n)$  is represented by a positive measure through  $m_n = \int x^n d\mu$ . On  $[0, 1]$ , positivity of the quadratic forms of squares and of  $X(1 - X)$  times squares suffices, and equivalent conditions are nonnegative finite differences of the moments [7]. Replacing real values by Hahn values changes two separate things: one must say what an infinite sum means, and positivity becomes lexicographic across infinitesimal scales. An ordinary polynomial has finite degree even when its coefficients are surreal, so it may be unable to tell a late infinitesimal negative weight from infinitely many earlier positive weights. This part makes that obstruction exact.

### 12.1 Finite scalar moment data and simple-root recurrences

**Lemma 12.1** (Finite signed moment sequences). *Let  $(a_n)_{n \geq 0}$  be a real sequence and  $\Omega \subseteq \mathbb{R}$ . The following are equivalent.*

(i) *There are finitely many distinct points  $x_1, \dots, x_r \in \Omega$  and real coefficients  $c_1, \dots, c_r$  with*

$$a_n = \sum_{j=1}^r c_j x_j^n \quad (n \geq 0). \quad (12.1)$$

(ii) *A monic polynomial  $r(Z)$  with distinct roots, all in  $\Omega$ , annihilates the sequence under the forward shift:  $r(E)a = 0$ ,  $(Ea)_n = a_{n+1}$ . The constant polynomial  $r = 1$  is allowed for the zero sequence.*

*After zero coefficients are removed the representation (12.1) is unique. We read  $0^0 = 1$  in moment formulas.*

*Proof.* For (i) $\Rightarrow$ (ii) take  $r(Z) = \prod_j (Z - x_j)$ ; each sequence  $(x_j^n)$  is killed by  $E - x_j$ , including  $(1, 0, 0, \dots)$  when  $x_j = 0$ . Conversely, let  $r$  have degree  $r_0$  and distinct roots  $x_1, \dots, x_{r_0}$ . Being monic, it determines all terms from the first  $r_0$ ; the Vandermonde matrix  $(x_j^n)_{0 \leq n < r_0, 1 \leq j \leq r_0}$  is invertible, so the  $c_j$  can be solved from those terms, and (12.1) then satisfies the same recurrence and initial data. For uniqueness subtract two representations and use the Vandermonde system on the union of their nodes.  $\square$

Equivalently,  $\sum_{n \geq 0} a_n z^n = \sum_j c_j / (1 - x_j z)$  as formal power series, a node 0 contributing the constant  $c_j$ . A rational generating function or finite Hankel rank would not suffice: repeated roots give sequences such as  $nb^{n-1}$  ( $b \neq 0$ ), the moments of the derivative functional  $p \mapsto p'(b)$  rather than of a finite signed point measure. No positivity of the  $c_j$  is assumed.

## 12.2 Assembling the coefficients into a measure

**Theorem 12.2** (Exact strong moment criterion; source 11). *Let  $\Omega \subseteq \mathbb{R}$  carry a countably separating  $\sigma$ -algebra containing its relative Borel sets. Let  $(m_n)_{n \geq 0} \subseteq K_{\mathbb{R}}$  with  $m_0 = 1$ , and put*

$$S_m = \bigcup_{n \geq 0} \text{supp } m_n, \quad a_n(\gamma) = [t^\gamma]m_n.$$

*There is a positive strong Hahn probability on  $\Omega$  with*

$$m_n = \int_{\Omega} x^n d\mu(x) \quad (n \geq 0) \quad (12.2)$$

*if and only if:*

- (a)  $S_m$  is well ordered;
- (b) for every  $\gamma \in S_m$  the sequence  $(a_n(\gamma))_{n \geq 0}$  has a finite signed representation

$$a_n(\gamma) = \sum_{x \in F_\gamma} c_{\gamma,x} x^n, \quad F_\gamma \subseteq \Omega \text{ finite}, \quad (12.3)$$

*equivalently satisfies the simple-root recurrence of Lemma 12.1 with roots in  $\Omega$ ;*

- (c) with  $c_{\gamma,x} = 0$  at missing nodes, every reassembled series

$$w_x = \sum_{\gamma \in S_m} c_{\gamma,x} t^\gamma \quad (12.4)$$

*is nonnegative in  $K_{\mathbb{R}}$ .*

*The representing positive strong measure is unique. More generally, conditions (a) and (b) alone give a unique signed strong measure with these moments, and (c) is exactly its positivity.*

*Proof.* Suppose a signed strong measure represents the moments. Theorem 4.2 supplies strongly summable point weights and finite coefficient measures as in (3.2); taking coefficients in the integral (Proposition 5.1) gives (12.3). Let  $S_w = \bigcup_x \text{supp } w_x$ ; certainly  $S_m \subseteq S_w$ . Conversely, at an exponent of  $S_w$  the coefficient measure is a nonzero finite signed point measure; if all its moments vanished, the Vandermonde argument of Lemma 12.1 would make it zero. So  $S_m = S_w$ , which gives (a), and the weights in (12.4) are the singleton weights of the measure; for a positive measure they are nonnegative, which gives (c).

Conversely, (a) makes every series (12.4) a Hahn series, and (b) says that at each exponent only finitely many of them have a nonzero coefficient. So the weight family is strongly summable, and Theorem 4.2 gives a signed strong measure whose moments agree with  $m_n$  coefficientwise by (12.3); its total mass is  $m_0 = 1$  and (c) makes it positive. Any other signed strong representing measure has finite coefficient measures with the same moment sequences, which Lemma 12.1 identifies; this proves uniqueness, even without positivity.  $\square$

*Remark 12.3* (Compactness is not responsible for this uniqueness). For ordinary real measures compact support is a familiar source of moment determinacy. The uniqueness above holds on an arbitrary ordinary real sample set, even an unbounded one, because every coefficient measure has *finite* support. It is a coefficientwise Vandermonde theorem, not an appeal to uniform polynomial approximation or to a real representation theorem.

**How to use the criterion, and what is not algorithmic.** The criterion separates three tasks: verify a common well order across the moments; solve a finite atomic problem at each exponent; inspect the leading coefficient of each reconstructed point weight. Positive Hankel matrices over  $K_{\mathbb{R}}$  do not substitute for the last task; the example of Section 14 satisfies much stronger positivity conditions and fails exactly there. With explicit finite recurrence data at one exponent the reconstruction is finite algebra (find the distinct real roots in  $\Omega$ , solve the Vandermonde system, verify the recurrence). But the theorem quantifies over all exponents and all moments: it is not a uniform decision algorithm from numerical access to coefficients or from a finite list of moments, and finitely many verified recurrence instances do not establish an infinite recurrence.

## 13 Finite visibility and valuation barriers

### 13.1 When finite polynomial tests do detect signs

For a signed strong measure with singleton weights  $(w_x)$ , the atoms *visible by scale*  $\gamma$  are

$$D_\gamma = \{x : w_x \neq 0, v(w_x) \leq \gamma\}, \quad (13.1)$$

the union of the finite coefficient supports  $F_\eta$  over  $\eta \leq \gamma$ . A fixed coefficient row is finite, but  $D_\gamma$  need not be.

**Theorem 13.1** (Finite visibility detects a negative atom; source 11). *Let  $\mu$  be a signed strong Hahn measure on  $\Omega \subseteq \mathbb{R}$  and  $L(P) = \int P d\mu$ . Suppose  $w_{x_0} < 0$ , put  $\gamma = v(w_{x_0})$ , and assume  $D_\gamma$  is finite. Then a real polynomial  $p$  of degree at most  $|D_\gamma| - 1$  satisfies  $L(p^2) < 0$ . Consequently, if every  $D_\gamma$  is finite, the following are equivalent:  $\mu$  is positive;  $L(p^2) \geq 0$  for all  $p \in \mathbb{R}[X]$ ;  $L(P^2) \geq 0$  for all  $P \in K_{\mathbb{R}}[X]$ .*

*Proof.* Take  $p(X) = \prod_{y \in D_\gamma \setminus \{x_0\}} (X - y)$ . Then  $p(x_0)$  is a nonzero real number, and every other atom that can contribute at an exponent at most  $\gamma$  is killed by  $p$ . The weight at  $x_0$  has no coefficient below  $\gamma$ , so

$$v(L(p^2)) = \gamma, \quad [t^\gamma]L(p^2) = [t^\gamma]w_{x_0} p(x_0)^2 < 0.$$

A positive measure is nonnegative on squares by strong summation of nonnegative values; conversely square positivity rules out every negative weight by the first assertion, and Theorem 4.2 applies.  $\square$

The same isolation, with the indicator of a cylinder containing  $x_0$  and no other point of  $D_\gamma$  in place of  $p^2$ , is source 10's proof of the cylinder form of the threshold (Theorem 15.2(a)).

**Corollary 13.2** (A necessary condition for hidden negative mass). *If a signed strong measure is nonnegative on all real polynomial squares, every negative atom has infinitely many distinct atoms of strictly smaller valuation.*

*Proof.* Only finitely many atoms have valuation exactly  $\gamma$ , since each contributes a nonzero coefficient at  $\gamma$ . If a negative atom of valuation  $\gamma$  had only finitely many earlier atoms,  $D_\gamma$  would be finite and Theorem 13.1 would produce a negative square.  $\square$

**Corollary 13.3** (The safe support range; source 11). *Let  $S = \bigcup_x \text{supp } w_x$ . If each element of  $S$  has only finitely many predecessors in  $S$ , equivalently  $\text{ot}(S) \leq \omega$ , then positivity on real polynomial squares implies positivity of the measure. In particular a globally left-finite support, with  $S \cap (-\infty, \gamma]$  finite for every  $\gamma \in \Gamma$ , has this property.*

*Proof.* For a proposed negative atom of valuation  $\gamma \in S$ , the exponents of  $S$  at or before  $\gamma$  form a finite set, each with a finite coefficient row, so  $D_\gamma$  is finite and Theorem 13.1 applies.  $\square$

A support of order type  $\omega$  may be bounded above in  $\Gamma$  and need not be globally left-finite; only the finite-predecessor condition at an actual leading exponent is used. The assumption is global: requiring each *individual* point weight to have left-finite support does not make the family left-finite. Conversely finite visibility is weaker than a global left-finite support, since infinitely many exponents attached to one fixed atom still give a finite  $D_\gamma$ . It is the number of spatially distinct atoms visible at a scale that controls the interpolation detector.

### 13.2 A barrier for every finite-degree polynomial

Let  $\nu$  be a positive strong measure with weights  $u_x$  and fix  $\gamma \in \Gamma$ . Its earlier atom set is

$$E_\gamma(\nu) = \{x : u_x > 0, v(u_x) < \gamma\}.$$

A signed strong measure  $\tau$  *starts at scale*  $\gamma$  if every nonzero singleton weight of  $\tau$  has valuation at least  $\gamma$ ; by Theorem 4.2 every coefficient of every weight of  $\tau$  then occurs at an exponent at least  $\gamma$ .

**Theorem 13.4** (Polynomial positivity barrier; source 11). *Let  $\Omega \subseteq \mathbb{R}$ , let  $\nu$  be positive with  $E_\gamma(\nu)$  infinite, let  $\tau$  be any signed strong measure starting at scale  $\gamma$ , and put  $L(P) = \int P d\nu + \int P d\tau$  for  $P \in K_{\mathbb{R}}[X]$ . For every nonzero polynomial  $P$  with  $P(x) \geq 0$  at every ordinary sample point  $x \in \Omega$ ,  $L(P) > 0$ . In particular  $L(P^2) > 0$  for every nonzero  $P \in K_{\mathbb{R}}[X]$ . No sign assumption is made on  $\tau$ .*

*Proof.* Let  $\delta$  be the least valuation of the nonzero coefficients of  $P$ ; the coefficient of  $t^\delta$  in  $P$  is a nonzero ordinary real polynomial  $p_\delta$ . At each real sample point  $v(P(x)) \geq \delta$ , with equality where  $p_\delta(x) \neq 0$ . A nonzero real polynomial has finitely many real zeros, so choose  $x \in E_\gamma(\nu)$  with  $p_\delta(x) \neq 0$ . Pointwise positivity gives  $P(x) > 0$ . All values  $u_y P(y)$  are nonnegative, and the chosen one has valuation  $v(u_x) + \delta < \gamma + \delta$ ; Lemma 2.3 gives  $\int P d\nu > 0$  with  $v(\int P d\nu) < \gamma + \delta$ . Every product contributing to  $\int P d\tau$  has valuation at least  $\gamma + \delta$ , and so does its sum, whatever cancellation occurs. The tail cannot change the sign of the earlier leading term, so  $L(P) > 0$ .  $\square$

The word “barrier” refers to a valuation inequality, not a topological limit. A finite-degree leading polynomial can annihilate only finitely many distinct early atoms, and allowing Hahn coefficients does not help: a finite-degree polynomial still has a nonzero ordinary coefficient polynomial at its least exponent.

**Corollary 13.5** (Hermitian and localizing forms; source 11). *Suppose  $\Omega \subseteq [0, 1]$  and the hypotheses of Theorem 13.4 hold. Extend  $L$  complex-linearly to  $K_{\mathbb{C}}[X]$ , and let  $P^*$  conjugate the coefficients of  $P$  while fixing  $X$ . For  $P \neq 0$  the four values*

$$L(P^*P), \quad L(XP^*P), \quad L((1-X)P^*P), \quad L(X(1-X)P^*P) \quad (13.2)$$

*are strictly positive elements of  $K_{\mathbb{R}}$ .*

*Proof.* At real  $x$ ,  $P^*(x)P(x)$  is a nonnegative real Hahn value. Writing  $P = A + iB$ ,  $P^*P = A^2 + B^2$  is a nonzero polynomial in  $K_{\mathbb{R}}[X]$ , and the four polynomials inside  $L$  are nonzero and nonnegative on  $\Omega$ . Apply Theorem 13.4 to each.  $\square$

**Corollary 13.6** (Zariski-dense early atoms; source 11). *For  $\Omega \subseteq \mathbb{R}^d$ , replace the infinitude of  $E_\gamma(\nu)$  by the condition that no nonzero real polynomial vanishes on all of  $E_\gamma(\nu)$ . Then Theorem 13.4 holds, with the same proof, for  $K_{\mathbb{R}}[X_1, \dots, X_d]$  and polynomials nonnegative on  $\Omega$ .*

*Proof.* The least-valuation coefficient of a nonzero multivariate Hahn polynomial is a nonzero ordinary polynomial, and the condition supplies an early atom where it does not vanish; the rest of the proof is unchanged.  $\square$

This is the multivariate replacement for “a polynomial has finitely many roots”. Infinitely many atoms on one algebraic curve would not suffice in higher dimension, and the corollary is a sufficient condition, not a classification of the intermediate cases.

### 13.3 A topological barrier for continuous functions

When the tail atoms lie in the ordinary closure of the earlier atom set, all nonnegative continuous functions can fail to see a negative event.

**Theorem 13.7** (Continuous-test barrier; source 11). *Let  $\Omega$  be a metrizable space with a countably separating measurable structure containing its Borel sets. Let  $\nu$  be a positive strong Hahn measure, let  $\tau$  start at scale  $\gamma$ , and assume  $\{x : \tau(\{x\}) \neq 0\} \subseteq \overline{E_\gamma(\nu)}$ . Then  $T(f) = \int f d\nu + \int f d\tau$  is nonnegative for every ordinary continuous  $f : \Omega \rightarrow \mathbb{R}$  with  $f \geq 0$ . If  $\Omega$  is compact and  $T(1) = 1$ , then*

$$-\|f\|_\infty \leq T(f) \leq \|f\|_\infty \quad (f \in C(\Omega, \mathbb{R})). \quad (13.3)$$

*Proof.* If  $f(x) > 0$  for some  $x \in E_\gamma(\nu)$ , the positive integral  $\int f d\nu$  has valuation less than  $\gamma$ , whereas  $v(\int f d\tau) \geq \gamma$ . Otherwise  $f$  vanishes on  $E_\gamma(\nu)$ , hence on its closure and at every atom of  $\tau$ , and  $T(f) = \int f d\nu \geq 0$ . On a compact space apply positivity to  $M \pm f$  with  $M = \|f\|_\infty \in \mathbb{R}$ .  $\square$

The bound (13.3) is an ordered-field inequality with an ordinary real sup norm. It does not make the target a real Banach space and does not assert continuity into the valuation topology or the surreal fine topology.

## 14 One weight pattern, two test classes

Sources 10 and 11 found, independently, the same weights. We print them once.

**Definition 14.1** (The hidden-negative weight pattern; sources 10 and 11). Let  $0 < \delta_1 < \delta_2 < \dots$  in  $\Gamma$  and  $\gamma \in \Gamma$  with  $\delta_n < \gamma$  for all  $n$ . For distinct points  $a$  (the *anchor*),  $o$  (the *negative atom*) and  $e_1, e_2, \dots$  (the *early atoms*) of a set  $X$ , the pattern is the weight family

$$w_{e_n} = t^{\delta_n}, \quad w_o = -t^\gamma, \quad w_a = 1 - \sum_{n \geq 1}^H t^{\delta_n} + t^\gamma, \quad (14.1)$$

with all other weights zero. The *standard pattern* has  $\Gamma = \mathbb{Q}$ ,  $\delta_n = q_n := n/(n+1)$  and  $\gamma = 1$ .

At each  $\delta_n$  exactly two weights ( $e_n$  and  $a$ ) have a nonzero coefficient, at  $\gamma$  two ( $o$  and  $a$ ), and at 0 one ( $a$ ). So the family is strongly summable, with sum 1 and global support

$$\{0\} \cup \{\delta_n : n \geq 1\} \cup \{\gamma\}, \quad \text{of order type } \omega + 1.$$

The anchor has leading term 1 and is positive, every early weight is positive, and  $w_o < 0$ . By Theorem 4.2 the pattern defines a normalized strong signed measure on all subsets of  $X$ , with mass  $-t^\gamma$  at  $o$ .

### 14.1 On Cantor space, with cylinder tests (source 10)

On binary Cantor space  $\{0,1\}^{\mathbb{N}}$  take the anchor  $a = (1,1,1,\dots)$ , the negative atom  $o = (0,0,0,\dots)$ , and the early atoms  $e_n$ , the path with its single 1 at position  $n$ . Let  $\mu_C$  be the strong signed measure of the pattern.

**Proposition 14.2** (Every finite-dimensional law is nonnegative; source 10). *For every cylinder  $[s]$ ,  $\mu_C([s]) \geq 0$ . Nevertheless no positive strong measure has these cylinder values.*

*Proof.* For  $N \geq 1$  the all-zero cylinder  $[0^N]$  contains  $o$  and exactly the  $e_n$  with  $n > N$ , but not  $a$ . Hence

$$\mu_C([0^N]) = \sum_{n>N}^H t^{\delta_n} - t^\gamma > 0, \quad (14.2)$$

with leading exponent  $\delta_{N+1} < \gamma$  and coefficient 1. A cylinder not containing  $o$  contains at most one  $e_n$ ; if it also contains  $a$ , its mass has constant coefficient 1 and is positive, and otherwise its mass is one  $t^{\delta_n}$  or zero. The empty-word cylinder has mass 1. So all cylinders, and all finite unions of cylinders, are nonnegative. The unique strong signed extension is  $\mu_C$  by Theorem 8.2, and  $\mu_C(\{o\}) = -t^\gamma < 0$ .  $\square$

The widths are at most two at every exponent. The failure is not unbounded width, lack of a common support, or a negative finite law. Each finite neighbourhood of  $o$  receives a positive contribution at an exponent below  $\gamma$ , but these contributions escape to later and later paths, and the singleton has none of them.

### 14.2 On the interval, with polynomial and continuous tests (source 11)

From here through Section 18 take the standard pattern on  $\Omega = [0,1] \subseteq \mathbb{R}$  with

$$a = \frac{3}{4}, \quad o = 0, \quad e_n = x_n := \frac{1}{n+1}, \quad q_n = \frac{n}{n+1} \quad (n \geq 1).$$

The points  $a, 0, x_1, x_2, \dots$  are distinct, and  $0 < q_1 < q_2 < \dots < 1$ . The associated signed strong measure on all subsets of  $[0,1]$  is

$$\mu_* = \left(1 - \sum_{n \geq 1}^H t^{q_n} + t\right) \delta_a + \sum_{n \geq 1}^H t^{q_n} \delta_{x_n} - t \delta_0, \quad (14.3)$$

a coefficientwise-finite strong sum of point measures, involving no countable additivity in a real norm. Split it as

$$\nu = \left(1 - \sum_{n \geq 1}^H t^{q_n}\right) \delta_a + \sum_{n \geq 1}^H t^{q_n} \delta_{x_n}, \quad \tau = t(\delta_a - \delta_0), \quad \mu_* = \nu + \tau. \quad (14.4)$$

Here  $\nu$  is a positive strong probability,  $\tau$  starts at scale 1 and has total mass zero,  $E_1(\nu) = \{a, x_1, x_2, \dots\}$  and  $0 \in \overline{E_1(\nu)}$  in the *ordinary* topology of the interval. For  $P \in K_{\mathbb{R}}[X]$ , integration against  $\mu_*$  is

$$L(P) = P(a) + \sum_{n \geq 1}^H t^{q_n} (P(x_n) - P(a)) + t(P(a) - P(0)); \quad (14.5)$$

for Hahn coefficients the middle sum is still strong by Lemma 2.3. The moments are  $m_0 = 1$  and

$$m_k = a^k + \sum_{n \geq 1}^H (x_n^k - a^k) t^{q_n} + a^k t \quad (k \geq 1), \quad (14.6)$$

with the coefficient measures read off directly:

| Exponent | Coefficient measure       | Moments at that exponent |
|----------|---------------------------|--------------------------|
| 0        | $\delta_a$                | $a^k$                    |
| $q_n$    | $\delta_{x_n} - \delta_a$ | $x_n^k - a^k$            |
| 1        | $\delta_a - \delta_0$     | $a^k - 0^k$              |

With  $0^0 = 1$ , every nonzero-exponent coefficient measure has total mass zero. At  $q_n$  the coefficient sequence is annihilated by  $(E - x_n)(E - a)$ , at 1 by  $E(E - a)$ : simple-root recurrences. So the failure below is not caused by a complicated coefficient moment problem, a multiple root, or an unbounded number of coefficient atoms.

**Theorem 14.3** (Full positivity but no positive strong representation; source 11). *The functional (14.5) has the following properties.*

- (i)  $L(1) = 1$ , and  $L(P) > 0$  for every nonzero  $P \in K_{\mathbb{R}}[X]$  with  $P(x) \geq 0$  at every ordinary  $x \in [0, 1]$ .
- (ii) Its complex-linear extension has all four values (13.2) strictly positive, for every nonzero  $P \in K_{\mathbb{C}}[X]$  of every finite degree.
- (iii) Its restriction to real-coefficient polynomials extends, by the same formula, to a positive unital  $\mathbb{R}$ -linear functional  $T : C([0, 1], \mathbb{R}) \rightarrow K_{\mathbb{R}}$  satisfying the order bound (13.3).
- (iv) There is no positive strong Hahn measure on the ordinary Borel space  $[0, 1]$  with moment sequence  $(m_k)$ : the unique signed strong representing measure is  $\mu_*$ , and  $\mu_*(\{0\}) = -t < 0$ .

*Proof.* The support and finite coefficient activity were checked after Definition 14.1. By (14.4) the construction falls under Theorem 13.4 with  $\gamma = 1$ , since infinitely many distinct atoms have positive weights of valuation less than 1; this gives (i), and Corollary 13.5 gives (ii). The atoms  $a$  and  $0$  of  $\tau$  lie in the ordinary closure of the earlier atom set, so Theorem 13.7 gives (iii); (14.5) makes sense for every ordinary continuous function by Proposition 5.1. Finally Theorem 12.2 gives uniqueness among signed strong measures. Directly: at exponent 1 the coefficient measure must have all the moments of  $\delta_a - \delta_0$ , so finite coefficient atomicity and Vandermonde uniqueness force it; at exponents below 1 the coefficient measures in the table have no atom at 0. So the weight at 0 has leading coefficient  $-1$  at exponent 1 and cannot be nonnegative.  $\square$

The fourth conclusion concerns the strong class on the stated ordinary sample space. It does not rule out generalized-limit probabilities, weaker additivity, representations with a different target, or measures on a sample space of nonreal field elements.

**Corollary 14.4** (All Hausdorff inequalities hold strictly; source 11). *For all ordinary integers  $k, n \geq 0$ ,*

$$\sum_{j=0}^n (-1)^j \binom{n}{j} m_{k+j} = L(X^k(1 - X)^n) > 0. \quad (14.7)$$

*In particular  $1 = m_0 > m_1 > m_2 > \cdots > 0$  and  $\text{st}(m_k) = a^k$ .*

*Proof.*  $X^k(1 - X)^n$  is nonzero and nonnegative on  $[0, 1]$ , so Theorem 14.3(i) applies;  $n = 1$  and  $n = 0$  give the stated inequalities, and the standard part is read from (14.6).  $\square$

For real sequences the non-strict finite-difference conditions characterize interval moment sequences [7]. Here they hold at every order and still produce no positive *strong Hahn* measure. This is an obstruction to one non-Archimedean extension of a representation theorem, not a counterexample to the classical real theorem.

**A continuous detector sequence and the missing limit.** For integers  $k \geq 2$  let  $f_k(x) = \max(1 - kx, 0)$  on  $[0, 1]$ . These nonnegative continuous functions decrease pointwise to  $\mathbf{1}_{\{0\}}$ , with  $f_k(a) = 0$  and  $f_k(x_n) > 0$  exactly for  $n \geq k$ . Hence

$$T(f_k) = \sum_{n \geq k}^H \left(1 - \frac{k}{n+1}\right) t^{q_n} - t, \quad v(T(f_k)) = q_k, \quad \text{lc}(T(f_k)) = \frac{1}{k+1} > 0. \quad (14.8)$$

The limiting indicator, integrated against  $\mu_*$ , gives  $-t$ . There is no contradiction: the positive terms do not become small relative to the scale  $t$ , since  $T(f_k) - \mu_*(\{0\})$  has valuation  $q_k < 1$  for every  $k$ , and a monotone-convergence step across this barrier is unavailable. The ordinary topology is what matters: a nonnegative continuous function positive at 0 is positive at infinitely many  $x_n$ , and one vanishing at every  $x_n$  vanishes at 0.

**Why finite truncations are not proofs of the example.** Let  $\mu_N$  retain only the early atoms with  $1 \leq n \leq N$ , with the anchor adjusted. It is normalized and still has mass  $-t$  at 0, but  $p_N(X) = (X - a) \prod_{n=1}^N (X - x_n)$  satisfies

$$\int p_N^2 d\mu_N = -t p_N(0)^2 < 0. \quad (14.9)$$

Once the next atom is included the same polynomial meets an earlier positive contribution:

$$v\left(\int p_N^2 d\mu_{N+1}\right) = q_{N+1}, \quad \text{lc}\left(\int p_N^2 d\mu_{N+1}\right) = p_N(x_{N+1})^2 > 0. \quad (14.10)$$

Every finite truncation eventually fails; the infinite theorem says that every fixed finite-degree polynomial meets a surviving earlier atom. Reversing those quantifiers would invalidate the proof. Source 11's check suite tests both (14.9) and (14.10), and source 10's suite tests the corresponding cutoff for  $\mu_C$ , to make the distinction visible.

*Remark 14.5* (The interval example fails in the coefficientwise class too). No positive *coefficientwise* Hahn measure on  $[0, 1]$  has the moments (14.6) either. A finite signed Borel measure on  $[0, 1]$  is determined by its moments (Weierstrass approximation and uniqueness in the Riesz representation), so every coefficientwise representing measure has exactly the coefficient measures of the table, and is  $\mu_*$  read coefficientwise. The null-ideal criterion (Theorem 19.2) detects the failure:  $\{0\}$  is null for  $|\mu_0| = \delta_a$  and for every  $|\mu_{q_n}| = \delta_{x_n} + \delta_a$ , so  $\{0\} \in \mathcal{J}_{<1}$ , while  $\mu_1^-(\{0\}) = 1 > 0$ . (This remark is added in the merge; neither source states it.)

## 15 The $\omega + 1$ threshold in the strong class

**Definition 15.1** (Finite initial segments). A well-ordered set  $S$  has *finite initial segments* if  $\{\delta \in S : \delta \leq \gamma\}$  is finite for every  $\gamma \in S$ ; equivalently  $\text{ot}(S) \leq \omega$ .

This is an internal condition on the active support and does not require  $S$  to be unbounded in  $\Gamma$ :  $\{n/(n+1) : n \geq 1\}$  has finite initial segments though it is bounded in  $\mathbb{Q}$ . The pattern of Definition 14.1 has the smallest possible infinite support order type that can exhibit hidden negative mass: an increasing sequence followed by one later exponent. Sources 10 and 11 proved this for two test classes; the threshold is the same, and it is stated once.

**Theorem 15.2** (The  $\omega + 1$  threshold for strong measures; sources 10 and 11).

(a) Cylinder tests (source 10). Let  $S \subseteq \Gamma_{\geq 0}$  be well ordered with  $0 \in S$ . The following are equivalent.

(i)  $S$  has finite initial segments.

- (ii) *On binary Cantor space, every normalized strong real signed measure with global support contained in  $S$  and nonnegative cylinder masses is positive on every Borel set.*
- (iii) *Every normalized nonnegative coherent binary cylinder family with combined support contained in  $S$  and satisfying (C2) has a positive strong extension.*

*For (i) $\Rightarrow$ (ii) the binary alphabet may be replaced by arbitrary finite alphabets.*

- (b) *Polynomial tests (source 11). Among signed strong Hahn measures on ordinary real sample sets whose polynomial moment functional is nonnegative on every real polynomial square but which have a negative event, the least possible order type of the global coefficient support is  $\omega + 1$ . It is attained by  $\mu_*$ , even with the stronger positivity properties of Theorem 14.3.*

*Proof.* (a) Assume (i) and let a strong signed measure  $\mu$  have nonnegative cylinder masses. Suppose some  $w_x < 0$  and let  $\gamma = v(w_x)$ . There are finitely many active exponents  $\delta \leq \gamma$ , each with finitely many coefficient atoms, so their union  $F$  is finite. Choose a cylinder containing  $x$  and no other point of  $F$ ; its coefficients at all exponents at most  $\gamma$  agree with those of  $w_x$ , and there are no exponents outside  $S$ . So its leading coefficient is the negative coefficient of  $w_x$  at  $\gamma$ , contradicting cylinder positivity. Every point weight is nonnegative, and Theorem 4.2 gives (ii); the argument works for any finite alphabets. The equivalence (ii) $\Leftrightarrow$ (iii) follows from Theorem 8.2: (C1) is automatic for a subset of  $S$  and the strong extension is unique. If (i) fails, choose  $\gamma \in S$  with infinitely many predecessors in  $S$ ; since  $S$  is well ordered it contains a strictly increasing sequence  $0 < \delta_1 < \delta_2 < \dots < \gamma$ . The pattern with these exponents has support in  $S$ , is normalized and strong, and by Proposition 14.2 has nonnegative cylinders and a negative singleton; so (ii) fails.

(b) If the global support has order type below  $\omega + 1$ , its order type is at most  $\omega$ , and Corollary 13.3 makes every singleton weight, hence (Theorem 4.2) every event, nonnegative. The measure  $\mu_*$  has global support of order type exactly  $\omega + 1$ , positive polynomial squares by Theorem 14.3, and a negative singleton.  $\square$

*Remark 15.3* (What the threshold is and is not). The proof of the positive direction is not an order-completeness argument: it does not pass to an infimum of positive test values, but isolates finitely many coefficient atoms, and the pattern shows exactly why this finite isolation fails when an active exponent has infinitely many predecessors. The order type  $\omega + 1$  is that of the *global* coefficient support of the measure, which by the proof of Theorem 12.2 is also the union of the supports of all its moments; it does not assert that a single moment of support type  $\omega$  causes or precludes the phenomenon. It is a normal-form *support* order type, not a Conway birthday or sign-sequence length of an individual number. Finally, the theorem quantifies over strong measures only: in the coefficientwise class two exponents suffice (Example 19.5, Section 2.6). A formalization must keep the data separate (coherence, well order of the combined support, boundedness of each width, signs of reconstructed point weights); collapsing the last into cylinder positivity would encode the false assertion refuted by Proposition 14.2.

## 16 Exact Hankel determinants and recurrence scales

### 16.1 A strongly summable Cauchy–Binet identity

For a finite set  $F$  of distinct ordinary real points write  $V(F)^2 = \prod_{\{x,y\} \subseteq F} (x - y)^2$ , independent of any ordering, and equal to 1 when  $|F| \leq 1$ . For an ordinary real polynomial  $g$  put

$$H_d^{(g)} = (L(g(X)X^{i+j}))_{0 \leq i,j \leq d}, \quad \Delta_d^{(g)} = \det H_d^{(g)},$$

and omit the superscript when  $g = 1$ .

**Lemma 16.1** (Strong atomic determinant formula; source 11). *For any signed strong Hahn measure on an ordinary real sample set,*

$$\Delta_d^{(g)} = \sum_{\substack{F \subseteq \Omega \\ |F|=d+1}}^H \left( \prod_{x \in F} w_x g(x) \right) V(F)^2, \quad (16.1)$$

*and the displayed family is strongly summable.*

*Proof.* Multiplying weights by the real values  $g(x)$  preserves strong summability; products of any fixed  $d + 1$  members are strongly summable by Lemma 2.3, and multiplication by the real number  $V(F)^2$  adds no exponent. With  $v_x = (1, x, \dots, x^d)^\top$  one has  $H_d^{(g)} = \sum_x w_x g(x) v_x v_x^\top$  coefficientwise. Expand the determinant multilinearly in its  $d + 1$  columns; terms with a repeated sample point vanish, and grouping the remaining ordered choices by their underlying set  $F$  gives one Vandermonde determinant and its transpose, that is  $V(F)^2$ . This is the finite Cauchy–Binet calculation, valid coefficientwise because every product family involved is strongly summable.  $\square$

The finite Cauchy–Binet mechanism is classical. What matters here is licensing the infinite atomic expansion and using its *unique* lowest-valuation term although one weight is negative.

## 16.2 The complete leading-term calculation

For  $d \geq 0$  put

$$Y_d = \{a, x_1, \dots, x_d\}, \quad s_d = \sum_{j=1}^d q_j, \quad C_d = V(Y_d)^2,$$

with  $Y_0 = \{a\}$ ,  $s_0 = 0$ ,  $C_0 = 1$ , and let  $\mathcal{H}_r = \sum_{j=1}^r 1/j$ .

**Theorem 16.2** (Every determinant scale and leading coefficient; source 11). *For the functional of Theorem 14.3,*

$$v(\Delta_d) = s_d = d + 1 - \mathcal{H}_{d+1}, \quad (16.2)$$

$$\text{lc}(\Delta_d) = C_d = \left[ \frac{\prod_{j=1}^d (3j-1) \prod_{j=1}^d (j-1)!}{4^d ((d+1)!)^d} \right]^2 > 0. \quad (16.3)$$

*For each of  $g = X$ ,  $1 - X$  and  $X(1 - X)$ ,*

$$v(\Delta_d^{(g)}) = s_d, \quad \text{lc}(\Delta_d^{(g)}) = C_d \prod_{y \in Y_d} g(y) > 0. \quad (16.4)$$

*Proof.* The valuations of the nonzero point weights are, in increasing order,  $0, q_1, q_2, \dots, 1$ , the last belonging to the negative atom at 0. Among all sets of  $d + 1$  distinct atoms,  $Y_d$  uniquely minimizes the sum of valuations, and its product of leading weight coefficients is 1; replacing any of these earliest valuations by a later one strictly increases the sum. So (16.1) has a unique lowest-valuation term, of valuation  $s_d$  and leading coefficient  $V(Y_d)^2$ . The valuation sum is  $\sum_{j=1}^d (1 - \frac{1}{j+1}) = d + 1 - \mathcal{H}_{d+1}$ . For the Vandermonde product use  $a - x_j = \frac{3j-1}{4(j+1)}$  and  $x_i - x_j = \frac{j-i}{(i+1)(j+1)}$  for  $1 \leq i < j \leq d$ : the numerators of the latter multiply to  $\prod_{j=1}^d (j-1)!$ , their denominators to  $((d+1)!)^{d-1}$ , and the anchor differences contribute another  $4^d (d+1)!$ ; squaring gives (16.3). Each listed localizer is a positive real number at every point of  $Y_d$  and leaves the corresponding valuations unchanged; elsewhere it can remove a contribution but not create an earlier one. The same unique minimum proves (16.4).  $\square$

| $d$ | $v(\Delta_d)$ | $\text{lc}(\Delta_d)$ |
|-----|---------------|-----------------------|
| 0   | 0             | 1                     |
| 1   | 1/2           | 1/16                  |
| 2   | 7/6           | 25/82944              |
| 3   | 23/12         | 25/764411904          |
| 4   | 163/60        | 121/3057647616000000  |

The valuations can exceed the location 1 of the negative atom. There is no inconsistency: a determinant uses a product of  $d + 1$  weights, and for each fixed size its earliest product uses the earliest  $d + 1$  atoms and avoids the negative one.

### 16.3 Orthogonal polynomials and loss of a common support

Because  $L$  is positive on nonzero squares there is a unique monic orthogonal polynomial  $\pi_d$  of each degree  $d$ , with  $h_d = L(\pi_d^2) > 0$ . With  $\Delta_{-1} = 1$ , finite Gram elimination (a unit triangular change of basis from monomials) gives

$$h_d = \frac{\Delta_d}{\Delta_{d-1}}, \quad \Delta_d = \prod_{j=0}^d h_j. \quad (16.5)$$

The monic three-term recurrence is

$$X\pi_d = \pi_{d+1} + \alpha_d\pi_d + \beta_d\pi_{d-1}, \quad \beta_d = \frac{h_d}{h_{d-1}} > 0 \quad (d \geq 1); \quad (16.6)$$

lower terms vanish by orthogonality (move  $X$  to the other factor), and pairing with  $\pi_{d-1}$  identifies  $\beta_d$ .

**Corollary 16.3** (Explicit Jacobi scales; source 11). *For  $d \geq 1$ ,*

$$v(h_d) = q_d = \frac{d}{d+1}, \quad v(\beta_d) = \frac{1}{d(d+1)}. \quad (16.7)$$

*The off-diagonal coefficient  $\sqrt{\beta_d}$  of the orthonormal Jacobi matrix has valuation  $1/(2d(d+1))$ . Each  $\beta_d$  is a Hahn series, but the family  $(\beta_d)_{d \geq 1}$  has no common well-ordered support and is not strongly summable.*

*Proof.* By (16.5) and Theorem 16.2,  $v(h_d) = s_d - s_{d-1} = q_d$ ; with  $h_0 = 1$  and  $q_0 = 0$ ,  $v(\beta_d) = q_d - q_{d-1} = 1/(d(d+1))$ . These leading exponents form a strictly decreasing sequence inside the union of the supports of the  $\beta_d$ .  $\square$

This is a concrete warning about infinite orthogonalization. The moment family has a common well-ordered support and every finite Gram process is valid, but the sequence of normalized recurrence data loses the common support. An infinite Jacobi matrix can still be defined entrywise; what fails is a description in which all its entries share one well-ordered Hahn support. No assertion about every other operator representation is intended.

## 17 Positive Gaussian quadrature at every finite order

The next theorem separates a finite algebraic success from the infinite strong-measure obstruction. Its mechanism is classical orthogonal-polynomial and spectral algebra, not a new spectral theorem. Source 11 reproved the finite spectral theorem over a real closed field for it; the

collection already contains that theorem, as `thm:spectral` of the spectral-theory report [14] for any set-sized real closed field  $F$  (every symmetric matrix over  $F$  has an orthonormal eigenbasis over  $F$  with eigenvalues in  $F$ ), and as `spec:thm:hermitian` for Hermitian matrices over  $\mathbb{C}((t^\Gamma))$  at *any* set-sized  $\Gamma$ , with a real orthogonal diagonalizer for real symmetric input. We cite them instead of reproving. Real closedness of  $F$  is used below also for the square roots in the orthonormalization; for  $F = \mathbb{R}((t^\Gamma))$  this holds when  $\Gamma$  is divisible.

**Theorem 17.1** (Finite Gaussian quadrature over a real closed field; source 11). *Let  $F$  be a real closed ordered field and  $L : F[X] \rightarrow F$  an  $F$ -linear map with  $L(1) = 1$  such that for every nonzero  $p \in F[X]$*

$$L(p^2) > 0, \quad L(Xp^2) > 0, \quad L((1-X)p^2) > 0. \quad (17.1)$$

*For each ordinary integer  $N \geq 1$  there are distinct  $z_1, \dots, z_N \in F$  with  $0 < z_j < 1$  and positive weights  $\lambda_1, \dots, \lambda_N \in F$  summing to one such that*

$$L(f) = \sum_{j=1}^N \lambda_j f(z_j) \quad (\deg f \leq 2N-1). \quad (17.2)$$

*The nodes are the roots of the monic orthogonal polynomial  $\pi_N$ .*

*Proof.* Equip  $F[X]$  with  $\langle p, q \rangle = L(pq)$ . Positive definiteness and square roots in  $F$  permit finite Gram–Schmidt; let  $e_0 = 1, e_1, \dots, e_{N-1}$  be orthonormal polynomials of the indicated degrees with positive leading coefficients  $\kappa_j$ . Compressing multiplication by  $X$  to their span gives the symmetric matrix  $J_{ij} = L(Xe_i e_j)$ ,  $0 \leq i, j < N$ . It is tridiagonal (move  $X$  to the lower-degree factor), with off-diagonal entries  $\kappa_j/\kappa_{j+1} > 0$ . For a nonzero vector  $c$  with  $p = \sum c_j e_j$ ,  $c^\top J c = L(Xp^2) > 0$  and  $c^\top (I - J)c = L((1-X)p^2) > 0$ . By the finite spectral theorem over  $F$  (`thm:spectral` of [14]),  $J$  has an orthonormal eigenbasis over  $F$  with eigenvalues in  $F$ , and these two inequalities put all eigenvalues in  $(0, 1)_F$ . An eigenvector of an irreducible tridiagonal matrix is determined by its first coordinate, so each eigenspace has dimension at most one;  $J$  being diagonalizable, its  $N$  eigenvalues are distinct and every normalized eigenvector has nonzero first coordinate. Choose an orthonormal eigenbasis  $u_1, \dots, u_N$  with eigenvalues  $z_j$  and put  $\lambda_j = (u_j)_0^2 > 0$ ; orthonormality gives  $\sum_j \lambda_j = 1$ . Expanding the characteristic determinants of the leading principal submatrices gives the monic recurrence (16.6) with the same initial data 1 and  $X - \alpha_0$ , so  $\det(XI - J) = \pi_N(X)$  and the eigenvalues are the roots of  $\pi_N$ . For  $0 \leq k < N$ ,  $J^k e_0$  is the coefficient vector of  $X^k$ , so  $L(r) = e_0^\top r(J) e_0 = \sum_j \lambda_j r(z_j)$  for  $\deg r < N$ . For  $\deg f \leq 2N-1$  write  $f = \pi_N q + r$  with  $\deg q, \deg r < N$ ; orthogonality gives  $L(\pi_N q) = 0$  and  $\pi_N(z_j) = 0$  gives  $f(z_j) = r(z_j)$ .  $\square$

For  $F = \mathbb{R}((t^\mathbb{Q}))$  and the example, all hypotheses hold by Theorem 14.3, so every finite quadrature problem has a solution with positive weights. This does not produce a strong measure on the ordinary sample space: the nodes lie in  $F$ , not necessarily in  $\mathbb{R}$ .

**Lemma 17.2** (Orthogonal polynomial from atomic subsets; source 11). *For a signed strong atomic measure with  $\Delta_{N-1} \neq 0$ , the monic polynomial orthogonal to all polynomials of degree less than  $N$  is*

$$\pi_N(X) = \frac{1}{\Delta_{N-1}} \sum_{\substack{F \subseteq \Omega \\ |F|=N}}^H \left( \prod_{x \in F} w_x \right) V(F)^2 \prod_{x \in F} (X - x), \quad (17.3)$$

*with all coefficient sums strong.*

*Proof.* Strong summability follows from fixed finite products of the weight family, since every coefficient of  $\prod_{x \in F} (X - x)$  is an ordinary real number. The numerator has leading coefficient  $\Delta_{N-1}$  by Lemma 16.1, so the polynomial is monic. Multiply the numerator by  $X^k$ ,  $0 \leq k < N$ ,

and apply  $L$ : this adds a sample point  $y$ ; terms with  $y \in F$  vanish, and grouping the rest by the  $(N + 1)$ -point union  $G = F \cup \{y\}$  gives

$$\left(\prod_{x \in G} w_x\right) V(G)^2 \sum_{y \in G} \frac{y^k}{\prod_{x \in G \setminus \{y\}} (y - x)},$$

whose last sum is the coefficient of  $X^N$  in the Lagrange interpolation of  $X^k$  at the  $N + 1$  nodes of  $G$ , hence zero since  $k < N$ . All groupings are strong by Lemma 2.3. The linear system for the lower coefficients of a monic orthogonal polynomial has matrix  $H_{N-1}$ , with nonzero determinant.  $\square$

**Theorem 17.3** (Standard parts and nonreal quadrature nodes; source 11). *For the example,*

$$\text{st}(\pi_N(X)) = (X - a) \prod_{n=1}^{N-1} (X - x_n), \quad (17.4)$$

*every coefficient of  $\pi_N$  being finite and the standard part being taken coefficientwise. The  $N$  Gaussian nodes can be labelled so that their standard parts are  $a, x_1, \dots, x_{N-1}$ . Every node lies in  $(0, a)_F$ ; in particular the node with standard part  $a$  is not an ordinary real number, for every  $N$ .*

*Proof.* In (17.3) the unique lowest-valuation subset is  $Y_{N-1}$ , as in Theorem 16.2. The denominator has valuation  $s_{N-1}$  and leading coefficient  $C_{N-1}$ ; every coefficient of the numerator has valuation at least  $s_{N-1}$ , with coefficient  $C_{N-1} \prod_{y \in Y_{N-1}} (X - y)$  there. Division gives finite coefficients and (17.4). The nodes lie between 0 and 1, and taking standard parts in  $\pi_N(X) = \prod_j (X - z_j)$  shows that the multiset of their standard parts is the root multiset of (17.4):  $N$  distinct ordinary points. For the upper bound, view the signed measure as supported on  $[0, a]$ ; its earlier positive atoms are still an infinite subset of that interval, so Theorem 13.4 gives  $L((a - X)p^2) > 0$  for  $p \neq 0$ , that is,  $aI - J$  is positive definite in the proof of Theorem 17.1, and every  $z_j < a$ . An ordinary real number equals its standard part, so the node with standard part  $a$  would be  $a$  itself, which is excluded.  $\square$

At  $N = 1$  the quadrature has weight 1 and node  $m_1 = a + \sum_n^H (x_n - a)t^{q_n} + at$ , infinitesimally below  $a$ . For larger  $N$  the residue nodes successively include more early atoms. This is a finite phenomenon, not an assertion that the finite quadratures converge to a positive strong measure on ordinary points.

## 18 Surcomplex Hermitian forms and actual surreal values

### 18.1 Positive forms without a positive strong real-point measure

Extend  $L$  to  $K_{\mathbb{C}}[X]$  and set  $\langle P, Q \rangle = L(P^*Q)$ , conjugate-linear in the first argument and linear in the second; it is Hermitian because  $L$  commutes with conjugation, and by Theorem 14.3,  $\langle P, P \rangle > 0$  for  $P \neq 0$ . It is a positive-definite algebraic Hermitian form with diagonal values in  $K_{\mathbb{R}}$ . The finite Cauchy–Schwarz calculation holds,

$$|\langle P, Q \rangle|^2 \leq \langle P, P \rangle \langle Q, Q \rangle, \quad (18.1)$$

by nonnegativity of the squared norm of  $Q - P\langle P, Q \rangle / \langle P, P \rangle$ , with equality exactly for linearly dependent  $P, Q$ ; no completion or Hilbert-space representation is needed. Multiplication by  $X$  is symmetric and satisfies  $\langle XP, P \rangle > 0$  and  $\langle (a - X)P, P \rangle > 0$  for  $P \neq 0$ ; all its finite cyclic compressions have positive spectral weights and eigenvalues in the real closed field, as in

Section 17. Yet its cyclic moments  $\langle 1, X^n \rangle$  have no positive strong scalar representing measure on the ordinary real interval. So no spectral-measure proposal that would necessarily produce such a scalar strong measure can represent these moments on that sample domain. This does not rule out different operator topologies, different notions of additivity, or spectra containing nonreal Hahn points. It is the strong-class, interval counterpart of the coefficientwise unitary example on the circle in the Herglotz report [16].

## 18.2 Embedding the construction into the surreals

With  $\Gamma = \mathbb{Q} \subseteq \mathbf{No}$  and Convention 2.1, the moments become

$$M_0 = 1, \quad M_k = a^k + \sum_{n \geq 1} (x_n^k - a^k) \omega^{-n/(n+1)} + a^k \omega^{-1} \quad (k \geq 1), \quad (18.2)$$

all sums being normal-form strong sums. The forced negative event is  $\mu_*(\{0\}) = -\omega^{-1}$ , the earlier weights are  $\omega^{-n/(n+1)}$ , and the anchor is  $1 - \sum_n \omega^{-n/(n+1)} + \omega^{-1}$ . The polynomial, determinant, quadrature and Hermitian conclusions transport along (2.1). The Cantor-space pattern is likewise a literal surreal example with  $t = \omega^{-1}$ .

**Corollary 18.1** (No positive surreal strong measure for the moments; source 11). *There is no positive  $\mathbf{No}$ -valued strongly countably additive measure on the ordinary Borel interval  $[0, 1]$  whose polynomial moments are (18.2), where “strongly” uses normal-form Hahn summability. The same construction gives positive-definite Hermitian moment forms over  $\mathbf{No}[i]$ .*

*Proof.* Suppose such a measure existed. Its domain is a set, so its range and the union of the exponent supports of its values are sets in a class theory with replacement. The rational span of the negatives of those exponents together with  $\mathbb{Q}$  is a set-sized divisible ordered subgroup  $\Gamma \subseteq \mathbf{No}$ , and every value lies in the image of (2.1). Pulling back gives a positive strong Hahn measure in  $\mathbb{R}((t^\Gamma))$  on the same ordinary sample space, and the coefficient reconstruction of Theorem 12.2 in this larger group forces the mass  $-t$  at 0, a contradiction. For Hermitian positivity over all of  $\mathbf{No}[i]$ : any one finite-degree polynomial has finitely many coefficients with set-sized supports; enlarge  $\Gamma$  to contain them. The barrier proof is valid for this arbitrary group (the early exponents stay below 1 and the leading ordinary coefficient polynomial still has finite degree), so the transported form is positive on every nonzero surcomplex polynomial.  $\square$

This localization uses neither a set of all surreals nor a proper-class sum.

## Part V

# Coefficientwise Hahn measures

This part is source 15’s. Every measure in it is a coefficientwise Hahn measure (Definition 2.6) and, unless it is proved, is *not* strong. Leading coefficients may be diffuse, standard parts may have infinitely many atoms, and two exponents suffice for hidden negative mass. Here the coefficientwise class answers, with a common well-ordered support *assumed*, part of source 10’s question about a weaker addition rule with convergent real coefficient sums (Question 26.2).

This requirement is natural when a surreal quantity records a hierarchy of exact asymptotic corrections to an ordinary probabilistic model: the leading coefficient is the ordinary law, later coefficients are successively smaller corrections, and finite signed coefficient measures permit integration, conditioning on events of nonzero probability and coefficientwise limits by ordinary measure theory. They do not require a strong sum of every countable family of event probabilities.

## 19 The signed extension test and null-ideal positivity

### 19.1 Uniform total variation is the exact signed extension test

Let  $E$  be a nonempty finite alphabet,  $\Omega = E^{\mathbb{N}}$  with its compact metric product topology, and  $\pi_n : \Omega \rightarrow E^n$  the first- $n$  projection. A cylinder function depends on finitely many coordinates; these continuous functions are uniformly dense in  $C(\Omega)$ . Finite signed or complex Borel measures on the compact metric space  $\Omega$  are automatically regular. Suppose  $\lambda_n(x) \in K$ ,  $x \in E^n$ , are consistent finite-dimensional weights whose supports all lie in one well-ordered  $W$ , and put

$$c_{\gamma,n}(x) = [t^\gamma] \lambda_n(x), \quad V_\gamma(n) = \sum_{x \in E^n} |c_{\gamma,n}(x)|. \quad (19.1)$$

**Theorem 19.1** (Coefficientwise signed extension test; source 15). *There is a coefficientwise Hahn measure on  $\Omega$  whose pushforwards under the  $\pi_n$  have weights  $\lambda_n$  if and only if*

$$\sup_{n \geq 1} V_\gamma(n) < \infty \quad \text{for every } \gamma \in W. \quad (19.2)$$

*The extension is unique, its coefficient support is contained in  $W$ , and*

$$\|\mu_\gamma\|_{\text{TV}} = \sup_n V_\gamma(n). \quad (19.3)$$

*This is an application of the classical bounded-functional representation of finite measures, not a new scalar extension theorem.*

*Proof.* If an extension exists, pushforward does not increase total variation, which gives necessity and  $V_\gamma(n) \leq \|\mu_\gamma\|_{\text{TV}}$ . For sufficiency fix  $\gamma$  and define, for a cylinder function  $f = g \circ \pi_n$ ,  $L_\gamma(f) = \sum_{x \in E^n} g(x) c_{\gamma,n}(x)$ . Consistency makes this independent of the level. Its norm on cylinder functions is exactly  $\sup_n V_\gamma(n)$ : the upper bound is immediate, and equality at level  $n$  is attained by choosing the sign, or complex phase, of each coefficient. It extends uniquely by uniform density to a bounded functional on  $C(\Omega)$ , and the Riesz–Markov theorem gives a unique finite regular signed or complex measure  $\mu_\gamma$  with that functional and norm. Do this for each  $\gamma \in W$ , set the other coefficients to zero, and use (2.2). A coefficient outside  $W$  of any competing extension vanishes on all cylinders, hence is zero by uniqueness, and the same uniqueness at each exponent proves uniqueness of the coefficientwise Hahn measure.  $\square$

### 19.2 The transfinite null-ideal positivity criterion

Theorem 19.1 is a signed statement: positivity of the leading coefficient on cylinders does not pass to arbitrary Borel sets. The following criterion decides positivity. It is stated on an arbitrary measurable space; its proof uses no topology, no countability of  $W$  and no divisibility of  $\Gamma$ . For a finite signed measure  $\nu$  write  $\nu^+, \nu^-$  for its Jordan parts and  $|\nu| = \nu^+ + \nu^-$ .

**Theorem 19.2** (Null-ideal positivity criterion; source 15, and independently source 18). *Let  $(X, \Sigma)$  be a measurable space,  $W \subseteq \Gamma$  a well-ordered set of any cardinality, and  $(\mu_\gamma)_{\gamma \in W}$  finite real signed measures on  $\Sigma$ . Put  $\mu(A) = \sum_{\gamma \in W} \mu_\gamma(A) t^\gamma \in K_{\mathbb{R}}$  and, for  $\gamma \in W$ ,*

$$\mathcal{J}_{<\gamma} = \{A \in \Sigma : |\mu_\delta|(A) = 0 \text{ for every } \delta \in W \text{ with } \delta < \gamma\}, \quad (19.4)$$

*a  $\sigma$ -ideal of  $\Sigma$  ( $\mathcal{J}_{<\min W} = \Sigma$ ). The following are equivalent.*

- (a)  $\mu(A) \geq 0$  for every  $A \in \Sigma$ .

(b) For every  $\gamma \in W$  and  $A \in \Sigma$ :

$$[|\mu_\delta|(A) = 0 \text{ for every } \delta < \gamma] \implies \mu_\gamma(A) \geq 0. \quad (19.5)$$

(c)  $\mu_\gamma^-(A) = 0$  for every  $\gamma \in W$  and every  $A \in \mathcal{J}_{<\gamma}$ .

If the predecessor set  $\{\delta \in W : \delta < \gamma\}$  is countable, enumerate it as  $(\beta_j)$  (finitely or countably) and put

$$\lambda_{<\gamma} = \sum_j 2^{-j} \frac{|\mu_{\beta_j}|}{1 + \|\mu_{\beta_j}\|_{\text{TV}}}, \quad (19.6)$$

with  $\lambda_{<\gamma} = 0$  for an empty predecessor set. Then the condition at  $\gamma$  in (c) is equivalent to  $\mu_\gamma^- \ll \lambda_{<\gamma}$ . No bound on a Radon–Nikodym derivative is required.

*Proof.* (a) $\implies$ (b): under the hypothesis of (19.5) every coefficient of  $\mu(A)$  before  $\gamma$  vanishes, so a negative  $\mu_\gamma(A)$  would be a negative leading coefficient. (b) $\implies$ (c):  $\mathcal{J}_{<\gamma}$  is closed under measurable subsets, so (b) gives  $\mu_\gamma(B) \geq 0$  for every measurable  $B \subseteq A \in \mathcal{J}_{<\gamma}$ , whence  $\mu_\gamma^-(A) = 0$ . (c) $\implies$ (b):  $\mu_\gamma(A) = \mu_\gamma^+(A) - \mu_\gamma^-(A) = \mu_\gamma^+(A) \geq 0$  for  $A \in \mathcal{J}_{<\gamma}$ .

(b) $\implies$ (a): take  $A$  with  $\mu(A) \neq 0$  and let  $\gamma$  be the least exponent with  $\mu_\gamma(A) \neq 0$ . We prove by transfinite induction on  $\delta \in W$ ,  $\delta < \gamma$ , that  $|\mu_\delta|(A) = 0$ . If this holds at all earlier indices, every measurable  $B \subseteq A$  is null for those earlier total variations, and (19.5) gives  $\mu_\delta(B) \geq 0$ . So the restriction of  $\mu_\delta$  to  $A$  is a positive measure of total mass  $\mu_\delta(A) = 0$ ; it is zero, and so is  $|\mu_\delta|(A)$ . Limit stages need nothing, since no measure is summed over the earlier indices. Now (19.5) at  $\gamma$  gives  $\mu_\gamma(A) \geq 0$ , and it is nonzero, hence positive; this is positivity of  $\mu(A)$  in the Hahn order.

Finally, (19.6) is a finite positive measure (total mass at most 1) whose null sets are exactly the common null sets of the  $|\mu_{\beta_j}|$ , since every weight is strictly positive; that is, its null sets form  $\mathcal{J}_{<\gamma}$ , and (c) at  $\gamma$  is  $\mu_\gamma^- \ll \lambda_{<\gamma}$ .  $\square$

*Remark 19.3* (Provenance and scope of the criterion). Source 15 stated the equivalence (a) $\Leftrightarrow$ (b) for finite signed Borel measures on a compact metric space, with the proof above. Source 18 of the same batch, now the report `surcomplex/hahn-herglotz-positivity` [16], proved (a) $\Leftrightarrow$ (c) and the control-measure form independently (`herg:thm:nullideal`), for real finite signed regular Borel measures on a compact Hausdorff space, using a Hahn decomposition for the necessity half; its standing hypothesis that  $\Gamma$  be divisible is not used in that proof. Both proofs work verbatim on any measurable space and any  $\Gamma$ , and the statement above covers both settings. The countable control measure is a corollary, not a substitute: source 18 gives (`herg:ex:uncountable`) a positive measure with uncountable well-ordered support, earlier coefficients Dirac masses and a final diffuse negative coefficient, for which  $\mu_\gamma^- \ll \lambda$  fails for every control measure  $\lambda$  built from countably many earlier coefficients; so the full null ideals are needed when predecessor sets are uncountable. For two exponents,  $W = \{0, \eta\}$  with  $\eta > 0$ , the criterion reads:  $\mu_0 + t^\eta \mu_\eta \geq 0$  if and only if  $\mu_0 \geq 0$  and  $\mu_\eta^- \ll \mu_0$ . The criterion permits negative coefficient measures (for an ordinary probability  $P$ ,  $P + t^\eta g P$  is positive for every real  $g \in L^1(P)$ , however negative or unbounded); it excludes a negative component concentrated on a set invisible to all earlier coefficients. The Herglotz report finds such a hidden negative atom already at two exponents on the circle (`herg:thm:negativeatom`). No matrix-valued version is claimed.

**Corollary 19.4** (Dominated corrections; source 15). *Let  $\mu$  be a real coefficientwise Hahn measure with  $0 \in W \subseteq \Gamma_{\geq 0}$ ,  $\mu_0 = P_0$  an ordinary probability, and  $\mu_\gamma \ll P_0$  for every  $\gamma > 0$ . Then  $\mu(A) \geq 0$  for every event  $A$ , and  $\mu(A) = 0$  if and only if  $P_0(A) = 0$ . If in addition  $\mu_\gamma(X) = 0$  for  $\gamma > 0$ , then  $\mu$  is a coefficientwise Hahn probability.*

*Proof.* If  $P_0(A) > 0$  it is the positive leading coefficient. If  $P_0(A) = 0$ , absolute continuity makes every coefficient vanish. Normalization is coefficientwise.  $\square$

**Example 19.5** (Positive cylinders, negative singleton, two exponents; source 15). Let  $P_0$  be the ordinary fair Bernoulli measure on  $\{0, 1\}^{\mathbb{N}}$ , choose distinct sequences  $x, y$ , let  $\alpha > 0$ , and put

$$\mu = P_0 + t^\alpha(\delta_y - \delta_x). \quad (19.7)$$

Every nonempty event depending on finitely many coordinates has strictly positive  $P_0$ -probability, so every finite-dimensional marginal of  $\mu$  is a positive normalized Hahn law; the marginals are consistent and their  $\alpha$ -coefficient variations are bounded by 2. Nonetheless  $\mu(\{x\}) = -t^\alpha < 0$ , since  $P_0$  is atomless. By Theorem 19.1 no other coefficientwise extension of these marginals exists, positive or not. In the language of Theorem 19.2:  $\{x\} \in \mathcal{J}_{<\alpha}$ , but  $\mu_\alpha^-(\{x\}) = 1$ .

The support  $\{0, \alpha\}$  has order type 2; the sample space and the tests are those of Theorem 15.2(a). This is Trap 1 of Section 2.6: the measure is not strong, because its leading coefficient is atomless. The Herglotz report [16] gives the corresponding two-exponent example with Toeplitz tests on the circle.

*Remark 19.6* (One exponent never suffices). With a single exponent the phenomenon cannot occur in either class: if  $\mu = \mu_0$  is an ordinary finite signed measure on the Borel sets of Cantor space and  $\mu_0([s]) \geq 0$  for every cylinder, then  $\mu_0 \geq 0$ , because every Borel set  $B$  is approximated by finite unions  $A$  of cylinders with  $|\mu_0|(A \triangle B)$  arbitrarily small. So the least support order type admitting nonnegative cylinders and a negative singleton is 2 in the coefficientwise class, against  $\omega + 1$  in the strong class. (This elementary remark is added in the merge.)

## 20 Interior Bernoulli laws: the exact square-summability theorem

Let  $S \subseteq \Gamma_{>0}$  be well ordered, and write

$$p_j = p_j^0 + \epsilon_j, \quad \epsilon_j = \sum_{\gamma \in S} a_{j,\gamma} t^\gamma, \quad (20.1)$$

with the *same*  $S$  for all coordinates  $j$ . The *row* at  $\gamma$  is the sequence  $(a_{j,\gamma})_{j \geq 1}$ . Let  $0 < \eta \leq \frac{1}{2}$ ,  $p_j^0 \in [\eta, 1 - \eta]$  and  $q_j^0 = 1 - p_j^0$ ; every  $p_j$  lies strictly between 0 and 1 in  $K_{\mathbb{R}}$ . Let  $P_n$  be the finite independent law (1.1) with these biases, and  $P_0$  the ordinary infinite product with coordinate biases  $p_j^0$ .

**Theorem 20.1** (Interior extension criterion; source 15). *The following are equivalent:*

- (i) *the finite laws  $P_n$  admit a coefficientwise signed Hahn extension;*
- (ii) *for every  $\gamma \in S$  the row*

$$b_\gamma = (b_{j,\gamma})_{j \geq 1}, \quad b_{j,\gamma} = \frac{a_{j,\gamma}}{\sqrt{p_j^0 q_j^0}}, \quad (20.2)$$

*belongs to  $\ell^2$ .*

*When they hold the extension is unique and is a coefficientwise Hahn probability. Its coefficients are supported in  $S^*$ , its constant coefficient is  $P_0$ , and*

$$\mu_\rho(A) = \int_A H_\rho dP_0, \quad H_\rho \in L^2(P_0). \quad (20.3)$$

*Its null Borel events are exactly the  $P_0$ -null events. No bound on  $\|b_\gamma\|_2$  uniform in  $\gamma$  is required.*

## 20.1 Orthogonal multilinear coefficients

Under  $P_0$  put  $Z_j = (X_j - p_j^0)/\sqrt{p_j^0 q_j^0}$ : independent, centered, of variance one. For finite  $I \subseteq \mathbb{N}$  put  $Z_I = \prod_{j \in I} Z_j$ ,  $Z_\emptyset = 1$ ; then

$$\mathbb{E}_0[Z_I Z_J] = \mathbf{1}_{I=J}. \quad (20.4)$$

For  $k \geq 1$  and  $u_1, \dots, u_k \in \ell^2$  set

$$Q_n(u_1, \dots, u_k) = \frac{1}{k!} \sum_{\substack{j_1, \dots, j_k \leq n \\ \text{pairwise distinct}}} \left( \prod_{r=1}^k u_r(j_r) \right) Z_{j_1} \cdots Z_{j_k}. \quad (20.5)$$

**Lemma 20.2** (Multilinear chaos bound; source 15).  $Q_n(u_1, \dots, u_k)$  converges in  $L^2(P_0)$  to a multilinear  $Q(u_1, \dots, u_k)$ , and

$$\|Q(u_1, \dots, u_k)\|_2 \leq \frac{1}{\sqrt{k!}} \prod_{r=1}^k \|u_r\|_2; \quad (20.6)$$

the same bound holds for  $Q_n$ , and conditional expectation onto the first  $n$  coordinates sends  $Q$  to  $Q_n$ .

*Proof.* Group (20.5) by the unordered set  $I = \{j_1, \dots, j_k\}$ : the coefficient of  $Z_I$  is the average of the  $k!$  corresponding products. The square of an average is at most the average of the squares, so by (20.4)

$$\|Q_n(u_1, \dots, u_k)\|_2^2 \leq \frac{1}{k!} \sum_{\substack{j_1, \dots, j_k \leq n \\ \text{pairwise distinct}}} \prod_{r=1}^k |u_r(j_r)|^2 \leq \frac{1}{k!} \prod_{r=1}^k \|u_r\|_2^2.$$

For finitely supported  $u_r$ ,  $Q_n$  eventually stabilizes. For general  $u_r$ , approximate by truncations and use multilinearity: a difference of two products is a sum of  $k$  multilinear terms each containing one tail, which the bound makes small in  $L^2$  uniformly in  $n$ . A term  $Z_I$  with an index beyond  $n$  has zero conditional expectation onto the first  $n$  coordinates, which proves the last claim for finitely supported inputs, and  $L^2$  continuity gives the general case.  $\square$

The finite likelihood ratio relative to  $P_0$  is

$$L_n = \prod_{j=1}^n \left( 1 + Z_j \sum_{\gamma \in S} b_{j,\gamma} t^\gamma \right), \quad (20.7)$$

since the one-coordinate ratio is  $1 + \epsilon_j/p_j^0$  at  $X_j = 1$  and  $1 - \epsilon_j/q_j^0$  at  $X_j = 0$ . At a positive exponent  $\rho$ ,

$$H_{\rho,n} := [t^\rho] L_n = \sum_{k \geq 1} \sum_{\substack{\gamma_1, \dots, \gamma_k \in S \\ \gamma_1 + \dots + \gamma_k = \rho}} Q_n(b_{\gamma_1}, \dots, b_{\gamma_k}), \quad (20.8)$$

a finite sum by Lemma 9.1. The factor  $1/k!$  compensates for ordering the distinct coordinate indices, so (20.8) handles repeated exponents and collisions such as  $\alpha + \alpha = 2\alpha$  correctly.

## 20.2 Sufficiency and positivity on all events

Assume every  $b_\gamma \in \ell^2$ . Put  $H_0 = 1$  and, for  $\rho > 0$  in  $S^*$ ,

$$H_\rho = \sum_{k \geq 1} \sum_{\substack{\gamma_1, \dots, \gamma_k \in S \\ \gamma_1 + \dots + \gamma_k = \rho}} Q(b_{\gamma_1}, \dots, b_{\gamma_k}), \quad \|H_\rho\|_2 \leq \sum_{k \geq 1} \sum_{\substack{\gamma_1 + \dots + \gamma_k = \rho \\ \gamma_r \in S}} \frac{1}{\sqrt{k!}} \prod_{r=1}^k \|b_{\gamma_r}\|_2 < \infty. \quad (20.9)$$

Set  $\mu_\rho = H_\rho P_0$  and assemble the Hahn series of measures. Lemma 20.2 gives  $\mathbb{E}_0[H_\rho \mid X_1, \dots, X_n] = H_{\rho,n}$ , so the marginal coefficients are exactly those of  $P_n$ ; each  $H_\rho$  with  $\rho > 0$  has mean zero, since every nonempty  $Z_I$  is centered, so the measure is normalized. All positive-exponent coefficients are absolutely continuous with respect to  $P_0$ , and Corollary 19.4 gives positivity on *all* Borel events and identifies the null ideal. This step is essential: Example 19.5 shows that cylinder positivity alone would not suffice. Uniqueness is Theorem 19.1. The bound in (20.9) is checked separately at each exponent; an estimate summing these bounds over all  $\rho$  is neither used nor generally available.

### 20.3 Necessity: the least bad exponent cannot be cancelled

**Lemma 20.3** (Uniform  $L^1$  lower bound; source 15). *For real  $c_1, \dots, c_n$  let  $T = \sum_j c_j Z_j$ ,  $V = \sum_j c_j^2$ , and  $C_\eta = \max\{3, \frac{1}{\eta(1-\eta)} - 3\}$ . Then*

$$\sqrt{V/C_\eta} \leq \mathbb{E}_0|T| \leq \sqrt{V}. \quad (20.10)$$

*Proof.* The upper bound is Cauchy–Schwarz. A direct calculation gives the fourth-power expectation  $\mathbb{E}_0 Z_j^4 = \frac{1}{p_j^0 q_j^0} - 3 \leq \frac{1}{\eta(1-\eta)} - 3$ . Independence and centering remove all terms of  $\mathbb{E}_0 T^4$  except fourth powers and paired indices:

$$\mathbb{E}_0 T^4 = \sum_j c_j^4 \mathbb{E}_0 Z_j^4 + 6 \sum_{i < j} c_i^2 c_j^2 \leq C_\eta V^2.$$

Hölder’s inequality gives  $V = \mathbb{E}_0|T|^2 \leq (\mathbb{E}_0|T|)^{2/3}(\mathbb{E}_0|T|^4)^{1/3}$ ; substituting and rearranging proves the lower bound when  $V > 0$ , and  $V = 0$  is immediate.  $\square$

Suppose a coefficientwise signed extension exists but some row is not square summable, and let  $\gamma$  be the least such exponent ( $S$  is well ordered). At  $\gamma$ , (20.8) splits as

$$H_{\gamma,n} = \sum_{j=1}^n b_{j,\gamma} Z_j + R_{\gamma,n}. \quad (20.11)$$

Every exponent in a word of length at least two summing to  $\gamma$  is strictly smaller than  $\gamma$ ; those rows are square summable by minimality, there are finitely many such words, and Lemma 20.2 gives  $\sup_n \|R_{\gamma,n}\|_2 < \infty$ . On the finite  $\sigma$ -algebra of the first  $n$  coordinates  $H_{\gamma,n}$  is the density of the  $\gamma$ -coefficient signed marginal relative to  $P_0$ , so  $V_\gamma(n) = \mathbb{E}_0|H_{\gamma,n}|$ , which Theorem 19.1 bounds uniformly. With the  $L^2$  bound on  $R_{\gamma,n}$  the first-order sums are bounded in  $L^1$  uniformly in  $n$ , and Lemma 20.3 forces  $\sup_n \sum_{j \leq n} b_{j,\gamma}^2 < \infty$ , a contradiction. This proves Theorem 20.1.

At a nonminimal exponent, lower-order perturbations contribute many nonlinear terms; the least-bad-exponent argument shows those already have uniformly bounded norm. It is this support-controlled induction, not a formal derivative at  $t = 0$ , that proves necessity for all Hahn exponents at once.

## 21 Deterministic coordinates: the exact absolute-summability theorem

Now let  $p_j^0 = 0$  and let each  $p_j = \epsilon_j$  in (20.1) be nonnegative; the coefficients  $a_{j,\gamma}$  need not all be nonnegative, only the first nonzero coefficient of each  $\epsilon_j$  must be positive. Let  $o = (0, 0, \dots)$  and let  $D \subseteq \Omega = \{0, 1\}^\mathbb{N}$  be the countable set of configurations with finitely many successes.

**Theorem 21.1** (Boundary extension criterion; source 15). *The finite independent laws with biases  $p_j = \epsilon_j \geq 0$  admit a coefficientwise signed extension if and only if*

$$a_\gamma = (a_{j,\gamma})_{j \geq 1} \in \ell^1 \quad \text{for every } \gamma \in S. \quad (21.1)$$

*The extension is then unique and positive; its constant coefficient is  $\delta_o$ , all its coefficients are supported on  $D$ , and its exponent support is contained in  $S^*$ .*

Give  $\Omega$  its compact abelian group structure under coordinatewise addition modulo 2, let  $e_j$  be the configuration with its single success at  $j$ , and put  $\nu_j = \delta_{e_j} - \delta_o$ , so  $\|\nu_j\|_{\text{TV}} = 2$ . Finite signed measures form a Banach algebra under convolution,  $\|\sigma * \tau\|_{\text{TV}} \leq \|\sigma\|_{\text{TV}} \|\tau\|_{\text{TV}}$  (for finite atomic measures by the triangle inequality on products of weights; in general from the product-measure and pushforward definition). The finite product law, placed on paths vanishing after  $n$ , is

$$\tilde{P}_n = \bigstar_{j=1}^n (\delta_o + \epsilon_j \nu_j), \quad (21.2)$$

each factor being exactly  $(1 - \epsilon_j)\delta_o + \epsilon_j\delta_{e_j}$ . For  $u_1, \dots, u_k \in \ell^1$  define

$$A(u_1, \dots, u_k) = \frac{1}{k!} \sum_{\substack{j_1, \dots, j_k \geq 1 \\ \text{pairwise distinct}}} \left( \prod_{r=1}^k u_r(j_r) \right) \nu_{j_1} * \dots * \nu_{j_k}, \quad (21.3)$$

$$\|A(u_1, \dots, u_k)\|_{\text{TV}} \leq \frac{2^k}{k!} \prod_{r=1}^k \|u_r\|_1, \quad (21.4)$$

an ordinary series absolutely convergent in total variation (bound each convolution norm by  $2^k$  and enlarge the distinct-index sum to the product of  $\ell^1$  sums); truncating all indices to  $j_r \leq n$  converges in total variation. Assuming (21.1), put  $\mu_o = \delta_o$  and

$$\mu_\rho = \sum_{k \geq 1} \sum_{\substack{\gamma_1 + \dots + \gamma_k = \rho \\ \gamma_r \in S}} A(a_{\gamma_1}, \dots, a_{\gamma_k}) \quad (\rho > 0), \quad (21.5)$$

$$\|\mu_\rho\|_{\text{TV}} \leq \sum_{k \geq 1} \sum_{\substack{\gamma_1 + \dots + \gamma_k = \rho \\ \gamma_r \in S}} \frac{2^k}{k!} \prod_{r=1}^k \|a_{\gamma_r}\|_1 < \infty, \quad (21.6)$$

the exponent-word sums being finite. These coefficients are total-variation limits of the coefficients of (21.2); projecting to the first  $n$  coordinates sends  $\nu_j$  to zero for  $j > n$ , so the limits have the prescribed marginals. Every nonempty convolution product has total mass zero, so  $\mu$  is normalized, and each term of (21.3) lives on finite-success configurations, a property preserved under total-variation limits. This proves signed existence with the stated support control.

## 21.1 Positivity via exact path weights

The leading law  $\delta_o$  cannot dominate the later coefficients, which put mass at paths other than  $o$ ; the interior positivity argument does not apply, and an all-orders atomic argument is needed. Identify a path in  $D$  with its finite set  $I \subseteq \mathbb{N}$  of success indices and define

$$w_I = \left( \prod_{i \in I} \epsilon_i \right) \prod_{j \notin I}^{\text{coef}} (1 - \epsilon_j). \quad (21.7)$$

Here  $\prod^{\text{coef}}$  means: expand each finite product, take the ordinary absolutely convergent sum of the scalar contributions at each fixed exponent, and assemble those coefficients into one Hahn

series. This exists: at exponent  $\rho$  there are finitely many exponent words, and for each word the scalar sum is bounded by the product of the corresponding  $\ell^1$  row norms divided by  $k!$ ; the common exponent support lies in  $S^*$ . The infinite product has constant term 1 and no negative exponents, so it is positive, and

$$w_I \geq 0 \quad (I \text{ finite}). \quad (21.8)$$

Total-variation convergence of the coefficient measures, evaluated on singletons, identifies the atomic masses:

$$\mu_\rho(\{I\}) = [t^\rho]w_I, \quad \sum_{I \text{ finite}} |[t^\rho]w_I| \leq \|\mu_\rho\|_{\text{TV}} < \infty, \quad (21.9)$$

and consequently, for every Borel  $B$ ,

$$\mu(B) = \sum_{I \in B \cap D}^{\text{coef}} w_I, \quad (21.10)$$

which is in general *not* a strong Hahn sum. To prove positivity, discard the zero  $w_I$ . If no nonzero weight remains in  $B$ ,  $\mu(B) = 0$ . Otherwise the leading exponents of the remaining weights form a nonempty subset of the well-ordered  $S^*$ , with a least member  $\rho$ ; all coefficients before  $\rho$  vanish, and at  $\rho$  every contributing atomic coefficient is nonnegative and at least one is positive. The absolute convergence in (21.9) makes their sum strictly positive, so  $\mu(B) > 0$ . This proves positivity on every Borel set, including arbitrary unions of finite-success paths and sets disjoint from  $D$ .

## 21.2 Necessity at the least non-absolutely-summable row

Suppose a coefficientwise signed extension exists and (21.1) fails; let  $\gamma$  be the least exponent with  $a_\gamma \notin \ell^1$ . At the first  $n$  coordinates its coefficient splits as

$$T_{\gamma,n} + R_{\gamma,n}, \quad T_{\gamma,n} = \sum_{j=1}^n a_{j,\gamma}(\delta_{e_j} - \delta_o). \quad (21.11)$$

All nonlinear exponent words at  $\gamma$  use smaller exponents, so by their absolute summability, Lemma 9.1 and (21.4) the remainder has uniformly bounded total variation. The  $n+1$  atoms  $o, e_1, \dots, e_n$  are distinct, so

$$\|T_{\gamma,n}\|_{\text{TV}} = \sum_{j=1}^n |a_{j,\gamma}| + \left| \sum_{j=1}^n a_{j,\gamma} \right| \geq \sum_{j=1}^n |a_{j,\gamma}|. \quad (21.12)$$

The signed extension test bounds  $\|T_{\gamma,n} + R_{\gamma,n}\|_{\text{TV}}$  uniformly in  $n$ , and with the remainder bound this bounds the left side of (21.12), contradicting the choice of  $\gamma$ . This completes Theorem 21.1.

At an interior law the first-order coefficient is a centered sum of independent scores, whose  $L^1$  norm is comparable to the  $\ell^2$  norm of the amplitudes; at a deterministic law it assigns masses to disjoint single-success paths, whose variation detects the  $\ell^1$  norm directly. Making the scale  $t^\gamma$  smaller cannot repair either divergence: the divergence is an ordinary real number at one coefficient.

*Remark 21.2* (The strong threshold at a deterministic baseline; sources 15 and 10). Under the hypotheses of Theorem 21.1, the finite Bernoulli laws admit a strong extension if and only if every row is finitely supported:

$$\{j : a_{j,\gamma} \neq 0\} \text{ is finite for every } \gamma \in S. \quad (21.13)$$

This is source 15's corollary "Finite rows are necessary and sufficient". With the common support  $S$  it is the special case of Corollary 10.3: here  $q_j = p_j = \epsilon_j$ , the union of the supports

lies in  $S$ , and  $(\epsilon_j)$  is strongly summable exactly when every row is finite. A signed strong extension is coefficientwise (Corollary 4.5) and hence, by the uniqueness in Theorem 21.1, is the positive one. Source 15's direct proof: finite rows are in  $\ell^1$ , and at a fixed output exponent finitely many exponent words use finitely many coordinates, so each  $\mu_\rho$  in (21.5) is finitely supported and Corollary 4.5 applies; conversely, at the least exponent  $\gamma$  with an infinitely supported row,  $\mu_\gamma = \sum_j a_{j,\gamma}(\delta_{e_j} - \delta_o) + R_\gamma$  with  $R_\gamma$  finitely supported (its rows are smaller and finite), so  $\mu_\gamma$  has infinitely many atoms, contradicting Theorem 4.2. For instance  $p_j = t^{j\alpha}$  gives an infinite strong probability with infinitesimal coordinate success probabilities at any rank, while  $p_j = c_j t^\alpha$  with infinitely many  $c_j > 0$  is never strong, even when  $\sum_j c_j < \infty$  and a coefficientwise extension exists. So three requirements occur in one problem: finite support per exponent row (strong addition),  $\ell^1$  per row (perturbing a point mass), and  $\ell^2$  per row (perturbing a nondegenerate product law).

## 22 A mixed interior–boundary classification

A tensor product of signed corrections is not positive coefficient by coefficient, so combining the two theorems needs care.

**Theorem 22.1** (Mixed extension criterion; source 15). *Let  $p_j = p_j^0 + \sum_{\gamma \in S} a_{j,\gamma} t^\gamma$  with  $p_j^0 \in \{0, 1\} \cup [\eta, 1 - \eta]$ ,  $0 < \eta \leq \frac{1}{2}$ , a common well-ordered  $S \subseteq \Gamma_{>0}$ , and  $0 \leq p_j \leq 1$  in  $K_{\mathbb{R}}$ . Put  $J_{\text{int}} = \{j : 0 < p_j^0 < 1\}$  and  $J_{\text{bd}} = \{j : p_j^0 \in \{0, 1\}\}$ . The finite independent laws admit a coefficientwise signed extension if and only if, for each  $\gamma \in S$ ,*

$$\sum_{j \in J_{\text{int}}} \frac{a_{j,\gamma}^2}{p_j^0(1 - p_j^0)} < \infty \quad \text{and} \quad \sum_{j \in J_{\text{bd}}} |a_{j,\gamma}| < \infty. \quad (22.1)$$

*The extension is then unique and positive, with exponent support in  $S^*$ .*

*Proof.* Flip every coordinate with  $p_j^0 = 1$ , replacing  $X_j$  by  $1 - X_j$ : its baseline becomes 0 and its success bias  $1 - p_j \geq 0$ ; the row changes sign and keeps its  $\ell^1$  norm. So assume all boundary baselines are 0. Necessity follows by marginalizing to the two coordinate subsequences: a pushforward of a finite coefficient measure is finite and the common support is preserved, so Theorems 20.1 and 21.1 apply (finite or empty subsequences satisfy the conditions automatically).

For sufficiency let  $\tau = \sum_\alpha \tau_\alpha t^\alpha$  be the interior extension and  $\beta = \sum_\delta \beta_\delta t^\delta$  the boundary extension, both with exponents in  $S^*$ , and define on the product of their sample spaces

$$\mu_\rho = \sum_{\substack{\alpha, \delta \in S^* \\ \alpha + \delta = \rho}} \tau_\alpha \otimes \beta_\delta, \quad \|\mu_\rho\|_{\text{TV}} \leq \sum_{\alpha + \delta = \rho} \|\tau_\alpha\|_{\text{TV}} \|\beta_\delta\|_{\text{TV}} < \infty, \quad (22.2)$$

a finite sum by Lemma 9.1. Its cylinder values are the products of the prescribed marginal weights; normalization and uniqueness follow. For positivity:  $\beta$  has nonnegative weights  $w_I$  on the countable set of finite-success boundary paths. For a Borel event  $A$  of the product let  $A_I$  be its interior section at  $I$ ; then

$$\mu(A) = \sum_I^{\text{coef}} w_I \tau(A_I), \quad (22.3)$$

absolutely convergent at every exponent (use the finite sum over  $\alpha + \delta = \rho$ ,  $|\tau_\alpha(A_I)| \leq \|\tau_\alpha\|_{\text{TV}}$  and  $\sum_I |w_I| \leq \|\beta\|_{\text{TV}}$ ). Each summand  $w_I \tau(A_I)$  is nonnegative with exponents in  $S^*$ , and the least-leading-exponent argument used for (21.10) gives  $\mu(A) \geq 0$ . Flipping back preserves positivity.  $\square$

## 23 Consequences, sharp examples, and the boundary of the theorems

### 23.1 Rigidity of a constant surreal bias

**Corollary 23.1** (No coefficientwise nonreal i.i.d. bias; source 15). *Let  $p \in K_{\mathbb{R}}$  with  $0 \leq p \leq 1$ . The consistent Bernoulli laws  $P_n(x) = p^{x_1+\dots+x_n}(1-p)^{n-(x_1+\dots+x_n)}$  have a coefficientwise signed extension to  $\{0,1\}^{\mathbb{N}}$  if and only if  $p$  is an ordinary real number. The extension is then the usual real Bernoulli probability, embedded as the exponent-zero coefficient.*

*Proof.* Write  $p = q + \epsilon$  with  $q \in [0,1] \cap \mathbb{R}$  its constant coefficient and  $\epsilon$  of positive support. If  $\epsilon = 0$  the classical product measure is the extension. Otherwise let  $\gamma = v(\epsilon)$  and  $a \neq 0$  its coefficient. If  $0 < q < 1$ , the row at  $\gamma$  is the constant nonzero row  $a/\sqrt{q(1-q)}$ , not in  $\ell^2$ , and Theorem 20.1 (with  $\eta = \min(q, 1-q)$ ) excludes an extension. If  $q = 0$ , positivity of  $p$  permits Theorem 21.1, and the constant row  $a$  is not in  $\ell^1$ . If  $q = 1$ , exchange success and failure.  $\square$

For  $p = t^\alpha$  the obstruction is visible at the first infinitesimal coefficient: the first- $n$  law has  $\alpha$ -coefficient  $-n$  at  $o$  and  $+1$  at each of  $e_1, \dots, e_n$ , of total variation exactly  $2n$ , and no finite signed measure has all these marginals. For  $p = q + at^\alpha$  with  $0 < q < 1$  the first coefficient variation is

$$V_\alpha(n) = \frac{|a|}{q(1-q)} \mathbb{E}|\text{Bin}(n, q) - nq|, \quad (23.1)$$

which Lemma 20.3 bounds above and below by positive constant multiples of  $\sqrt{n}$ ; no central limit theorem is needed.

The corollary holds for surreal biases through Convention 2.1. More generally, a purported coefficientwise normal-form measure with values in **No** and a common support can be placed in a set-sized Hahn workspace by taking the additive subgroup generated by that support, so enlarging the exponent workspace cannot evade this obstruction. The i.i.d. law with real bias  $p \in (0,1)$  exists coefficientwise but not strongly (Section 2.6, Trap 2).

### 23.2 Examples distinguishing the thresholds

Fix  $\alpha > 0$ . The following conclusions are exact.

| Coordinate biases                                 | Infinite extension conclusion                                                                                                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $p_j = \frac{1}{2} + t^\alpha/j$                  | A coefficientwise Hahn probability exists, since $\sum j^{-2} < \infty$ . It is not strong: its constant coefficient is an atomless Bernoulli product, contrary to Theorem 4.2.                             |
| $p_j = \frac{1}{2} + t^\alpha/\sqrt{j}$           | No coefficientwise signed extension, since $\sum j^{-1} = \infty$ .                                                                                                                                         |
| $p_j = t^\alpha/j^2$                              | A coefficientwise Hahn probability exists; it is not strong: at the single-success path $e_j$ the coefficient of $t^\alpha$ is $j^{-2}$ , so countably many disjoint singletons contribute at one exponent. |
| $p_j = t^\alpha/j$                                | No coefficientwise signed extension.                                                                                                                                                                        |
| $p_j = t^{j\alpha}$                               | A strong Hahn probability exists (Remark 21.2).                                                                                                                                                             |
| $p_j = p \in K_{\mathbb{R}} \setminus \mathbb{R}$ | No coefficientwise signed i.i.d. extension.                                                                                                                                                                 |

**Genuine higher-rank examples.** Take  $\Gamma = \mathbb{Z}\alpha \oplus \mathbb{Z}\beta$  ordered so that  $0 < n\alpha < \beta$  for every integer  $n \geq 1$  (a surreal realization is  $\alpha = 1$ ,  $\beta = \omega$ ), and  $p_j = \frac{1}{2} + t^\alpha/j + t^\beta/j^2$ . Both rows satisfy the interior criterion, so an extension exists. At exponent  $\beta$  only the length-one word  $(\beta)$  contributes, since no finite multiple of  $\alpha$  equals  $\beta$ ; at  $\beta + 2\alpha$  the three orderings of  $(\beta, \alpha, \alpha)$

contribute, with the factorial normalization of (20.5). There is no limiting passage from  $n\alpha$  to  $\beta$ . More generally one can prescribe arbitrarily large finite values of  $\|b_\gamma\|_2$  at successive exponents, or use an uncountable well-ordered  $S$  in a suitable group; Theorem 20.1 applies row by row, and no common real-analytic perturbation parameter is required.

### 23.3 Why uniform separation from 0 and 1 matters

The weighted  $\ell^2$  condition is sufficient at an interior baseline even without uniform separation (the construction uses only independence and variance one), but its necessity used the uniform bound on  $\mathbb{E}_0 Z_j^4$ .

**Proposition 23.2** (A universal  $\ell^1$  sufficient condition; source 15). *Let  $0 < p_j^0 < 1$  be arbitrary, with no uniform separation from the endpoints, and suppose every row  $a_\gamma$  in (20.1) is in  $\ell^1$ . Then the finite Bernoulli laws admit a unique coefficientwise Hahn probability extension whose coefficients are absolutely continuous with respect to  $P_0$ .*

*Proof.* For each exponent define the centered score  $h_{j,\gamma} = a_{j,\gamma}(X_j - p_j^0)/(p_j^0(1 - p_j^0))$ ; a two-point calculation gives  $\mathbb{E}_0|h_{j,\gamma}| = 2|a_{j,\gamma}|$ . For distinct coordinates, independence makes the  $L^1$  norm of a product of scores the product of their  $L^1$  norms, so for each exponent word the distinct-coordinate expansion converges absolutely in  $L^1$ , with bound  $\frac{2^k}{k!} \prod_r \|a_{\gamma_r}\|_1$ . Finitely many words occur at each output exponent; their sums are  $L^1(P_0)$  densities with the prescribed cylinder coefficients and mean zero at positive exponents. Absolute continuity gives positivity as in Corollary 19.4, and scalar uniqueness gives uniqueness.  $\square$

**Example 23.3** (Failure of unrestricted weighted- $\ell^2$  necessity; source 15). Let  $p_j^0 = 2^{-4j}$  and  $\epsilon_j = 2^{-j}t^\alpha$ . The row is in  $\ell^1$ , so Proposition 23.2 gives an extension, but  $\sum_j 2^{-2j}/(2^{-4j}(1 - 2^{-4j})) = \sum_j 2^{2j}/(1 - 2^{-4j}) = +\infty$ . The equivalence of Theorem 20.1 is false without its uniform interior hypothesis.

### 23.4 No ordinary continuity in the field order

For disjoint events each coefficient obeys ordinary countable additivity. It does not follow that the partial sums of their Hahn-valued probabilities converge in an order or valuation topology, nor that an ordinary decreasing limit of real leading coefficients eventually falls below a positive infinitesimal tolerance; these are different statements with different quantifiers. The proofs of this part use only finite total variation at each coefficient and support-controlled assembly, and they are not identified with convergence in the surreal fine topology or with an ultrafilter sum.

## 24 Complex coefficients, conditioning, and observables

**Proposition 24.1** (Surcomplex analogue; source 15). *Let  $a_{j,\gamma} \in \mathbb{C}$ , take the values of the finite Bernoulli-form weights in  $K_{\mathbb{C}}$ , and keep real baseline probabilities  $p_j^0$ . For  $p_j^0 \in [\eta, 1 - \eta]$ , a coefficientwise complex extension exists if and only if each row  $(a_{j,\gamma}/\sqrt{p_j^0(1 - p_j^0)})_j$  is in complex  $\ell^2$ ; at the baseline  $p_j^0 = 0$  the exact condition is  $a_\gamma \in \ell^1(\mathbb{C})$ . These extensions are unique and have the support control and norm bounds of the real theorems. No positivity assertion is made for complex weights.*

*Proof.* Orthogonality (20.4) gives the same  $L^2$  estimate with squared moduli, so interior sufficiency is unchanged. For necessity, if  $T = \sum_j c_j Z_j$  with complex  $c_j$ , its real or imaginary part has variance at least  $\frac{1}{2} \sum_j |c_j|^2$ , and Lemma 20.3 for that part gives  $\mathbb{E}_0|T| \geq (2C_\eta)^{-1/2} (\sum_j |c_j|^2)^{1/2}$ ;

the least-bad-exponent argument is unchanged. At the endpoint the convolution series converges absolutely in the Banach space of finite complex measures with the same bound, and the first-order total variation is  $\sum_j |a_{j,\gamma}| + |\sum_j a_{j,\gamma}|$ . Uniqueness is the complex case of Theorem 19.1.  $\square$

**Proposition 24.2** (Conditioning on infinitesimal events; source 15). *Let  $\mu$  be a coefficientwise Hahn probability and  $B$  an event with  $\mu(B) > 0$ . Then  $\mu(A \mid B) = \mu(A \cap B)/\mu(B)$  is again a coefficientwise Hahn probability on the same space.*

*Proof.* Each  $A \mapsto \mu_\gamma(A \cap B)$  is a finite signed measure with the original common support, and  $c = \mu(B)^{-1}$  is a Hahn series. The sum of two well-ordered supports is well ordered with finitely many contributing pairs per exponent, so every coefficient of  $c\mu(\cdot \cap B)$  is a finite linear combination of finite signed measures, with a common well-ordered support. Positivity and the bound 1 follow from  $0 \leq \mu(A \cap B) \leq \mu(B)$ ; normalization is immediate; and any apparent negative-exponent coefficient must vanish, since the values lie in  $[0, 1]_{K_{\mathbb{R}}}$ .  $\square$

**Example 24.3** (The location of one rare success; source 15). Let  $p_j = c_j t^\alpha$  with  $c_j > 0$  and  $C = \sum_j c_j < \infty$ . The endpoint theorem gives a coefficientwise probability on finite-success paths. For the event  $B$  of at least one success,

$$\mu(B) = 1 - \prod_j^{\text{coef}} (1 - c_j t^\alpha) = C t^\alpha + \text{terms of larger exponent},$$

so conditioning on  $B$  is legitimate, and

$$\text{st } \mu(\cdot \mid B) = \sum_{j=1}^{\infty} \frac{c_j}{C} \delta_{e_j}. \quad (24.1)$$

Indeed the  $\alpha$ -coefficient of  $\mu(\cdot \cap B)$  is exactly  $\sum_j c_j \delta_{e_j}$ , since paths with two or more successes first occur at exponent at least  $2\alpha$ , and multiplication by  $\mu(B)^{-1}$  gives (24.1) at exponent zero. The standard part is an ordinary countably additive probability with infinitely many atoms, not a strong sum of constant series; this is Trap 2 of Section 2.6.

**Bounded ordinary observables.** For a bounded ordinary complex Borel function  $f$  and a coefficientwise Hahn measure  $\mu$  define

$$\int_X f d\mu = \sum_{\gamma \in W} \left( \int_X f d\mu_\gamma \right) t^\gamma \in K_{\mathbb{C}}, \quad (24.2)$$

well defined because its support lies in  $W$  and  $|\int f d\mu_\gamma| \leq \|f\|_\infty \|\mu_\gamma\|_{\text{TV}}$ . Finite products of coordinate indicators have the probabilities (1.1), by the marginal identities already proved. If a uniformly bounded sequence of ordinary observables converges pointwise, dominated convergence applies separately to every coefficient measure; the conclusion is coefficientwise convergence of the integrals, not convergence in the fine surreal topology. Integrating arbitrary Hahn-valued functions would need further joint support and integrability conditions and is not asserted. (For strong measures the integral (5.1) is defined for every scalar function, bounded or not.)

## Part VI

# Scope, questions, verification, and provenance

## 25 What is settled, and what is not claimed

### 25.1 What the report settles

**Strong class.** On countably separated sample spaces the representation problem for strong Hahn measures is solved: atomicity and one common support follow from the axiom (Theorem 4.2), the axiom is equivalent to finite atomicity of every coefficient (Corollary 4.3), and a strong measure is exactly a coefficientwise one with finitely atomic coefficients (Corollary 4.5). The extension problem is solved for coherent cylinder data over finite alphabets (Theorem 8.2) and for independent finite-state coordinates (Theorem 10.1), and the moment problem on ordinary real sample sets (Theorem 12.2). Positivity is a separate condition in each: the least support order type at which positivity of cylinder tests or of polynomial squares can fail to imply positivity of events is  $\omega + 1$  (Theorem 15.2), attained while the coefficient measures use only one or two nodes per scale. Several independent positivity tests survive the interval example: pointwise polynomial positivity, all Hermitian and localizing Hankel forms, all Hausdorff differences, and ordinary continuous functions; they quantify over every finite degree and every Hahn coefficient choice, and the obstruction is exposed by reconstructing one discontinuous event, not by a missed high-degree polynomial. The finite Gaussian theory stays intact because it is algebraic and admits nonreal nodes.

**Coefficientwise class.** Finite-dimensional consistency is not enough: consistent positive laws with bounded signed coefficient variations need not have a positive Borel extension (Example 19.5). For independent perturbations on a common well-ordered exponent set the exact sequence-space conditions are identified, and positivity is proved rather than assumed, by domination in the interior and by nonnegative exact path weights at the boundary (Theorems 20.1–22.1).

**A practical rule (source 15).** A model treating Hahn coefficients as finite ordinary measures can carry infinitely many independent infinitesimal perturbations, but the amplitudes must decay in the sequence space appropriate to the baseline; a constant nonzero infinitesimal correction repeated at every coordinate does not decay, and no smaller valuation scale repairs it. A model insisting on strong addition across every disjoint countable family pays more: each coefficient becomes finitely atomic, diffuse leading laws are excluded, and symmetry forces the atoms onto finite orbits. Strong countable additivity is therefore unsuitable as a direct replacement for ordinary additivity in diffuse stochastic processes, though it is consistent and useful for scale-separated atomic models such as  $p_j = t^{j\alpha}$ . Neither conclusion decrees a unique convention for surreal probability.

### 25.2 Non-claims, source by source

Every limitation stated by a source is kept. The table lists them, with the place where the report keeps each.

| No.              | Non-claim                                                                                                                                                                                                                                 | Kept at              |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <i>Source 10</i> |                                                                                                                                                                                                                                           |                      |
| 10.1             | The negative results concern (SH) only; they do not rule out finitely additive surreal probabilities, generalized-limit constructions, coefficientwise real-summed measures, or ultrametrically continuous measures on other event rings. | Sec. 5, Part V       |
| 10.2             | Not a first non-Archimedean probability theory; no previously named conjecture solved; its questions are its own, not from a named list.                                                                                                  | Title page, Sec. 26  |
| 10.3             | Not “a first non-Archimedean Kolmogorov theorem” (Ludkovsky–Khrennikov); here extended to source 15’s theorems.                                                                                                                           | Title page, Sec. 1.6 |
| 10.4             | The Benci–Horsten–Wenmackers continuity axiom is different; atomicity does not apply to their theory merely because its values are non-Archimedean.                                                                                       | Sec. 1.6             |
| 10.5             | Priority is provisional; archive interiors at the pin were unread; no theorem in an unread archive is declared absent; complete repository coverage is not claimed.                                                                       | Sec. 28              |
| 10.6             | Atomicity does not mean countably many atoms.                                                                                                                                                                                             | Rem. 4.8             |
| 10.7             | The extension to $\mathcal{P}(X)$ is not an extension theorem for Lebesgue measure.                                                                                                                                                       | After Thm. 4.2       |
| 10.8             | The integral is not Lebesgue integration and does not cover Hahn-valued integrands.                                                                                                                                                       | Sec. 5               |
| 10.9             | Lebesgue measure is not strong; the finite shadow does not exclude an ordinary real probability embedded by a different addition rule.                                                                                                    | Ex. 5.3              |
| 10.10            | Theorems are not to be transferred to more permissive rules without checking; a weaker rule gives up the finite-contribution property.                                                                                                    | Sec. 5               |
| 10.11            | Nothing is Lean-checked; finite checks cannot certify well ordering, width bounds or countable additivity.                                                                                                                                | Sec. 27              |
| 10.12            | Hahn fields and their summability calculus, Neumann’s lemma, the scalar Boolean-algebra argument, binary randomization and surreal normal forms are background, not claims.                                                               | Secs. 3, 9, 28       |
| 10.13            | The equal-infinitesimal-weight obstruction is prior (physics report).                                                                                                                                                                     | Sec. 1.5             |
| 10.14            | The criteria are not decision procedures on arbitrary coefficient oracles.                                                                                                                                                                | Question 26.3        |
| 10.15            | No topology on <b>No</b> is used and nothing is asserted about a measure on the full power class of <b>No</b> .                                                                                                                           | Sec. 7               |
| 10.16            | The normal-form map preserves the designated strong sums, not arbitrary topological convergence.                                                                                                                                          | Conv. 2.1            |
| 10.17            | The coarse-graining example does not contradict preservation of strong measures under pushforward.                                                                                                                                        | Sec. 11              |
| 10.18            | Not a replacement for all probability theories.                                                                                                                                                                                           | Sec. 1.6             |
| <i>Source 11</i> |                                                                                                                                                                                                                                           |                      |
| 11.1             | No certified priority, no named published open problem solved, not refereed, not formalized in Lean.                                                                                                                                      | Title page           |
| 11.2             | The hypotheses concern strong Hahn additivity; nothing refutes Benci–Horsten–Wenmackers or Brickhill–Horsten; Bottazzi’s and Kaiser’s settings are different; no reason to reject non-Archimedean integration.                            | Sec. 1.6             |
| 11.3             | The finite spectral theorem, Cauchy–Binet, Vandermonde interpolation and Gaussian quadrature are not new; the $n/(n+1)$ support is not new.                                                                                               | Secs. 16, 17, 1.5    |
| 11.4             | The moment criterion is not a decision algorithm; finitely many verified recurrence instances do not establish an infinite recurrence.                                                                                                    | Sec. 12              |
| 11.5             | Not every Hahn-valued function is integrable.                                                                                                                                                                                             | Sec. 5               |

| No.              | Non-claim                                                                                                                                                                                                                                                                                     | Kept at            |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 11.6             | The nonexistence in Theorem 14.3 concerns strong measures on the ordinary sample space; generalized limits, weaker additivity, other targets and nonreal sample points are not excluded, and a positive moment functional is not shown to lack every non-Archimedean integral representation. | After Thm. 14.3    |
| 11.7             | Not a counterexample to the classical real moment theorem.                                                                                                                                                                                                                                    | After Cor. 14.4    |
| 11.8             | The order bound is not continuity in the valuation or fine topology; the barrier is a valuation inequality, not a limit.                                                                                                                                                                      | Sec. 13            |
| 11.9             | Finite truncations are not proofs of the example.                                                                                                                                                                                                                                             | Sec. 14            |
| 11.10            | No assertion about every operator representation (Jacobi), nor about other operator topologies, additivity notions or spectra with nonreal Hahn points (Hermitian forms); no completion or Hilbert space.                                                                                     | Secs. 16, 18       |
| 11.11            | $\omega + 1$ is a support order type, not a birthday, and concerns the common family, not a single moment.                                                                                                                                                                                    | Rem. 15.3          |
| 11.12            | The quadratures give no strong measure on ordinary points and are not claimed to converge to one.                                                                                                                                                                                             | Sec. 17            |
| 11.13            | The localization uses no set of all surreals and no proper-class sum.                                                                                                                                                                                                                         | Sec. 18            |
| 11.14            | Repository search was an unusable index; negative searches are not evidence; the binary archives were not inspected.                                                                                                                                                                          | Sec. 28            |
| 11.15            | The multivariate barrier is a sufficient condition, not a classification.                                                                                                                                                                                                                     | Cor. 13.6          |
| 11.16            | The finite checks cannot prove the support theorem, quantify over all supports or degrees, or certify quadrature existence.                                                                                                                                                                   | Sec. 27            |
| 11.17            | Its further questions are its own formulations, not named open problems.                                                                                                                                                                                                                      | Sec. 26            |
| <i>Source 15</i> |                                                                                                                                                                                                                                                                                               |                    |
| 15.1             | Nothing excludes Benci–Horsten–Wenmackers, finite additivity, hyperfinite or ultrafilter probability; there is no identification with fine-topology convergence or ultrafilter sums; no unique convention is decreed.                                                                         | Secs. 1.6, 23, 25  |
| 15.2             | The $\ell^2$ threshold phenomenon is Kakutani’s; Riesz–Markov, Neumann’s lemma, the scalar bounded-variation extension, orthogonal-product estimates and lexicographic probability are classical inputs.                                                                                      | Thm. 19.1, Sec. 28 |
| 15.3             | Not Lean-verified, not peer reviewed; no named historical conjecture resolved.                                                                                                                                                                                                                | Title page         |
| 15.4             | Uniform interior separation is needed for $\ell^2$ necessity.                                                                                                                                                                                                                                 | Ex. 23.3           |
| 15.5             | No order or valuation continuity is supplied.                                                                                                                                                                                                                                                 | Sec. 23            |
| 15.6             | Not a classification of all non-Archimedean measures.                                                                                                                                                                                                                                         | Rem. 4.8           |
| 15.7             | No positivity claim for complex weights; no Hahn-valued integrands; coefficientwise convergence of integrals is not fine-topology convergence.                                                                                                                                                | Sec. 24            |
| 15.8             | Archives were not unpacked and historical manuscripts not read; the repository may have changed; searches are not certificates of absence.                                                                                                                                                    | Sec. 28            |
| 15.9             | Its check suite uses integer exponents and verifies no arbitrary-rank support lemma, transfinite argument, Riesz representation, infinite $L^2$ limit or all-event positivity.                                                                                                                | Sec. 27            |
| 15.10            | “Regular” never meant Brickhill–Horsten regularity.                                                                                                                                                                                                                                           | Conv. 2.11         |
| 15.11            | The results do not silently cover near-boundary baselines, several vanishing residues, Markov dependence, or definitions without a global well-ordered support.                                                                                                                               | Sec. 26            |
| 15.12            | “Candidate” qualifies originality, not a hidden omitted hypothesis.                                                                                                                                                                                                                           | Sec. 28            |
| 15.13            | Nothing contradicts classical exchangeability theory.                                                                                                                                                                                                                                         | After Cor. 6.3     |
| 15.14            | No summation of the per-exponent bounds over all exponents is used or available.                                                                                                                                                                                                              | Sec. 20            |
| 15.15            | Powers of $t^\alpha$ are never inferred to be cofinal.                                                                                                                                                                                                                                        | Rem. 9.2           |

Two limitations belong to the merge. The strong spine now has two independent derivations (sources 10 and 11), but neither was audited against the literature on vector measures with finite coefficient incidence, so it remains a candidate contribution. Corollary 4.5 and Remarks 14.5 and 19.6 were added in the merge to reconcile the sources; they are elementary consequences of the sources' theorems, and they are marked where they appear.

## 26 Open questions

The questions below were formulated by the three sources, not taken from a named list of open problems. Where a question is answered in part, the answer and its source are recorded and the question stays open, re-scoped.

**Question 26.1** (Standard Borel alphabets; source 10). Replace the finite alphabets of Theorem 8.2 by standard Borel coordinate spaces. Atomicity still holds (Theorem 4.2), but a finite-width cylinder tree no longer describes the extension problem directly; a finite atomic compatibility criterion would have to say how point locations are recovered from the chosen cylinder family. Open.

**Question 26.2** (A weaker addition rule; source 10). Find the exact extension theorem for an addition rule that permits convergent ordinary real coefficient sums, with the correct uniform support and variation conditions. *Answered in part* by source 15: for coherent data on  $E^{\mathbb{N}}$ ,  $E$  finite, with one common well-ordered support assumed, per-exponent bounded variation is exact (Theorem 19.1) and the null-ideal criterion decides positivity (Theorem 19.2). Open without the common-support assumption, which is source 15's item "definitions without a global well-ordered support" (Question 26.10).

**Question 26.3** (Effective versions; sources 10 and 11). The criteria of Theorems 8.2, 10.1 and 12.2 are necessary and sufficient mathematical conditions, not decision procedures on arbitrary coefficient oracles; finite-prefix verification cannot decide well ordering or eventual width bounds. Which restricted representations with certificates for these infinite conditions make them decidable? Open.

**Question 26.4** (Positive functionals with extra continuity; source 11). Classify positive functionals on  $C(\Omega, \mathbb{R})$  with Hahn values under continuity conditions strong enough to recover a positive representing object but weak enough to include ordinary diffuse probability. Theorem 4.2 shows that the strong axiom on all Borel events cannot include diffuse laws. In the coefficientwise class, per-exponent boundedness of the coefficient functionals gives a unique signed representing coefficientwise measure (the Riesz–Markov step of Theorem 19.1, applied exponent by exponent), and Theorem 19.2 decides whether that measure is positive. But positivity of the functional does not suffice: the functional  $T$  of Theorem 14.3(iii) is positive on  $C([0, 1], \mathbb{R})$  with per-exponent bounded coefficients, and its representing measure is not positive in either class (Remark 14.5). A continuity condition on the functional alone that guarantees a positive representing measure is not established. Open, re-scoped.

**Question 26.5** (Hahn-valued sample points; source 11). Study measures on Hahn-valued sample points. The finite-coefficient Vandermonde argument does not directly reduce moments to finite measures at ordinary nodes, since a nonreal sample point has its own exponent expansion and mixes scales under powers; the positive quadratures of Theorem 17.3 give concrete data. No general existence or uniqueness claim is made. The collection's Prony report [17] treats the different finite problem of reconstructing a finite weighted configuration with Hahn-valued nodes from its power moments. Open.

**Question 26.6** (Multivariate separation; source 11). Which algebraic sets of earlier atoms permit a prescribed later signed tail to remain positive on a given polynomial cone? Corollary 13.6 gives a sufficient condition only. Open.

**Question 26.7** (Near-boundary baselines; source 15). When  $p_j^0$  approaches 0 or 1 without reaching it, Theorem 19.1 remains the exact abstract answer, but no weighted- $\ell^2$  condition is necessary (Example 23.3). Find an intrinsic sequence-space description distinguishing rare large scores from uniformly nondegenerate ones and controlling all nonlinear coefficient terms. Open.

**Question 26.8** (Several vanishing residues; source 15). Extend the coefficientwise thresholds to finite alphabets with several vanishing residue probabilities. In the strong class this is answered, for arbitrary finite alphabets and residue patterns, by Theorem 10.1 (source 10). Open in the coefficientwise class.

**Question 26.9** (Markov dependence; source 15). Replace independence by dependence governed by a transition kernel. Open in both classes. The collection's report on finite-state Hahn Markov generators [18] excludes path measures and does not treat this extension problem.

**Question 26.10** (No global well-ordered support; source 15). Find definitions of countable additivity without a global well-ordered support across all events. For the strong axiom (SH) the question is answered: on countably separated spaces the common support is automatic (Theorem 4.2, sources 10 and 11), and even finite coefficient atomicity alone forces it (Corollary 4.3). Open for coefficientwise and other rules.

## 27 The three check suites

Each source delivered a standard-library Python program using exact rational arithmetic; they are kept byte-identical under `code/`, with their recorded outputs under `data/`. They check finite identities that detect algebraic mistakes. They do not prove any infinite statement, cannot establish that a support is well ordered or that infinitely many widths are bounded, and certify neither novelty nor Lean correctness.

| Program                                                      | Assertions | What it checks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>10-strong-hahn-measures-verify.py</code>               | 5,135      | The scalar nonvanishing probability bound behind the randomized selection (30); finite coefficient trees, width monotonicity and reconstruction (3,600); finite independent products and first-error recovery (408); finite truncations of the Cantor-space pattern, including that the cutoff reveals the negative atom (1,035); the rank-two coarse-graining coefficient witness $\frac{1}{2}$ along descending $\gamma_n$ (62). Exponents are lexicographic rational tuples.                                                                                                                                      |
| <code>11-hahn-moments-verify.py</code>                       | 682        | Determinants against the Cauchy–Binet expansion (68) and predicted valuations and leading coefficients (68); square and localizer positivity for 96 seeded polynomials with finite Hahn coefficients and four localizers (384); simple-root coefficient recurrences (90); negative detectors for finite truncations (8) and restoration by the next atom (8); tent-function leading terms (8); the closed Vandermonde formula through $d = 12$ (13); harmonic-number valuations (13); Jacobi valuation differences (12); normalization and the negative singleton (10). Rational exponents; seed 20260922.           |
| <code>15-infinite-products-of-probabilities-verify.py</code> | 1,009      | Finite positivity (186), normalization (15) and marginal consistency (93); the ordered-word/ $(1/k!)$ identity against direct products (128); the multilinear $L^2$ bound (39); first-order variations at interior and boundary baselines (8, 8, 5) and the two-sided bounds of Lemma 20.3 at the fair baseline (8, recorded under the name <code>interior_moment_bounds</code> ); the cylinders of Example 19.5 (510) and its normalization (8); and one check ( <code>counterexample_singleton_formula</code> ) that only asserts that a hard-coded value is negative, so is tautological. Integer exponents only. |

All three were rerun on a scratch copy of this directory while the report was written (Python 3.14.4). All assertions passed with the recorded counts; the outputs agree with the delivered records except for the recorded Python version (source 10) and line endings (source 11). Two of the programs write files, so run them on a copy: `10-strong-hahn-measures-verify.py` writes `verification.json` and `verification.txt` into the `data/` directory next to its own `code/` directory (unprefixed names, beside the prefixed records), and `11-hahn-moments-verify.py` writes `verification_results.json` and `verification_summary.txt` into its own directory unless `--output-dir` is given. `15-infinite-products-of-probabilities-verify.py` prints to standard output and writes a file only with `--output`. The delivered build scripts (`code/*-build.sh`) are kept as delivered; they refer to the source packages’ own layouts and do not build this report. The file `15-infinite-products-of-probabilities-build_summary.json` is source 15’s hand-recorded note on its own 23-page build; no program generates it.

## 28 Provenance, audit, and attribution

### 28.1 Repository snapshot and what the sources inspected

All three sources inspected VladimirReshetnikov/Surreal at the snapshot `4cf691c7d951e037739d32d9f5c387dcce724f3c` (22 September 2026) and wrote no repository files. Source 10 read the root and catalogue READMEs, directory listings, and a prior physics text;

its code search found no Kolmogorov match while an API search reported incomplete indexing. Source 11 read the root README, the catalogue, directory listings and the opening of the spectral-theory article; its search index returned no hits even for a control term known to occur, so it treated searches as unusable. Source 15 read the root README, the catalogue, the opening of docs/FORMALIZATION.md and directory listings; its indexed searches for Kolmogorov and probability returned no matches. At the pin the catalogue listed 26 reports and docs/new held 18 archives whose interiors none of the sources could fully read; source 10 named surreal\_symbolic\_dynamics.zip and the infinite-spectral archives as possible overlaps. Those archives have since been placed as reports of the same batch as the three sources. A search of the collection made while this report was written finds no other treatment of Hahn-valued measures than the Herglotz report [16], no occurrence of Kolmogorov outside this report, and no probability or invariant-measure content in the symbolic-dynamics report (surcomplex/expanding-polynomial-dynamics); the infinite-dimensional spectral report explicitly supplies no spectral measures. The delivered audit records 10-strong-hahn-measures-provenance.md and 15-infinite-products-of-probabilities-RESEARCH\_AUDIT.md are kept as written at the pin; source 11's audit was an appendix of its article and is summarized here.

## 28.2 Stale statements corrected

- (1) Source 15 reported that indexed searches for probability returned no matches. At the pin the physics report already contained `phys:sub:probability` and `phys:ex:summable`, on the failure of equal infinitesimal weights and the need for a summation theory; its scope table lists a probability theory among things to be supplied, and the spectral-theory report states that no measure-valued probability theory is supplied. These are cited (Section 1.5). Source 15's statement that the catalogue listed no probability report was correct at the pin; the collection now contains this report and the Herglotz report.
- (2) Source 11 reproved the finite spectral theorem over a real closed field; the collection has it (`thm:spectral`, and `spec:thm:hermitian` at any  $\Gamma$ ), and it is cited (Section 17).
- (3) Source 11 found no repository matches for moment-related terms and treated its index as unusable. The collection does use “moment” in other senses (contour, residue, Schoenberg and inverse moments in other surcomplex reports, and now Fourier moments in the Herglotz report and finite-configuration power moments in the Prony report); none is a Hahn-valued measure moment problem (Convention 2.11).
- (4) Source 10 cited the physics discussion as Section 13.3 of a PDF; it is now cited by its labels in `physics/surreal-scalars-and-spacetime`.
- (5) Source 10's statement that the randomized selection argument is needed for the coefficientwise criterion is withdrawn (Remark 4.4).
- (6) Source 15 did not cite Ludkovsky and Khrennikov; source 10's warning against “a first non-Archimedean Kolmogorov theorem” is extended to source 15's extension theorems (Section 1.6).

## 28.3 A result-by-result attribution ledger

| Result                                                                                     | Status and source                                                  |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Hahn arithmetic, strong sums, $n/(n+1)$ example, Neumann lemma, normal forms, localization | Standard or prior collection material [12, 8, 9, 10]; not claimed. |

| Result                                                                                                    | Status and source                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Scalar lemma (Lemma 3.2)                                                                                  | Elementary; proved identically by 10 and 11; 15's compact-metric special case is Remark 3.3.                                       |
| Automatic common support, atomic classification, coefficient test (Lemma 4.1, Theorem 4.2, Corollary 4.3) | Candidate contribution, found independently by 10 and 11 (and, with a common support assumed, by 15); deterministic proof from 11. |
| Finite shadow, cardinality, countable-separation example                                                  | 10 (shadow also 11).                                                                                                               |
| Finite-orbit rigidity, exchangeability                                                                    | 15, re-derived on countably separated spaces; transitive case also 11.                                                             |
| Cylinder criterion, positivity test, independent criterion, coarse-graining, enlargement                  | Candidate contributions of 10.                                                                                                     |
| Moment criterion, visibility, barriers, interval example, Hausdorff inequalities                          | Candidate contributions of 11.                                                                                                     |
| Weight pattern and $\omega + 1$ threshold                                                                 | Found independently by 10 (cylinders) and 11 (polynomials).                                                                        |
| Hankel scales, Jacobi scales, residue nodes                                                               | Exact consequences for the example, 11; Cauchy–Binet, Heine, Gaussian quadrature and the spectral theorem are classical.           |
| Coefficientwise extension test                                                                            | Riesz–Markov per exponent; 15; not new as a scalar theorem.                                                                        |
| Null-ideal criterion                                                                                      | Candidate formulation; 15, and independently 18.                                                                                   |
| $\ell^2$ , $\ell^1$ , mixed thresholds, i.i.d. corollary, near-boundary example                           | Candidate contributions of 15; the $\ell^2$ phenomenon for real products is Kakutani's [6].                                        |
| Two-axiom equivalence, one-exponent remark, interval example in the coefficientwise class                 | Added in the merge; elementary.                                                                                                    |
| Named open conjecture; priority; refereeing; Lean                                                         | None claimed.                                                                                                                      |

## 28.4 A formalization path

Nothing here is formalized. Source 10 suggests: the scalar lemma via the  $\sigma$ -ideal of hereditarily null sets and its Boolean quotient, or by reusing finite atomic scalar measure results; the coefficient tree lemma from finite branching, monotone natural-number widths and stabilization of bounded monotone sequences; the Hahn stages against an interface providing coefficient maps, strong-family regrouping, positive-support word finiteness and geometric inversion. Source 15's dependency order for the coefficientwise class is: scalar signed-measure extension; finite-product coefficient identities; the multilinear norm bounds; common-support Hahn assembly; and the separate positivity arguments.

## 29 Reproducing the document and the checks

The directory contains this standalone L<sup>A</sup>T<sub>E</sub>X source with an embedded bibliography, its compiled PDF, a README, the three delivered programs and build scripts under `code/`, their recorded outputs under `data/`, and the two delivered audit records. No source manuscript is redistributed. From a copy of the directory:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
python code/10-strong-hahn-measures-verify.py
python code/11-hahn-moments-verify.py --output-dir rerun-11
```

```
python code/15-infinite-products-of-probabilities-verify.py \  
    --output rerun-15.json
```

Python 3.10 or later suffices; only the standard library is used.

## References

- [1] V. Benci, L. Horsten and S. Wenmackers, *Non-Archimedean probability*, Milan Journal of Mathematics **81** (2013), 121–151. doi:10.1007/s00032-012-0191-x; preprint arXiv:1106.1524, Section 3.1 used for the axiom comparison.
- [2] H. Brickhill and L. Horsten, *Popper functions, lexicographical probability, and non-Archimedean probability*, preprint (2016), arXiv:1608.02850.
- [3] S. Ludkovsky and A. Khrennikov, *Stochastic processes on non-Archimedean spaces with values in non-Archimedean fields*, preprint (2001), arXiv:math/0110305; Definitions 2.1–2.4 and Theorem 2.15.2 of the retrieved version.
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- [7] K. Schmüdgen, *Ten lectures on the moment problem*, preprint (2020), arXiv:2008.12698; Lecture 1, Theorems 1.9, 1.16 and 1.18.
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- [10] O. Bournez and Q. Guilmant, *Surreal fields stable under exponential and logarithmic functions*, preprint (2022), arXiv:2201.08199; Section 2.3 on Hahn fields and normal forms.
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- [12] VladimirReshetnikov/Surreal project, *Foundations*, report docs/foundations-and-computation/foundations/; labels found:eq:summability, found:ex:boundedrankone, found:sub:positivesupport, found:eq:workspaces, found:thm:workspace.
- [13] VladimirReshetnikov/Surreal project, *Surreal scalars and spacetime*, report docs/physics/surreal-scalars-and-spacetime/; labels phys:ex:summable, phys:sub:probability.
- [14] VladimirReshetnikov/Surreal project, *Surcomplex spectral theory: singular values, infinitesimal rank, and multiscale stability*, report docs/surcomplex/spectral-theory/; theorems thm:spectral and spec:thm:hermitian.
- [15] VladimirReshetnikov/Surreal project, *Dynamics and normal forms*, report docs/surcomplex/dynamics-and-normal-forms/; dyn:thm:qpdensity, dyn:subsec:qpdensity and non-claim N75.

- [16] VladimirReshetnikov/Surreal project, report  
docs/surcomplex/hahn-herglotz-positivity/, from source 18 of the same batch  
(*Positivity without a positive measure: Hahn–Herglotz normalization and strict moment hierarchies in surcomplex analysis*).
- [17] VladimirReshetnikov/Surreal project, report  
docs/surcomplex/prony-reconstruction-at-surreal-scales/.
- [18] VladimirReshetnikov/Surreal project, report  
docs/surreal/markov-generators-at-every-scale/.